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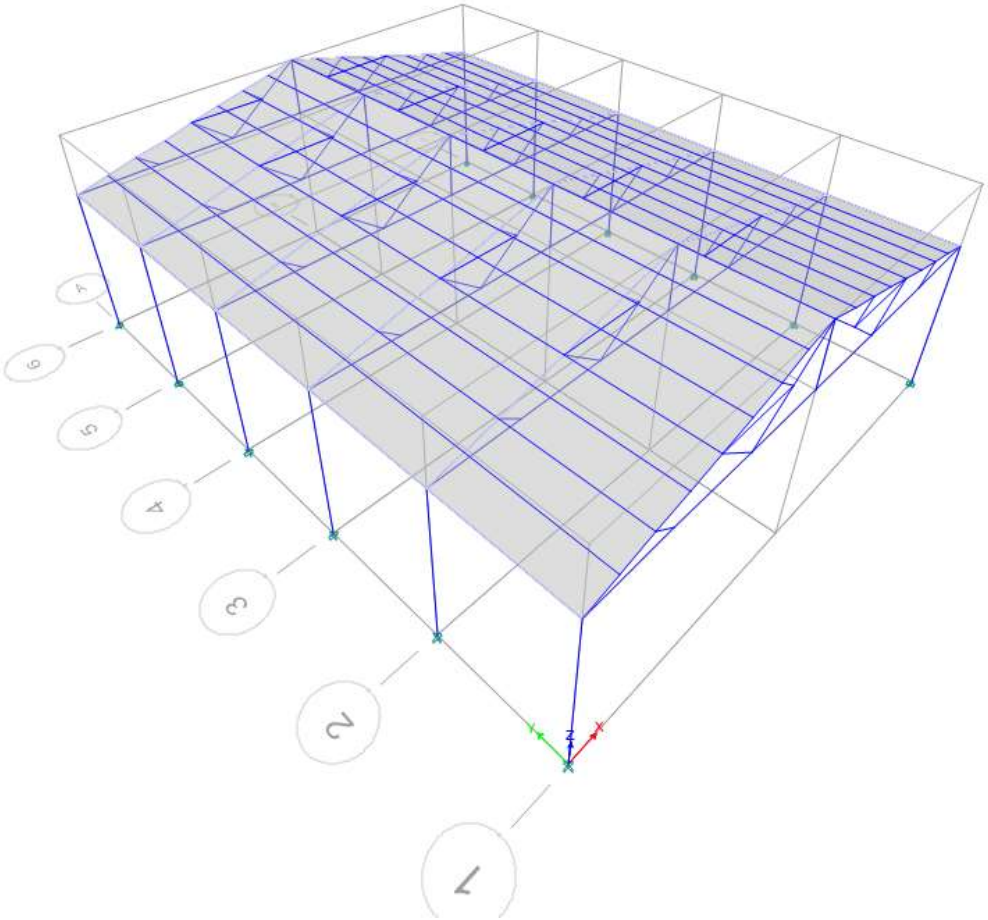

Project Report

Model File: Basir sir assignment, Revision 0
1/4/2026

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1 Structure Data

This chapter provides model geometry information, including items such as story levels, point coordinates, and element connectivity.

1.1 Story Data

Table 1.1 - Story Definitions

Tower	Name	Height ft	Master Story	Similar To	Splice Story	Splice Height ft	Color
T1	Story2	6	Yes	None	No		Green
T1	Story1	15	No	Story2	No		Cyan

1.2 Grid Data

Table 1.2 - Grid Definitions - General

Tower	Name	Type	Ux ft	Uy ft	Rz deg	Story Range	User Top Story	User Bottom Story	Bubble Size in	Color
T1	G1	Cartesian	0	0	0	Default			60	Gray6

Table 1.3 - Grid Definitions - Grid Lines

Name	Grid Line Type	ID	Ordinate ft	Angle deg	X1 ft	Y1 ft	X2 ft	Y2 ft	Bubble Location	Visible
G1	X (Cartesian)	A	0						End	Yes
G1	X (Cartesian)	B	24						End	Yes
G1	X (Cartesian)	C	48						End	Yes
G1	Y (Cartesian)	1	0						Start	Yes
G1	Y (Cartesian)	2	12						Start	Yes
G1	Y (Cartesian)	3	24						Start	Yes
G1	Y (Cartesian)	4	36						Start	Yes
G1	Y (Cartesian)	5	48						Start	Yes
G1	Y (Cartesian)	6	60						Start	Yes

1.3 Point Coordinates

Table 1.4 - Point Bays

Label	Is Auto Point	X ft	Y ft	DZBelow ft
1	No	0	0	0
2	No	48	0	0
3	No	24	0	0
4	No	8	0	0
5	No	16	0	0
6	No	6	0	4.5
7	No	12	0	3
8	No	18	0	1.5
9	No	20	0	3
10	No	40	0	0
11	No	32	0	0
12	No	42	0	4.5
13	No	36	0	3
14	No	30	0	1.5
15	No	28	0	3

Label	Is Auto Point	X ft	Y ft	DZBelow ft
16	No	3	0	5.25
17	No	9	0	3.75
18	No	15	0	2.25
19	No	21	0	0.75
20	No	45	0	5.25
21	No	39	0	3.75
22	No	33	0	2.25
23	No	27	0	0.75
24	No	0	12	0
25	No	48	12	0
26	No	24	12	0
27	No	8	12	0
28	No	16	12	0
29	No	6	12	4.5
30	No	12	12	3
31	No	18	12	1.5
32	No	20	12	3
33	No	40	12	0
34	No	32	12	0
35	No	42	12	4.5
36	No	36	12	3
37	No	30	12	1.5
38	No	28	12	3
39	No	3	12	5.25
40	No	9	12	3.75
41	No	15	12	2.25
42	No	21	12	0.75
43	No	45	12	5.25
44	No	39	12	3.75
45	No	33	12	2.25
46	No	27	12	0.75
47	No	0	24	0
48	No	48	24	0
49	No	24	24	0
50	No	8	24	0
51	No	16	24	0
52	No	6	24	4.5
53	No	12	24	3
54	No	18	24	1.5
55	No	20	24	3
56	No	40	24	0
57	No	32	24	0
58	No	42	24	4.5
59	No	36	24	3
60	No	30	24	1.5
61	No	28	24	3
62	No	3	24	5.25
63	No	9	24	3.75
64	No	15	24	2.25
65	No	21	24	0.75
66	No	45	24	5.25
67	No	39	24	3.75
68	No	33	24	2.25
69	No	27	24	0.75

Label	Is Auto Point	X ft	Y ft	DZBelow ft
70	No	0	36	0
71	No	48	36	0
72	No	24	36	0
73	No	8	36	0
74	No	16	36	0
75	No	6	36	4.5
76	No	12	36	3
77	No	18	36	1.5
78	No	20	36	3
79	No	40	36	0
80	No	32	36	0
81	No	42	36	4.5
82	No	36	36	3
83	No	30	36	1.5
84	No	28	36	3
85	No	3	36	5.25
86	No	9	36	3.75
87	No	15	36	2.25
88	No	21	36	0.75
89	No	45	36	5.25
90	No	39	36	3.75
91	No	33	36	2.25
92	No	27	36	0.75
93	No	0	48	0
94	No	48	48	0
95	No	24	48	0
96	No	8	48	0
97	No	16	48	0
98	No	6	48	4.5
99	No	12	48	3
100	No	18	48	1.5
101	No	20	48	3
102	No	40	48	0
103	No	32	48	0
104	No	42	48	4.5
105	No	36	48	3
106	No	30	48	1.5
107	No	28	48	3
108	No	3	48	5.25
109	No	9	48	3.75
110	No	15	48	2.25
111	No	21	48	0.75
112	No	45	48	5.25
113	No	39	48	3.75
114	No	33	48	2.25
115	No	27	48	0.75
116	No	0	60	0
117	No	48	60	0
118	No	24	60	0
119	No	8	60	0
120	No	16	60	0
121	No	6	60	4.5
122	No	12	60	3
123	No	18	60	1.5

Label	Is Auto Point	X ft	Y ft	DZBelow ft
124	No	20	60	3
125	No	40	60	0
126	No	32	60	0
127	No	42	60	4.5
128	No	36	60	3
129	No	30	60	1.5
130	No	28	60	3
131	No	3	60	5.25
132	No	9	60	3.75
133	No	15	60	2.25
134	No	21	60	0.75
135	No	45	60	5.25
136	No	39	60	3.75
137	No	33	60	2.25
138	No	27	60	0.75

1.4 Line Connectivity

Table 1.5 - Column Bays

Label	PointBayI	PointBayJ	IEndStory
C1	1	1	Below
C2	2	2	Below
C3	3	3	Below
C4	24	24	Below
C5	25	25	Below
C6	26	26	Below
C7	47	47	Below
C8	48	48	Below
C9	49	49	Below
C10	70	70	Below
C11	71	71	Below
C12	72	72	Below
C13	93	93	Below
C14	94	94	Below
C15	95	95	Below
C16	116	116	Below
C17	117	117	Below
C18	118	118	Below

Table 1.6 - Beam Bays

Label	PointBayI	PointBayJ
B3	1	4
B4	4	5
B5	5	3
B10	7	9
B11	10	2
B12	11	10
B13	3	11
B18	15	13
B19	1	16
B20	16	6
B21	6	17
B22	17	7

Label	PointBayI	PointBayJ
B23	7	18
B24	18	8
B25	8	19
B26	19	3
B27	2	20
B28	20	12
B29	12	21
B30	21	13
B31	13	22
B32	22	14
B33	14	23
B34	23	3
B35	24	27
B36	27	28
B37	28	26
B38	30	32
B39	33	25
B40	34	33
B41	26	34
B42	38	36
B43	24	39
B44	39	29
B45	29	40
B46	40	30
B47	30	41
B48	41	31
B49	31	42
B50	42	26
B51	25	43
B52	43	35
B53	35	44
B54	44	36
B55	36	45
B56	45	37
B57	37	46
B58	46	26
B59	47	50
B60	50	51
B61	51	49
B62	53	55
B63	56	48
B64	57	56
B65	49	57
B66	61	59
B67	47	62
B68	62	52
B69	52	63
B70	63	53
B71	53	64
B72	64	54
B73	54	65
B74	65	49
B75	48	66
B76	66	58
B77	58	67
B78	67	59

Label	PointBayI	PointBayJ
B79	59	68
B80	68	60
B81	60	69
B82	69	49
B83	70	73
B84	73	74
B85	74	72
B86	76	78
B87	79	71
B88	80	79
B89	72	80
B90	84	82
B91	70	85
B92	85	75
B93	75	86
B94	86	76
B95	76	87
B96	87	77
B97	77	88
B98	88	72
B99	71	89
B100	89	81
B101	81	90
B102	90	82
B103	82	91
B104	91	83
B105	83	92
B106	92	72
B107	93	96
B108	96	97
B109	97	95
B110	99	101
B111	102	94
B112	103	102
B113	95	103
B114	107	105
B115	93	108
B116	108	98
B117	98	109
B118	109	99
B119	99	110
B120	110	100
B121	100	111
B122	111	95
B123	94	112
B124	112	104
B125	104	113
B126	113	105
B127	105	114
B128	114	106
B129	106	115
B130	115	95
B131	116	119
B132	119	120
B133	120	118
B134	122	124

Label	PointBayI	PointBayJ
B135	125	117
B136	126	125
B137	118	126
B138	130	128
B139	116	131
B140	131	121
B141	121	132
B142	132	122
B143	122	133
B144	133	123
B145	123	134
B146	134	118
B147	117	135
B148	135	127
B149	127	136
B150	136	128
B151	128	137
B152	137	129
B153	129	138
B154	138	118
B155	1	24
B156	24	47
B157	47	70
B158	70	93
B159	93	116
B160	2	25
B161	25	48
B162	48	71
B163	71	94
B164	94	117
B165	3	26
B166	26	49
B167	49	72
B168	72	95
B169	95	118
B170	6	29
B171	29	52
B172	52	75
B173	75	98
B174	98	121
B175	7	30
B176	30	53
B177	53	76
B178	76	99
B179	99	122
B180	8	31
B181	31	54
B182	54	77
B183	77	100
B184	100	123
B185	12	35
B186	35	58
B187	58	81
B188	81	104
B189	104	127
B190	13	36

Label	PointBayI	PointBayJ
B191	36	59
B192	59	82
B193	82	105
B194	105	128
B195	14	37
B196	37	60
B197	60	83
B198	83	106
B199	106	129
B200	16	39
B201	39	62
B202	62	85
B203	85	108
B204	108	131
B205	17	40
B206	40	63
B207	63	86
B208	86	109
B209	109	132
B210	18	41
B211	41	64
B212	64	87
B213	87	110
B214	110	133
B215	19	42
B216	42	65
B217	65	88
B218	88	111
B219	111	134
B220	20	43
B221	43	66
B222	66	89
B223	89	112
B224	112	135
B225	21	44
B226	44	67
B227	67	90
B228	90	113
B229	113	136
B230	22	45
B231	45	68
B232	68	91
B233	91	114
B234	114	137
B235	23	46
B236	46	69
B237	69	92
B238	92	115
B239	115	138

Table 1.7 - Brace Bays

Label	PointBayI	PointBayJ	IEndStory
D1	4	6	Below
D3	5	7	Below
D5	5	9	Below
D6	9	3	Same

Label	PointBayI	PointBayJ	IEndStory
D7	9	8	Same
D8	10	12	Below
D9	11	13	Below
D10	11	15	Below
D11	15	3	Same
D12	15	14	Same
D13	27	29	Below
D14	28	30	Below
D15	28	32	Below
D16	32	26	Same
D17	32	31	Same
D18	33	35	Below
D19	34	36	Below
D20	34	38	Below
D21	38	26	Same
D22	38	37	Same
D23	50	52	Below
D24	51	53	Below
D25	51	55	Below
D26	55	49	Same
D27	55	54	Same
D28	56	58	Below
D29	57	59	Below
D30	57	61	Below
D31	61	49	Same
D32	61	60	Same
D33	73	75	Below
D34	74	76	Below
D35	74	78	Below
D36	78	72	Same
D37	78	77	Same
D38	79	81	Below
D39	80	82	Below
D40	80	84	Below
D41	84	72	Same
D42	84	83	Same
D43	96	98	Below
D44	97	99	Below
D45	97	101	Below
D46	101	95	Same
D47	101	100	Same
D48	102	104	Below
D49	103	105	Below
D50	103	107	Below
D51	107	95	Same
D52	107	106	Same
D53	119	121	Below
D54	120	122	Below
D55	120	124	Below
D56	124	118	Same
D57	124	123	Same
D58	125	127	Below
D59	126	128	Below
D60	126	130	Below
D61	130	118	Same
D62	130	129	Same

1.5 Area Connectivity

Table 1.8 - Floor Bays

Label	NumPoints	PointNumber	PointBay
F1	4	1	24
F1		2	1
F1		3	16
F1		4	39
F2	4	1	47
F2		2	24
F2		3	39
F2		4	62
F3	4	1	70
F3		2	47
F3		3	62
F3		4	85
F4	4	1	93
F4		2	70
F4		3	85
F4		4	108
F5	4	1	116
F5		2	93
F5		3	108
F5		4	131
F6	4	1	39
F6		2	16
F6		3	6
F6		4	29
F7	4	1	62
F7		2	39
F7		3	29
F7		4	52
F8	4	1	85
F8		2	62
F8		3	52
F8		4	75
F9	4	1	108
F9		2	85
F9		3	75
F9		4	98
F10	4	1	131
F10		2	108
F10		3	98
F10		4	121
F11	4	1	29
F11		2	6
F11		3	17
F11		4	40
F12	4	1	52
F12		2	29
F12		3	40
F12		4	63
F13	4	1	75
F13		2	52
F13		3	63
F13		4	86
F14	4	1	98

Label	NumPoints	PointNumber	PointBay
F14		2	75
F14		3	86
F14		4	109
F15	4	1	121
F15		2	98
F15		3	109
F15		4	132
F16	4	1	40
F16		2	17
F16		3	7
F16		4	30
F17	4	1	63
F17		2	40
F17		3	30
F17		4	53
F18	4	1	86
F18		2	63
F18		3	53
F18		4	76
F19	4	1	109
F19		2	86
F19		3	76
F19		4	99
F20	4	1	132
F20		2	109
F20		3	99
F20		4	122
F21	4	1	30
F21		2	7
F21		3	18
F21		4	41
F22	4	1	53
F22		2	30
F22		3	41
F22		4	64
F23	4	1	76
F23		2	53
F23		3	64
F23		4	87
F24	4	1	99
F24		2	76
F24		3	87
F24		4	110
F25	4	1	122
F25		2	99
F25		3	110
F25		4	133
F26	4	1	41
F26		2	18
F26		3	8
F26		4	31
F27	4	1	64
F27		2	41
F27		3	31
F27		4	54
F28	4	1	87

Label	NumPoints	PointNumber	PointBay
F28		2	64
F28		3	54
F28		4	77
F29	4	1	110
F29		2	87
F29		3	77
F29		4	100
F30	4	1	133
F30		2	110
F30		3	100
F30		4	123
F31	4	1	31
F31		2	8
F31		3	19
F31		4	42
F32	4	1	54
F32		2	31
F32		3	42
F32		4	65
F33	4	1	77
F33		2	54
F33		3	65
F33		4	88
F34	4	1	100
F34		2	77
F34		3	88
F34		4	111
F35	4	1	123
F35		2	100
F35		3	111
F35		4	134
F36	4	1	42
F36		2	19
F36		3	3
F36		4	26
F37	4	1	65
F37		2	42
F37		3	26
F37		4	49
F38	4	1	88
F38		2	65
F38		3	49
F38		4	72
F39	4	1	111
F39		2	88
F39		3	72
F39		4	95
F40	4	1	134
F40		2	111
F40		3	95
F40		4	118
F81	4	1	43
F81		2	20
F81		3	2
F81		4	25
F82	4	1	66

Label	NumPoints	PointNumber	PointBay
F82		2	43
F82		3	25
F82		4	48
F83	4	1	89
F83		2	66
F83		3	48
F83		4	71
F84	4	1	112
F84		2	89
F84		3	71
F84		4	94
F85	4	1	135
F85		2	112
F85		3	94
F85		4	117
F86	4	1	35
F86		2	12
F86		3	20
F86		4	43
F87	4	1	58
F87		2	35
F87		3	43
F87		4	66
F88	4	1	81
F88		2	58
F88		3	66
F88		4	89
F89	4	1	104
F89		2	81
F89		3	89
F89		4	112
F90	4	1	127
F90		2	104
F90		3	112
F90		4	135
F91	4	1	44
F91		2	21
F91		3	12
F91		4	35
F92	4	1	67
F92		2	44
F92		3	35
F92		4	58
F93	4	1	90
F93		2	67
F93		3	58
F93		4	81
F94	4	1	113
F94		2	90
F94		3	81
F94		4	104
F95	4	1	136
F95		2	113
F95		3	104
F95		4	127
F96	4	1	36

Label	NumPoints	PointNumber	PointBay
F96		2	13
F96		3	21
F96		4	44
F97	4	1	59
F97		2	36
F97		3	44
F97		4	67
F98	4	1	82
F98		2	59
F98		3	67
F98		4	90
F99	4	1	105
F99		2	82
F99		3	90
F99		4	113
F100	4	1	128
F100		2	105
F100		3	113
F100		4	136
F101	4	1	45
F101		2	22
F101		3	13
F101		4	36
F102	4	1	68
F102		2	45
F102		3	36
F102		4	59
F103	4	1	91
F103		2	68
F103		3	59
F103		4	82
F104	4	1	114
F104		2	91
F104		3	82
F104		4	105
F105	4	1	137
F105		2	114
F105		3	105
F105		4	128
F106	4	1	37
F106		2	14
F106		3	22
F106		4	45
F107	4	1	60
F107		2	37
F107		3	45
F107		4	68
F108	4	1	83
F108		2	60
F108		3	68
F108		4	91
F109	4	1	106
F109		2	83
F109		3	91
F109		4	114
F110	4	1	129

Label	NumPoints	PointNumber	PointBay
F110		2	106
F110		3	114
F110		4	137
F111	4	1	46
F111		2	23
F111		3	14
F111		4	37
F112	4	1	69
F112		2	46
F112		3	37
F112		4	60
F113	4	1	92
F113		2	69
F113		3	60
F113		4	83
F114	4	1	115
F114		2	92
F114		3	83
F114		4	106
F115	4	1	138
F115		2	115
F115		3	106
F115		4	129
F116	4	1	26
F116		2	3
F116		3	23
F116		4	46
F117	4	1	49
F117		2	26
F117		3	46
F117		4	69
F118	4	1	72
F118		2	49
F118		3	69
F118		4	92
F119	4	1	95
F119		2	72
F119		3	92
F119		4	115
F120	4	1	118
F120		2	95
F120		3	115
F120		4	138

1.6 Mass

Table 1.9 - Mass Source Definition (Part 1 of 2)

Name	Is Default	Include Lateral Mass?	Include Vertical Mass?	Lump Mass?	Source Self Mass?	Source Added Mass?	Source Load Patterns?	Move Mass Centroid?	Move Ratio X	Move Ratio Y
MsSrc1	Yes	Yes	No	Yes	Yes	Yes	No	No		

Table 1.9 - Mass Source Definition (Part 2 of 2)

Name	Load Pattern	Multiplier

Name	Load Pattern	Multiplier
MsSrc1		

Table 1.10 - Mass Summary by Story

Story	UX lb-s2/ft	UY lb-s2/ft	UZ lb-s2/ft
Story2	424.36	424.36	0
Story1	678.1	678.1	0
Base	82.2	82.2	0

Table 1.11 - Mass Summary by Group

Group	Self Mass lb-s2/ft	Self Weight kip	Mass X lb-s2/ft	Mass Y lb-s2/ft	Mass Z lb-s2/ft
All	1184.66	0	1184.66	1184.66	0

2 Properties

This chapter provides property information for materials, frame sections, shell sections, and links.

2.1 Materials

Table 2.1 - Material Properties - General

Material	Type	SymType	Grade	Color	Notes
Concrete	Concrete	Isotropic	f'c 4000 psi	Magenta	
Rebar	Rebar	Uniaxial	Grade 60	Yellow	
Steel	Steel	Isotropic	Grade 50	Red	
Tendon	Tendon	Uniaxial	Grade 270	Gray8Dark	

2.2 Frame Sections

Table 2.2 - Frame Section Property Definitions - Summary (Part 1 of 4)

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
Column		Auto Select						
HSS101.6X101.6X12.7	Steel	Steel Tube	Red	6.01	21	11.89	11.89	
HSS101.6X101.6X3.2	Steel	Steel Tube	Yellow	1.77	6.92	4.4	4.4	
HSS101.6X101.6X4.8	Steel	Steel Tube	Gray8Dark	2.57	9.99	6.2	6.2	
HSS101.6X101.6X6.4	Steel	Steel Tube	Blue	3.36	12.81	7.81	7.81	
HSS101.6X101.6X7.9	Steel	Steel Tube	Green	4.11	15.3	9.13	9.13	
HSS101.6X101.6X9.5	Steel	Steel Tube	Cyan	4.77	17.49	10.31	10.31	
HSS101.6X50.8X3.2	Steel	Steel Tube	Cyan	1.3	2.2	2.64	0.9	
HSS101.6X50.8X4.8	Steel	Steel Tube	Red	1.89	3.08	3.65	1.22	
HSS101.6X50.8X6.4	Steel	Steel Tube	Gray8Dark	2.43	3.82	4.49	1.48	
HSS101.6X50.8X7.9	Steel	Steel Tube	Blue	2.95	4.4	5.14	1.67	
HSS101.6X50.8X9.5	Steel	Steel Tube	Green	3.39	4.83	5.6	1.8	
HSS101.6X63.5X3.2	Steel	Steel Tube	Cyan	1.42	3.22	3.1	1.49	
HSS101.6X63.5X4.8	Steel	Steel Tube	Red	2.06	4.59	4.3	2.06	
HSS101.6X63.5X6.4	Steel	Steel Tube	Magenta	2.67	5.79	5.31	2.52	
HSS101.6X63.5X7.9	Steel	Steel Tube	Yellow	3.22	6.78	6.13	2.88	
HSS101.6X63.5X9.5	Steel	Steel Tube	Gray8Dark	3.74	7.57	6.78	3.17	
HSS101.6X76.2X3.2	Steel	Steel Tube	Blue	1.54	4.37	3.53	2.27	
HSS101.6X76.2X4.8	Steel	Steel Tube	Green	2.25	6.27	4.93	3.17	
HSS101.6X76.2X6.4	Steel	Steel Tube	Cyan	2.91	7.95	6.15	3.92	
HSS101.6X76.2X7.9	Steel	Steel Tube	Red	3.52	9.42	7.14	4.52	
HSS101.6X76.2X9.5	Steel	Steel Tube	Magenta	4.09	10.6	7.93	5.02	
HSS114.3X114.3X12.7	Steel	Steel Tube	Cyan	6.94	31.23	18.09	18.09	
HSS114.3X114.3X3.2	Steel	Steel Tube	Magenta	2	9.92	6.34	6.34	
HSS114.3X114.3X4.8	Steel	Steel Tube	Yellow	2.93	14.39	9.01	9.01	
HSS114.3X114.3X6.4	Steel	Steel Tube	Gray8Dark	3.84	18.5	11.41	11.41	
HSS114.3X114.3X7.9	Steel	Steel Tube	Blue	4.68	22.3	13.5	13.5	
HSS114.3X114.3X9.5	Steel	Steel Tube	Green	5.49	25.71	15.3	15.3	
HSS127X101.6X12.7	Steel	Steel Tube	Blue	6.94	30.27	21.19	14.9	
HSS127X101.6X3.2	Steel	Steel Tube	Red	2	9.66	7.42	5.26	
HSS127X101.6X4.8	Steel	Steel Tube	Magenta	2.93	14.01	10.6	7.47	
HSS127X101.6X6.4	Steel	Steel Tube	Yellow	3.84	17.99	13.41	9.47	
HSS127X101.6X7.9	Steel	Steel Tube	Gray8Dark	4.68	21.69	15.81	11.1	
HSS127X101.6X9.5	Steel	Steel Tube	Blue	5.49	24.99	17.9	12.59	
HSS127X127X12.7	Steel	Steel Tube	Gray8Dark	7.87	44.69	25.95	25.95	
HSS127X127X3.2	Steel	Steel Tube	Green	2.23	13.69	8.79	8.79	
HSS127X127X4.8	Steel	Steel Tube	Cyan	3.29	19.89	12.59	12.59	
HSS127X127X6.4	Steel	Steel Tube	Red	4.29	25.71	16	16	
HSS127X127X7.9	Steel	Steel Tube	Magenta	5.25	31.23	19	19	
HSS127X127X9.5	Steel	Steel Tube	Yellow	6.18	36.04	21.69	21.69	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
HSS127X50.8X3.2	Steel	Steel Tube	Red	1.54	2.96	4.66	1.1	
HSS127X50.8X4.8	Steel	Steel Tube	Magenta	2.25	4.16	6.51	1.51	
HSS127X50.8X6.4	Steel	Steel Tube	Yellow	2.91	5.17	8.07	1.84	
HSS127X50.8X7.9	Steel	Steel Tube	Gray8Dark	3.52	5.98	9.35	2.1	
HSS127X50.8X9.5	Steel	Steel Tube	Blue	4.09	6.61	10.4	2.28	
HSS127X63.5X3.2	Steel	Steel Tube	Green	1.64	4.4	5.33	1.82	
HSS127X63.5X4.8	Steel	Steel Tube	Cyan	2.4	6.27	7.52	2.52	
HSS127X63.5X6.4	Steel	Steel Tube	Red	3.15	7.93	9.39	3.12	
HSS127X76.2X12.7	Steel	Steel Tube	Cyan	6.01	17.61	16.41	7.18	
HSS127X76.2X3.2	Steel	Steel Tube	Magenta	1.77	6.03	6.03	2.74	
HSS127X76.2X4.8	Steel	Steel Tube	Yellow	2.57	8.65	8.53	3.84	
HSS127X76.2X6.4	Steel	Steel Tube	Gray8Dark	3.36	11	10.69	4.81	
HSS127X76.2X7.9	Steel	Steel Tube	Blue	4.11	13.09	12.59	5.6	
HSS127X76.2X9.5	Steel	Steel Tube	Green	4.77	14.9	14.1	6.25	
HSS139.7X139.7X3.2	Steel	Steel Tube	Blue	2.46	18.31	11.8	11.8	
HSS139.7X139.7X4.8	Steel	Steel Tube	Green	3.63	26.67	17.01	17.01	
HSS139.7X139.7X6.4	Steel	Steel Tube	Cyan	4.77	34.84	21.69	21.69	
HSS139.7X139.7X7.9	Steel	Steel Tube	Red	5.84	42.28	25.95	25.95	
HSS139.7X139.7X9.5	Steel	Steel Tube	Magenta	6.88	49.01	29.79	29.79	
HSS152.4X101.6X12.7	Steel	Steel Tube	Blue	7.87	40.36	34.12	17.8	
HSS152.4X101.6X3.2	Steel	Steel Tube	Cyan	2.23	12.59	11.41	6.15	
HSS152.4X101.6X4.8	Steel	Steel Tube	Red	3.29	18.21	16.41	8.77	
HSS152.4X101.6X6.4	Steel	Steel Tube	Magenta	4.29	23.59	20.9	11.1	
HSS152.4X101.6X7.9	Steel	Steel Tube	Yellow	5.25	28.35	24.75	13.19	
HSS152.4X101.6X9.5	Steel	Steel Tube	Gray8Dark	6.18	32.91	28.35	14.9	
HSS152.4X127X12.7	Steel	Steel Tube	Gray8Dark	8.8	59.82	41.08	30.75	
HSS152.4X127X3.2	Steel	Steel Tube	Green	2.46	17.99	13.41	10.21	
HSS152.4X127X4.8	Steel	Steel Tube	Cyan	3.63	26.19	19.29	14.61	
HSS152.4X127X6.4	Steel	Steel Tube	Red	4.77	34.12	24.75	18.69	
HSS152.4X127X7.9	Steel	Steel Tube	Magenta	5.84	41.32	29.55	22.3	
HSS152.4X127X9.5	Steel	Steel Tube	Yellow	6.88	48.05	33.88	25.47	
HSS152.4X152.4X12.7	Steel	Steel Tube	Yellow	9.73	81.2	48.29	48.29	
HSS152.4X152.4X15.9	Steel	Steel Tube	Gray8Dark	11.7	94.9	55.26	55.26	
HSS152.4X152.4X3.2	Steel	Steel Tube	Blue	2.7	23.9	15.5	15.5	
HSS152.4X152.4X4.8	Steel	Steel Tube	Green	3.98	35.08	22.3	22.3	
HSS152.4X152.4X6.4	Steel	Steel Tube	Cyan	5.24	45.65	28.59	28.59	
HSS152.4X152.4X7.9	Steel	Steel Tube	Red	6.43	55.5	34.36	34.36	
HSS152.4X152.4X9.5	Steel	Steel Tube	Magenta	7.58	64.63	39.4	39.4	
HSS152.4X50.8X3.2	Steel	Steel Tube	Yellow	1.77	3.72	7.42	1.31	
HSS152.4X50.8X4.8	Steel	Steel Tube	Gray8Dark	2.57	5.24	10.5	1.8	
HSS152.4X50.8X6.4	Steel	Steel Tube	Blue	3.36	6.56	13.09	2.21	
HSS152.4X50.8X7.9	Steel	Steel Tube	Green	4.11	7.59	15.3	2.52	
HSS152.4X50.8X9.5	Steel	Steel Tube	Cyan	4.77	8.41	17.11	2.76	
HSS152.4X76.2X12.7	Steel	Steel Tube	Green	6.94	23.09	26.91	8.7	
HSS152.4X76.2X3.2	Steel	Steel Tube	Red	2	7.74	9.44	3.22	
HSS152.4X76.2X4.8	Steel	Steel Tube	Magenta	2.93	11.1	13.41	4.54	
HSS152.4X76.2X6.4	Steel	Steel Tube	Yellow	3.84	14.2	17.01	5.69	
HSS152.4X76.2X7.9	Steel	Steel Tube	Gray8Dark	4.68	16.89	20.11	6.68	
HSS152.4X76.2X9.5	Steel	Steel Tube	Blue	5.49	19.29	22.7	7.47	
HSS177.8X101.6X12.7	Steel	Steel Tube	Magenta	8.8	50.45	50.69	20.71	
HSS177.8X101.6X3.2	Steel	Steel Tube	Gray8Dark	2.46	15.59	16.6	7.04	
HSS177.8X101.6X4.8	Steel	Steel Tube	Blue	3.63	22.7	23.81	9.99	
HSS177.8X101.6X6.4	Steel	Steel Tube	Green	4.77	29.31	30.51	12.81	
HSS177.8X101.6X7.9	Steel	Steel Tube	Cyan	5.84	35.32	36.52	15.21	
HSS177.8X101.6X9.5	Steel	Steel Tube	Red	6.88	41.08	41.8	17.3	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
HSS177.8X127X12.7	Steel	Steel Tube	Red	9.73	75.92	60.54	35.56	
HSS177.8X127X3.2	Steel	Steel Tube	Yellow	2.7	22.51	19.29	11.6	
HSS177.8X127X4.8	Steel	Steel Tube	Gray8Dark	3.98	32.91	27.87	16.6	
HSS177.8X127X6.4	Steel	Steel Tube	Blue	5.24	43	35.8	21.31	
HSS177.8X127X7.9	Steel	Steel Tube	Green	6.43	52.13	43	25.47	
HSS177.8X127X9.5	Steel	Steel Tube	Cyan	7.58	60.54	49.49	29.31	
HSS177.8X177.8X12.7	Steel	Steel Tube	Cyan	11.59	133.1	80.48	80.48	
HSS177.8X177.8X15.9	Steel	Steel Tube	Red	14	158.09	93.46	93.46	
HSS177.8X177.8X3.2	Steel	Steel Tube	Magenta	3.16	38.2	24.75	24.75	
HSS177.8X177.8X4.8	Steel	Steel Tube	Yellow	4.67	56.22	36.04	36.04	
HSS177.8X177.8X6.4	Steel	Steel Tube	Gray8Dark	6.17	73.52	46.61	46.61	
HSS177.8X177.8X7.9	Steel	Steel Tube	Blue	7.6	89.61	56.22	56.22	
HSS177.8X177.8X9.5	Steel	Steel Tube	Green	8.97	104.99	65.11	65.11	
HSS177.8X50.8X3.2	Steel	Steel Tube	Blue	2	4.52	11.1	1.52	
HSS177.8X50.8X4.8	Steel	Steel Tube	Green	2.93	6.34	15.69	2.1	
HSS177.8X50.8X6.4	Steel	Steel Tube	Cyan	3.84	7.95	19.8	2.57	
HSS177.8X76.2X12.7	Steel	Steel Tube	Yellow	7.87	28.59	40.6	10.21	
HSS177.8X76.2X3.2	Steel	Steel Tube	Red	2.23	9.49	13.79	3.7	
HSS177.8X76.2X4.8	Steel	Steel Tube	Magenta	3.29	13.69	19.8	5.24	
HSS177.8X76.2X6.4	Steel	Steel Tube	Yellow	4.29	17.49	25.23	6.61	
HSS177.8X76.2X7.9	Steel	Steel Tube	Red	5.25	20.9	29.79	7.74	
HSS177.8X76.2X9.5	Steel	Steel Tube	Magenta	6.18	23.9	34.12	8.72	
HSS203.2X101.6X12.7	Steel	Steel Tube	Gray8Dark	9.73	61.02	71.84	23.59	
HSS203.2X101.6X15.9	Steel	Steel Tube	Blue	11.7	70.39	81.93	26.67	
HSS203.2X101.6X3.2	Steel	Steel Tube	Green	2.7	18.69	22.9	7.9	
HSS203.2X101.6X4.8	Steel	Steel Tube	Cyan	3.98	27.15	33.15	11.29	
HSS203.2X101.6X6.4	Steel	Steel Tube	Red	5.24	35.32	42.52	14.39	
HSS203.2X101.6X7.9	Steel	Steel Tube	Magenta	6.43	42.52	50.93	17.2	
HSS203.2X101.6X9.5	Steel	Steel Tube	Yellow	7.58	49.25	58.62	19.6	
HSS203.2X152.4X12.7	Steel	Steel Tube	Yellow	11.59	127.09	98.26	62.47	
HSS203.2X152.4X15.9	Steel	Steel Tube	Gray8Dark	14	149.92	114.12	72.32	
HSS203.2X152.4X4.8	Steel	Steel Tube	Green	4.67	53.82	43.73	28.11	
HSS203.2X152.4X6.4	Steel	Steel Tube	Cyan	6.17	70.39	56.7	36.52	
HSS203.2X152.4X7.9	Steel	Steel Tube	Red	7.6	85.77	68.23	43.73	
HSS203.2X152.4X9.5	Steel	Steel Tube	Magenta	8.97	99.94	79.04	50.69	
HSS203.2X203.2X12.7	Steel	Steel Tube	Yellow	13.5	203.97	124.93	124.93	
HSS203.2X203.2X15.9	Steel	Steel Tube	Gray8Dark	16.43	245.06	146.07	146.07	
HSS203.2X203.2X3.2	Steel	Steel Tube	Blue	3.63	57.42	37.48	37.48	
HSS203.2X203.2X4.8	Steel	Steel Tube	Green	5.36	84.57	54.3	54.3	
HSS203.2X203.2X6.4	Steel	Steel Tube	Cyan	7.1	111	70.63	70.63	
HSS203.2X203.2X7.9	Steel	Steel Tube	Red	8.76	135.98	85.53	85.53	
HSS203.2X203.2X9.5	Steel	Steel Tube	Magenta	10.4	160.01	99.94	99.94	
HSS203.2X50.8X3.2	Steel	Steel Tube	Magenta	2.23	5.31	15.69	1.72	
HSS203.2X50.8X4.8	Steel	Steel Tube	Yellow	3.29	7.47	22.39	2.39	
HSS203.2X50.8X6.4	Steel	Steel Tube	Gray8Dark	4.29	9.37	28.59	2.93	
HSS203.2X50.8X7.9	Steel	Steel Tube	Blue	5.25	10.91	33.64	3.39	
HSS203.2X50.8X9.5	Steel	Steel Tube	Green	6.18	12.11	38.2	3.72	
HSS203.2X76.2X12.7	Steel	Steel Tube	Blue	8.8	34.36	58.62	11.7	
HSS203.2X76.2X3.2	Steel	Steel Tube	Cyan	2.46	11.29	19.29	4.2	
HSS203.2X76.2X4.8	Steel	Steel Tube	Red	3.63	16.19	27.87	5.93	
HSS203.2X76.2X6.4	Steel	Steel Tube	Magenta	4.77	20.81	35.56	7.5	
HSS203.2X76.2X7.9	Steel	Steel Tube	Yellow	5.84	24.99	42.28	8.82	
HSS203.2X76.2X9.5	Steel	Steel Tube	Gray8Dark	6.88	28.59	48.53	9.99	
HSS228.6X127X12.7	Steel	Steel Tube	Magenta	11.59	109.07	115.08	45.17	
HSS228.6X127X15.9	Steel	Steel Tube	Yellow	14	128.05	133.1	51.89	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
HSS228.6X127X4.8	Steel	Steel Tube	Blue	4.67	46.85	51.17	20.71	
HSS228.6X127X6.4	Steel	Steel Tube	Green	6.17	61.26	66.07	26.67	
HSS228.6X127X7.9	Steel	Steel Tube	Cyan	7.6	74.48	79.76	31.95	
HSS228.6X127X9.5	Steel	Steel Tube	Red	8.97	86.97	92.5	36.76	
HSS228.6X177.8X12.7	Steel	Steel Tube	Red	13.5	197.01	148.96	99.94	
HSS228.6X177.8X15.9	Steel	Steel Tube	Magenta	16.43	234.97	173.94	117	
HSS228.6X177.8X4.8	Steel	Steel Tube	Gray8Dark	5.36	81.69	64.63	44.21	
HSS228.6X177.8X6.4	Steel	Steel Tube	Blue	7.1	106.91	84.09	57.18	
HSS228.6X177.8X7.9	Steel	Steel Tube	Green	8.76	130.94	102.11	69.19	
HSS228.6X177.8X9.5	Steel	Steel Tube	Cyan	10.4	154	118.92	80.48	
HSS228.6X228.6X12.7	Steel	Steel Tube	Cyan	15.3	295.51	183.07	183.07	
HSS228.6X228.6X15.9	Steel	Steel Tube	Red	18.76	355.57	215.99	215.99	
HSS228.6X228.6X3.2	Steel	Steel Tube	Yellow	4.09	81.93	53.58	53.58	
HSS228.6X228.6X6.4	Steel	Steel Tube	Gray8Dark	8.03	159.05	102.11	102.11	
HSS228.6X228.6X7.9	Steel	Steel Tube	Blue	9.92	196.04	123.97	123.97	
HSS228.6X228.6X74.8	Steel	Steel Tube	Magenta	6.06	121.09	78.08	78.08	
HSS228.6X228.6X9.5	Steel	Steel Tube	Green	11.8	230.88	145.11	145.11	
HSS228.6X76.2X12.7	Steel	Steel Tube	Gray8Dark	9.73	39.88	80.72	13.19	
HSS228.6X76.2X4.8	Steel	Steel Tube	Cyan	3.98	18.91	37.72	6.63	
HSS228.6X76.2X6.4	Steel	Steel Tube	Red	5.24	24.27	48.29	8.38	
HSS228.6X76.2X7.9	Steel	Steel Tube	Magenta	6.43	28.83	57.66	9.87	
HSS228.6X76.2X9.5	Steel	Steel Tube	Yellow	7.58	33.15	66.31	11.2	
HSS254X101.6X12.7	Steel	Steel Tube	Yellow	11.59	82.65	129.01	29.55	
HSS254X101.6X15.9	Steel	Steel Tube	Gray8Dark	14	95.62	148.96	33.39	
HSS254X101.6X3.2	Steel	Steel Tube	Blue	3.16	24.99	39.88	9.66	
HSS254X101.6X4.8	Steel	Steel Tube	Green	4.67	36.52	57.9	13.91	
HSS254X101.6X6.4	Steel	Steel Tube	Cyan	6.17	47.33	74.72	17.71	
HSS254X101.6X7.9	Steel	Steel Tube	Red	7.6	57.42	90.09	21.19	
HSS254X101.6X9.5	Steel	Steel Tube	Magenta	8.97	66.55	104.03	24.27	
HSS254X127X4.8	Steel	Steel Tube	Blue	5.02	54.06	66.31	22.7	
HSS254X127X6.4	Steel	Steel Tube	Green	6.63	70.63	85.77	29.31	
HSS254X127X7.9	Steel	Steel Tube	Cyan	8.17	86.01	104.03	35.32	
HSS254X127X9.5	Steel	Steel Tube	Red	9.67	99.94	119.89	40.6	
HSS254X152.4X12.7	Steel	Steel Tube	Green	13.5	176.1	171.06	76.88	
HSS254X152.4X15.9	Steel	Steel Tube	Cyan	16.43	209.02	201.09	89.37	
HSS254X152.4X4.8	Steel	Steel Tube	Magenta	5.36	73.76	74.72	34.12	
HSS254X152.4X6.4	Steel	Steel Tube	Yellow	7.1	96.58	96.82	44.21	
HSS254X152.4X7.9	Steel	Steel Tube	Gray8Dark	8.76	117.96	117.96	53.34	
HSS254X152.4X9.5	Steel	Steel Tube	Blue	10.4	139.11	136.94	61.74	
HSS254X203.2X12.7	Steel	Steel Tube	Green	15.3	288.3	214.06	151.12	
HSS254X203.2X15.9	Steel	Steel Tube	Cyan	18.76	345.96	252.26	178.03	
HSS254X203.2X4.8	Steel	Steel Tube	Red	6.06	117.96	91.3	65.11	
HSS254X203.2X6.4	Steel	Steel Tube	Yellow	8.03	154.96	118.92	84.81	
HSS254X203.2X7.9	Steel	Steel Tube	Gray8Dark	9.92	190.04	145.11	103.07	
HSS254X203.2X9.5	Steel	Steel Tube	Blue	11.8	223.91	168.9	119.89	
HSS254X254X12.7	Steel	Steel Tube	Blue	17.21	410.83	257.07	257.07	
HSS254X254X15.9	Steel	Steel Tube	Green	20.93	497.32	305.12	305.12	
HSS254X254X4.8	Steel	Steel Tube	Red	6.76	166.97	108.11	108.11	
HSS254X254X6.4	Steel	Steel Tube	Magenta	8.96	220.07	141.03	141.03	
HSS254X254X7.9	Steel	Steel Tube	Yellow	11.1	271.48	172.02	172.02	
HSS254X254X9.5	Steel	Steel Tube	Gray8Dark	13.21	319.53	202.05	202.05	
HSS254X50.8X3.2	Steel	Steel Tube	Yellow	2.7	6.9	28.59	2.14	
HSS254X50.8X4.8	Steel	Steel Tube	Gray8Dark	3.98	9.73	41.08	2.98	
HSS254X50.8X6.4	Steel	Steel Tube	Blue	5.24	12.2	52.61	3.68	
HSS254X50.8X7.9	Steel	Steel Tube	Green	6.43	14.2	62.71	4.23	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
HSS254X50.8X9.5	Steel	Steel Tube	Cyan	7.58	15.9	71.59	4.71	
HSS254X76.2X3.2	Steel	Steel Tube	Red	2.93	14.9	34.12	5.17	
HSS254X76.2X4.8	Steel	Steel Tube	Magenta	4.32	21.5	49.49	7.33	
HSS254X76.2X6.4	Steel	Steel Tube	Yellow	5.7	27.63	63.67	9.27	
HSS254X76.2X7.9	Steel	Steel Tube	Gray8Dark	7.01	32.91	76.4	11	
HSS254X76.2X9.5	Steel	Steel Tube	Blue	8.28	37.72	87.93	12.4	
HSS254X88.9X12.7	Steel	Steel Tube	Gray8Dark	11.1	63.19	117.96	21.41	
HSS254X88.9X3.2	Steel	Steel Tube	Green	3.04	19.8	37	7.23	
HSS254X88.9X4.8	Steel	Steel Tube	Cyan	4.5	28.59	53.58	10.31	
HSS254X88.9X6.4	Steel	Steel Tube	Red	5.94	37	69.19	13.09	
HSS254X88.9X7.9	Steel	Steel Tube	Magenta	7.3	44.69	83.13	15.59	
HSS254X88.9X9.5	Steel	Steel Tube	Yellow	8.62	51.41	96.1	17.8	
HSS304.8X101.6X12.7	Steel	Steel Tube	Blue	13.5	104.99	209.98	35.32	
HSS304.8X101.6X15.9	Steel	Steel Tube	Green	16.43	122.05	245.06	40.36	
HSS304.8X101.6X4.8	Steel	Steel Tube	Red	5.36	46.13	91.78	16.41	
HSS304.8X101.6X6.4	Steel	Steel Tube	Magenta	7.1	59.82	118.92	21	
HSS304.8X101.6X7.9	Steel	Steel Tube	Yellow	8.76	72.32	143.91	25.23	
HSS304.8X101.6X9.5	Steel	Steel Tube	Gray8Dark	10.4	84.09	167.94	28.83	
HSS304.8X152.4X12.7	Steel	Steel Tube	Gray8Dark	15.3	227.04	271.48	91.06	
HSS304.8X152.4X15.9	Steel	Steel Tube	Blue	18.76	271.48	321.94	106.91	
HSS304.8X152.4X4.8	Steel	Steel Tube	Cyan	6.06	94.66	116.04	39.88	
HSS304.8X152.4X6.4	Steel	Steel Tube	Red	8.03	123.97	151.12	51.89	
HSS304.8X152.4X7.9	Steel	Steel Tube	Magenta	9.92	152.08	184.03	62.71	
HSS304.8X152.4X9.5	Steel	Steel Tube	Yellow	11.8	178.03	215.02	72.8	
HSS304.8X203.2X12.7	Steel	Steel Tube	Yellow	17.21	377.19	333.95	178.03	
HSS304.8X203.2X15.9	Steel	Steel Tube	Gray8Dark	20.93	454.07	396.41	209.98	
HSS304.8X203.2X4.8	Steel	Steel Tube	Green	6.76	153.04	140.07	75.68	
HSS304.8X203.2X6.4	Steel	Steel Tube	Cyan	8.96	202.05	184.03	98.74	
HSS304.8X203.2X7.9	Steel	Steel Tube	Red	11.1	247.46	223.91	119.89	
HSS304.8X203.2X9.5	Steel	Steel Tube	Magenta	13.21	293.11	261.87	140.07	
HSS304.8X254X12.7	Steel	Steel Tube	Red	19.07	545.37	394.01	297.91	
HSS304.8X254X6.4	Steel	Steel Tube	Blue	9.9	288.3	215.99	164.09	
HSS304.8X254X7.9	Steel	Steel Tube	Green	12.2	355.57	264.28	199.89	
HSS304.8X254X9.5	Steel	Steel Tube	Cyan	14.6	420.44	309.92	234	
HSS304.8X304.8X12.7	Steel	Steel Tube	Green	20.93	727.96	456.48	456.48	
HSS304.8X304.8X15.9	Steel	Steel Tube	Cyan	25.73	884.12	547.77	547.77	
HSS304.8X304.8X4.8	Steel	Steel Tube	Magenta	8.15	290.7	189.08	189.08	
HSS304.8X304.8X6.4	Steel	Steel Tube	Yellow	10.8	384.4	247.46	247.46	
HSS304.8X304.8X7.9	Steel	Steel Tube	Gray8Dark	13.41	473.29	305.12	305.12	
HSS304.8X304.8X9.5	Steel	Steel Tube	Blue	15.97	562.19	357.97	357.97	
HSS304.8X50.8X4.8	Steel	Steel Tube	Cyan	4.67	11.99	67.51	3.56	
HSS304.8X50.8X6.4	Steel	Steel Tube	Red	6.17	15.11	86.97	4.42	
HSS304.8X50.8X7.9	Steel	Steel Tube	Magenta	7.6	17.61	104.03	5.09	
HSS304.8X76.2X4.8	Steel	Steel Tube	Yellow	5.02	26.91	79.52	8.72	
HSS304.8X76.2X6.4	Steel	Steel Tube	Gray8Dark	6.63	34.6	103.07	11.1	
HSS304.8X76.2X7.9	Steel	Steel Tube	Blue	8.17	41.32	123.97	13.09	
HSS304.8X88.9X7.9	Steel	Steel Tube	Green	8.46	55.98	134.06	18.6	
HSS304.8X88.9X9.5	Steel	Steel Tube	Cyan	10	64.63	155.92	21.31	
HSS355.6X101.6X12.7	Steel	Steel Tube	Blue	15.3	127.09	317.13	41.08	
HSS355.6X101.6X15.9	Steel	Steel Tube	Green	18.76	147.99	372.39	47.09	
HSS355.6X101.6X4.8	Steel	Steel Tube	Red	6.06	55.74	136.94	19	
HSS355.6X101.6X6.4	Steel	Steel Tube	Magenta	8.03	72.32	178.03	24.51	
HSS355.6X101.6X7.9	Steel	Steel Tube	Yellow	9.92	87.69	215.99	29.31	
HSS355.6X101.6X9.5	Steel	Steel Tube	Gray8Dark	11.8	102.11	252.26	33.64	
HSS355.6X152.4X12.7	Steel	Steel Tube	Gray8Dark	17.21	278.69	401.22	104.99	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
HSS355.6X152.4X15.9	Steel	Steel Tube	Blue	20.93	333.95	478.1	123.97	
HSS355.6X152.4X4.8	Steel	Steel Tube	Cyan	6.76	116.04	170.1	45.89	
HSS355.6X152.4X6.4	Steel	Steel Tube	Red	8.96	152.08	221.99	59.58	
HSS355.6X152.4X7.9	Steel	Steel Tube	Magenta	11.1	185.95	271.48	72.32	
HSS355.6X152.4X9.5	Steel	Steel Tube	Yellow	13.21	219.11	317.13	84.09	
HSS355.6X254X12.7	Steel	Steel Tube	Magenta	20.93	684.72	574.2	341.16	
HSS355.6X254X15.9	Steel	Steel Tube	Yellow	25.73	831.27	687.12	406.02	
HSS355.6X254X6.4	Steel	Steel Tube	Green	10.8	362.78	309.92	185.95	
HSS355.6X254X7.9	Steel	Steel Tube	Cyan	13.41	446.87	379.6	227.04	
HSS355.6X254X9.5	Steel	Steel Tube	Red	15.97	528.55	446.87	266.68	
HSS355.6X355.6X12.7	Steel	Steel Tube	Green	24.65	1170.02	742.38	742.38	
HSS355.6X355.6X15.9	Steel	Steel Tube	Cyan	30.23	1429.49	896.14	896.14	
HSS355.6X355.6X7.9	Steel	Steel Tube	Gray8Dark	15.66	759.19	490.11	490.11	
HSS355.6X355.6X9.5	Steel	Steel Tube	Blue	18.76	900.94	576.6	576.6	
HSS406.4X101.6X12.7	Steel	Steel Tube	Blue	17.21	149.92	454.07	47.09	
HSS406.4X101.6X15.9	Steel	Steel Tube	Green	20.93	173.94	538.16	54.06	
HSS406.4X101.6X4.8	Steel	Steel Tube	Red	6.76	65.59	192.92	21.5	
HSS406.4X101.6X6.4	Steel	Steel Tube	Magenta	8.96	85.29	252.26	27.63	
HSS406.4X101.6X7.9	Steel	Steel Tube	Yellow	11.1	103.07	307.52	33.15	
HSS406.4X101.6X9.5	Steel	Steel Tube	Gray8Dark	13.21	119.89	360.38	38.2	
HSS406.4X203.2X12.7	Steel	Steel Tube	Yellow	20.93	562.19	679.91	229.92	
HSS406.4X203.2X15.9	Steel	Steel Tube	Gray8Dark	25.73	679.91	814.45	273.89	
HSS406.4X203.2X6.4	Steel	Steel Tube	Cyan	10.8	300.31	367.58	127.09	
HSS406.4X203.2X7.9	Steel	Steel Tube	Red	13.41	369.99	451.67	154.96	
HSS406.4X203.2X9.5	Steel	Steel Tube	Magenta	15.97	434.85	530.95	180.91	
HSS406.4X304.8X12.7	Steel	Steel Tube	Cyan	24.65	1119.57	903.34	581.41	
HSS406.4X304.8X15.9	Steel	Steel Tube	Red	30.23	1369.43	1090.74	699.13	
HSS406.4X304.8X7.9	Steel	Steel Tube	Blue	15.66	727.96	595.82	384.4	
HSS406.4X304.8X9.5	Steel	Steel Tube	Green	18.76	862.5	701.53	451.67	
HSS406.4X406.4X12.7	Steel	Steel Tube	Gray8Dark	28.37	1770.65	1129.18	1129.18	
HSS406.4X406.4X15.9	Steel	Steel Tube	Blue	35.03	2169.47	1369.43	1369.43	
HSS406.4X406.4X7.9	Steel	Steel Tube	Magenta	18.14	1141.19	739.97	739.97	
HSS406.4X406.4X9.5	Steel	Steel Tube	Yellow	21.55	1350.21	872.11	872.11	
HSS457.2X152.4X12.7	Steel	Steel Tube	Magenta	20.93	386.8	768.8	134.06	
HSS457.2X152.4X15.9	Steel	Steel Tube	Yellow	25.73	461.28	922.56	158.09	
HSS457.2X152.4X6.4	Steel	Steel Tube	Green	10.8	209.98	418.04	75.2	
HSS457.2X152.4X7.9	Steel	Steel Tube	Cyan	13.41	257.07	514.14	91.3	
HSS457.2X152.4X9.5	Steel	Steel Tube	Red	15.97	302.72	603.03	105.95	
HSS50.8X25.4X3.2	Steel	Steel Tube	Cyan	0.61	0.24	0.28	0.09	
HSS50.8X25.4X4.8	Steel	Steel Tube	Red	0.84	0.3	0.35	0.11	
HSS50.8X38.1X3.2	Steel	Steel Tube	Magenta	0.72	0.49	0.38	0.25	
HSS50.8X38.1X4.8	Steel	Steel Tube	Yellow	1.02	0.66	0.49	0.31	
HSS50.8X50.8X3.2	Steel	Steel Tube	Gray8Dark	0.84	0.8	0.49	0.49	
HSS50.8X50.8X4.8	Steel	Steel Tube	Blue	1.19	1.09	0.64	0.64	
HSS50.8X50.8X6.4	Steel	Steel Tube	Green	1.51	1.31	0.75	0.75	
HSS508X101.6X12.7	Steel	Steel Tube	Cyan	20.93	195.08	838.48	58.62	
HSS508X101.6X6.4	Steel	Steel Tube	Gray8Dark	10.8	111	458.88	34.36	
HSS508X101.6X7.9	Steel	Steel Tube	Blue	13.41	134.06	559.78	41.08	
HSS508X101.6X9.5	Steel	Steel Tube	Green	15.97	155.92	655.89	47.57	
HSS508X203.2X12.7	Steel	Steel Tube	Yellow	24.65	756.79	1189.24	283.5	
HSS508X203.2X15.9	Steel	Steel Tube	Gray8Dark	30.23	915.36	1439.1	338.75	
HSS508X203.2X7.9	Steel	Steel Tube	Red	15.66	494.92	785.62	189.08	
HSS508X203.2X9.5	Steel	Steel Tube	Magenta	18.76	586.21	924.97	221.99	
HSS508X304.8X12.7	Steel	Steel Tube	Cyan	28.37	1540.01	1549.62	703.94	
HSS508X304.8X15.9	Steel	Steel Tube	Red	35.03	1890.78	1881.17	850.49	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
HSS508X304.8X7.9	Steel	Steel Tube	Blue	18.14	997.04	1009.05	463.68	
HSS508X304.8X9.5	Steel	Steel Tube	Green	21.55	1179.63	1198.85	547.77	
HSS57.2X50.8X3.2	Steel	Steel Tube	Cyan	0.9	0.96	0.65	0.54	
HSS57.2X50.8X4.8	Steel	Steel Tube	Red	1.28	1.32	0.86	0.71	
HSS57.2X57.2X3.2	Steel	Steel Tube	Magenta	0.96	1.15	0.71	0.71	
HSS57.2X57.2X4.8	Steel	Steel Tube	Yellow	1.37	1.6	0.95	0.95	
HSS57.2X57.2X6.4	Steel	Steel Tube	Gray8Dark	1.74	1.96	1.13	1.13	
HSS63.5X25.4X3.2	Steel	Steel Tube	Blue	0.72	0.32	0.5	0.12	
HSS63.5X25.4X4.8	Steel	Steel Tube	Green	1.02	0.41	0.65	0.14	
HSS63.5X38.1X3.2	Steel	Steel Tube	Cyan	0.84	0.69	0.67	0.3	
HSS63.5X38.1X4.8	Steel	Steel Tube	Red	1.19	0.93	0.88	0.39	
HSS63.5X38.1X6.4	Steel	Steel Tube	Magenta	1.51	1.1	1.03	0.45	
HSS63.5X50.8X3.2	Steel	Steel Tube	Yellow	0.96	1.12	0.83	0.59	
HSS63.5X50.8X4.8	Steel	Steel Tube	Gray8Dark	1.37	1.55	1.12	0.79	
HSS63.5X50.8X6.4	Steel	Steel Tube	Blue	1.74	1.9	1.33	0.93	
HSS63.5X63.5X3.2	Steel	Steel Tube	Gray8Dark	1.07	1.61	1	1	
HSS63.5X63.5X4.8	Steel	Steel Tube	Blue	1.54	2.25	1.35	1.35	
HSS63.5X63.5X6.4	Steel	Steel Tube	Green	1.97	2.79	1.63	1.63	
HSS63.5X63.5X7.9	Steel	Steel Tube	Cyan	2.36	3.2	1.82	1.82	
HSS76.2X25.4X3.2	Steel	Steel Tube	Red	0.84	0.41	0.82	0.14	
HSS76.2X25.4X4.8	Steel	Steel Tube	Magenta	1.19	0.53	1.07	0.17	
HSS76.2X38.1X3.2	Steel	Steel Tube	Yellow	0.96	0.89	1.06	0.36	
HSS76.2X38.1X4.8	Steel	Steel Tube	Gray8Dark	1.37	1.21	1.42	0.47	
HSS76.2X38.1X6.4	Steel	Steel Tube	Blue	1.74	1.44	1.68	0.54	
HSS76.2X50.8X3.2	Steel	Steel Tube	Green	1.07	1.47	1.3	0.69	
HSS76.2X50.8X4.8	Steel	Steel Tube	Cyan	1.54	2.05	1.77	0.93	
HSS76.2X50.8X6.4	Steel	Steel Tube	Red	1.97	2.52	2.13	1.11	
HSS76.2X50.8X7.9	Steel	Steel Tube	Magenta	2.36	2.86	2.38	1.24	
HSS76.2X63.5X3.2	Steel	Steel Tube	Yellow	1.19	2.13	1.54	1.16	
HSS76.2X63.5X4.8	Steel	Steel Tube	Gray8Dark	1.71	3	2.11	1.59	
HSS76.2X63.5X6.4	Steel	Steel Tube	Blue	2.22	3.75	2.57	1.93	
HSS76.2X63.5X7.9	Steel	Steel Tube	Green	2.64	4.35	2.93	2.18	
HSS76.2X76.2X3.2	Steel	Steel Tube	Cyan	1.3	2.83	1.78	1.78	
HSS76.2X76.2X4.8	Steel	Steel Tube	Red	1.89	4.04	2.45	2.45	
HSS76.2X76.2X6.4	Steel	Steel Tube	Magenta	2.43	5.07	3.03	3.03	
HSS76.2X76.2X7.9	Steel	Steel Tube	Yellow	2.95	5.93	3.46	3.46	
HSS76.2X76.2X9.5	Steel	Steel Tube	Gray8Dark	3.39	6.63	3.77	3.77	
HSS88.9X38.1X3.2	Steel	Steel Tube	Blue	1.07	1.09	1.57	0.41	
HSS88.9X38.1X4.8	Steel	Steel Tube	Green	1.54	1.49	2.12	0.54	
HSS88.9X38.1X6.4	Steel	Steel Tube	Cyan	1.97	1.79	2.55	0.64	
HSS88.9X50.8X3.2	Steel	Steel Tube	Red	1.19	1.83	1.9	0.8	
HSS88.9X50.8X4.8	Steel	Steel Tube	Magenta	1.71	2.55	2.62	1.08	
HSS88.9X50.8X6.4	Steel	Steel Tube	Yellow	2.22	3.17	3.17	1.3	
HSS88.9X63.5X3.2	Steel	Steel Tube	Gray8Dark	1.3	2.67	2.23	1.33	
HSS88.9X63.5X4.8	Steel	Steel Tube	Blue	1.89	3.77	3.09	1.83	
HSS88.9X63.5X6.4	Steel	Steel Tube	Green	2.43	4.76	3.79	2.23	
HSS88.9X63.5X7.9	Steel	Steel Tube	Cyan	2.95	5.53	4.34	2.55	
HSS88.9X63.5X9.5	Steel	Steel Tube	Red	3.39	6.15	4.75	2.76	
HSS88.9X88.9X3.2	Steel	Steel Tube	Magenta	1.54	4.59	2.91	2.91	
HSS88.9X88.9X4.8	Steel	Steel Tube	Yellow	2.25	6.56	4.06	4.06	
HSS88.9X88.9X6.4	Steel	Steel Tube	Gray8Dark	2.91	8.36	5.05	5.05	
HSS88.9X88.9X7.9	Steel	Steel Tube	Blue	3.52	9.9	5.84	5.84	
HSS88.9X88.9X9.5	Steel	Steel Tube	Green	4.09	11.2	6.49	6.49	
L102X102X11.1	Steel	Steel Angle	Red	3.3	0.22	4.93	4.93	2.96
L102X102X12.7	Steel	Steel Angle	Magenta	3.75	0.32	5.53	5.53	3.3

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
L102X102X15.9	Steel	Steel Angle	Yellow	4.6	0.61	6.63	6.63	3.91
L102X102X19	Steel	Steel Angle	Gray8Dark	5.44	1.02	7.62	7.62	4.43
L102X102X6.4	Steel	Steel Angle	Blue	1.94	0.04	3	3	1.82
L102X102X7.9	Steel	Steel Angle	Green	2.4	0.08	3.68	3.68	2.22
L102X102X9.5	Steel	Steel Angle	Cyan	2.87	0.14	4.32	4.32	2.6
L102X76X12.7	Steel	Steel Angle	Green	3.26	0.28	5.02	2.4	2.03
L102X76X15.9	Steel	Steel Angle	Cyan	3.98	0.53	6.01	2.86	2.37
L102X76X6.4	Steel	Steel Angle	Yellow	1.69	0.04	2.74	1.33	1.14
L102X76X7.9	Steel	Steel Angle	Gray8Dark	2.09	0.07	3.36	1.62	1.39
L102X76X9.5	Steel	Steel Angle	Blue	2.5	0.12	3.94	1.89	1.62
L102X89X12.7	Steel	Steel Angle	Gray8Dark	3.5	0.3	5.31	3.77	2.63
L102X89X6.4	Steel	Steel Angle	Red	1.81	0.04	2.88	2.07	1.46
L102X89X7.9	Steel	Steel Angle	Magenta	2.25	0.08	3.53	2.52	1.79
L102X89X9.5	Steel	Steel Angle	Yellow	2.68	0.13	4.16	2.96	2.09
L127X127X11.1	Steel	Steel Angle	Gray8Dark	4.22	0.28	9.99	9.99	5.9
L127X127X12.7	Steel	Steel Angle	Blue	4.79	0.42	11.29	11.29	6.62
L127X127X15.9	Steel	Steel Angle	Green	5.91	0.79	13.6	13.6	7.95
L127X127X19	Steel	Steel Angle	Cyan	6.98	1.33	15.69	15.69	9.14
L127X127X22.2	Steel	Steel Angle	Red	8	2.07	17.8	17.8	10.17
L127X127X7.9	Steel	Steel Angle	Magenta	3.07	0.11	7.45	7.45	4.38
L127X127X9.5	Steel	Steel Angle	Yellow	3.64	0.18	8.77	8.77	5.16
L127X76X11.1	Steel	Steel Angle	Red	3.32	0.22	8.41	2.29	2.52
L127X76X12.7	Steel	Steel Angle	Magenta	3.75	0.32	9.44	2.55	2.81
L127X76X6.4	Steel	Steel Angle	Blue	1.94	0.04	5.09	1.41	1.56
L127X76X7.9	Steel	Steel Angle	Green	2.4	0.08	6.25	1.72	1.91
L127X76X9.5	Steel	Steel Angle	Cyan	2.87	0.14	7.35	2.01	2.23
L127X89X12.7	Steel	Steel Angle	Green	4	0.34	9.99	4.01	3.68
L127X89X15.9	Steel	Steel Angle	Cyan	4.93	0.65	11.99	4.81	4.36
L127X89X19	Steel	Steel Angle	Red	5.84	1.09	13.91	5.53	4.95
L127X89X6.4	Steel	Steel Angle	Yellow	2.08	0.05	5.36	2.2	2.02
L127X89X7.9	Steel	Steel Angle	Gray8Dark	2.56	0.09	6.58	2.69	2.47
L127X89X9.5	Steel	Steel Angle	Blue	3.05	0.15	7.76	3.15	2.9
L152X102X11.1	Steel	Steel Angle	Cyan	4.19	0.28	15.4	5.55	5.42
L152X102X12.7	Steel	Steel Angle	Red	4.74	0.41	17.3	6.22	6.07
L152X102X14.3	Steel	Steel Angle	Magenta	5.32	0.57	19.2	6.87	6.68
L152X102X15.9	Steel	Steel Angle	Yellow	5.86	0.78	21	7.47	7.26
L152X102X19	Steel	Steel Angle	Gray8Dark	6.94	1.31	24.51	8.63	8.33
L152X102X22.2	Steel	Steel Angle	Blue	8	2.03	27.63	9.71	9.23
L152X102X7.9	Steel	Steel Angle	Blue	3.02	0.1	11.41	4.13	4.03
L152X102X9.5	Steel	Steel Angle	Green	3.61	0.18	13.41	4.85	4.74
L152X152X11.1	Steel	Steel Angle	Red	5.08	0.34	17.61	17.61	10.37
L152X152X12.7	Steel	Steel Angle	Magenta	5.77	0.5	19.89	19.89	11.67
L152X152X14.3	Steel	Steel Angle	Yellow	6.45	0.7	22.01	22.01	12.93
L152X152X15.9	Steel	Steel Angle	Gray8Dark	7.13	0.96	24.03	24.03	14.13
L152X152X19	Steel	Steel Angle	Blue	8.46	1.61	28.11	28.11	16.38
L152X152X22.2	Steel	Steel Angle	Green	9.75	2.5	31.95	31.95	18.38
L152X152X25.4	Steel	Steel Angle	Cyan	11.01	3.68	35.32	35.32	20.24
L152X152X7.9	Steel	Steel Angle	Green	3.67	0.13	13	13	7.64
L152X152X9.5	Steel	Steel Angle	Cyan	4.39	0.22	15.4	15.4	9.04
L152X89X12.7	Steel	Steel Angle	Gray8Dark	4.5	0.39	16.6	4.23	4.78
L152X89X7.9	Steel	Steel Angle	Magenta	2.88	0.1	10.91	2.83	3.19
L152X89X9.5	Steel	Steel Angle	Yellow	3.44	0.17	12.9	3.34	3.75
L178X102X11.1	Steel	Steel Angle	Magenta	4.63	0.31	23.59	5.79	6.79
L178X102X12.7	Steel	Steel Angle	Yellow	5.25	0.46	26.67	6.49	7.62
L178X102X15.9	Steel	Steel Angle	Gray8Dark	6.49	0.87	32.43	7.78	9.14

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
L178X102X19	Steel	Steel Angle	Blue	7.73	1.47	37.72	9.01	10.5
L178X102X9.5	Steel	Steel Angle	Red	4	0.2	20.49	5.07	5.94
L203X102X11.1	Steel	Steel Angle	Cyan	5.12	0.34	34.12	6.03	8.15
L203X102X12.7	Steel	Steel Angle	Red	5.8	0.5	38.68	6.75	9.15
L203X102X14.3	Steel	Steel Angle	Magenta	6.49	0.7	43	7.45	10.1
L203X102X15.9	Steel	Steel Angle	Yellow	7.16	0.96	47.09	8.12	11
L203X102X19	Steel	Steel Angle	Gray8Dark	8.49	1.61	55.02	9.37	12.66
L203X102X22.2	Steel	Steel Angle	Blue	9.8	2.5	62.71	10.5	14.09
L203X102X25.4	Steel	Steel Angle	Green	11.1	3.68	69.67	11.6	15.38
L203X152X11.1	Steel	Steel Angle	Cyan	5.98	0.4	39.4	19.29	16.07
L203X152X12.7	Steel	Steel Angle	Red	6.8	0.58	44.45	21.69	18.13
L203X152X14.3	Steel	Steel Angle	Magenta	7.61	0.82	49.49	24.03	20.12
L203X152X15.9	Steel	Steel Angle	Yellow	8.42	1.12	54.3	26.43	22.04
L203X152X19	Steel	Steel Angle	Gray8Dark	10	1.9	63.43	30.75	25.68
L203X152X22.2	Steel	Steel Angle	Blue	11.5	2.96	72.32	34.84	28.95
L203X152X25.4	Steel	Steel Angle	Green	13.1	4.35	80.96	38.68	32.06
L203X203X12.7	Steel	Steel Angle	Cyan	7.84	0.68	48.77	48.77	28.72
L203X203X14.3	Steel	Steel Angle	Red	8.77	0.96	54.3	54.3	31.96
L203X203X15.9	Steel	Steel Angle	Magenta	9.69	1.3	59.58	59.58	35.11
L203X203X19	Steel	Steel Angle	Yellow	11.5	2.21	69.91	69.91	41.14
L203X203X22.2	Steel	Steel Angle	Gray8Dark	13.3	3.46	79.76	79.76	46.64
L203X203X25.4	Steel	Steel Angle	Blue	15.1	5.07	89.13	89.13	51.96
L203X203X28.6	Steel	Steel Angle	Green	16.74	7.14	98.02	98.02	56.93
L51X51X3.2	Steel	Steel Angle	Blue	0.49	0.002931	0.19	0.19	0.11
L51X51X4.8	Steel	Steel Angle	Green	0.72	0.01	0.27	0.27	0.16
L51X51X6.4	Steel	Steel Angle	Cyan	0.94	0.02	0.35	0.35	0.2
L51X51X7.9	Steel	Steel Angle	Red	1.16	0.04	0.41	0.41	0.24
L51X51X9.5	Steel	Steel Angle	Magenta	1.37	0.07	0.48	0.48	0.27
L64X38X4.8	Steel	Steel Angle	Yellow	0.72	0.01	0.46	0.13	0.14
L64X38X6.4	Steel	Steel Angle	Gray8Dark	0.95	0.02	0.59	0.16	0.18
L64X51X4.8	Steel	Steel Angle	Blue	0.82	0.01	0.51	0.29	0.23
L64X51X6.4	Steel	Steel Angle	Green	1.07	0.02	0.66	0.37	0.29
L64X51X7.9	Steel	Steel Angle	Cyan	1.32	0.04	0.79	0.45	0.34
L64X51X9.5	Steel	Steel Angle	Red	1.55	0.07	0.91	0.51	0.39
L64X64X12.7	Steel	Steel Angle	Green	2.26	0.19	1.22	1.22	0.69
L64X64X4.8	Steel	Steel Angle	Magenta	0.9	0.01	0.54	0.54	0.32
L64X64X6.4	Steel	Steel Angle	Yellow	1.19	0.03	0.69	0.69	0.41
L64X64X7.9	Steel	Steel Angle	Gray8Dark	1.46	0.05	0.84	0.84	0.5
L64X64X9.5	Steel	Steel Angle	Blue	1.74	0.08	0.97	0.97	0.57
L76X51X12.7	Steel	Steel Angle	Gray8Dark	2.26	0.19	1.92	0.67	0.62
L76X51X4.8	Steel	Steel Angle	Cyan	0.92	0.01	0.85	0.31	0.29
L76X51X6.4	Steel	Steel Angle	Red	1.2	0.03	1.09	0.39	0.38
L76X51X7.9	Steel	Steel Angle	Magenta	1.48	0.05	1.32	0.47	0.45
L76X51X9.5	Steel	Steel Angle	Yellow	1.75	0.09	1.54	0.54	0.52
L76X64X11.1	Steel	Steel Angle	Magenta	2.22	0.15	1.87	1.17	0.85
L76X64X12.7	Steel	Steel Angle	Yellow	2.5	0.21	2.07	1.29	0.94
L76X64X4.8	Steel	Steel Angle	Blue	1	0.01	0.9	0.57	0.42
L76X64X6.4	Steel	Steel Angle	Green	1.32	0.03	1.16	0.74	0.55
L76X64X7.9	Steel	Steel Angle	Cyan	1.63	0.06	1.41	0.89	0.66
L76X64X9.5	Steel	Steel Angle	Red	1.94	0.09	1.65	1.03	0.76
L76X76X11.1	Steel	Steel Angle	Red	2.43	0.16	1.98	1.98	1.16
L76X76X12.7	Steel	Steel Angle	Magenta	2.76	0.23	2.2	2.2	1.27
L76X76X4.8	Steel	Steel Angle	Gray8Dark	1.09	0.01	0.95	0.95	0.56
L76X76X6.4	Steel	Steel Angle	Blue	1.44	0.03	1.23	1.23	0.73
L76X76X7.9	Steel	Steel Angle	Green	1.78	0.06	1.5	1.5	0.89

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
L76X76X9.5	Steel	Steel Angle	Cyan	2.11	0.1	1.75	1.75	1.03
L89X64X12.7	Steel	Steel Angle	Green	2.77	0.23	3.24	1.36	1.19
L89X64X6.4	Steel	Steel Angle	Yellow	1.45	0.03	1.81	0.78	0.69
L89X64X7.9	Steel	Steel Angle	Gray8Dark	1.78	0.06	2.2	0.94	0.83
L89X64X9.5	Steel	Steel Angle	Blue	2.12	0.1	2.57	1.09	0.96
L89X76X11.1	Steel	Steel Angle	Yellow	2.67	0.18	3.1	2.09	1.48
L89X76X12.7	Steel	Steel Angle	Gray8Dark	3.02	0.26	3.46	2.32	1.64
L89X76X6.4	Steel	Steel Angle	Cyan	1.58	0.04	1.92	1.3	0.93
L89X76X7.9	Steel	Steel Angle	Red	1.95	0.07	2.33	1.58	1.13
L89X76X9.5	Steel	Steel Angle	Magenta	2.33	0.11	2.74	1.84	1.31
L89X89X11.1	Steel	Steel Angle	Red	2.88	0.19	3.24	3.24	1.9
L89X89X12.7	Steel	Steel Angle	Magenta	3.26	0.28	3.63	3.63	2.11
L89X89X6.4	Steel	Steel Angle	Blue	1.71	0.04	2	2	1.18
L89X89X7.9	Steel	Steel Angle	Green	2.09	0.07	2.45	2.45	1.44
L89X89X9.5	Steel	Steel Angle	Cyan	2.5	0.12	2.86	2.86	1.68
Rufter		Auto Select						
Truss		Auto Select						
W1000X222	Steel	Steel I/Wide Flange	Yellow	43.87	9.37	9802.24	228.96	
W1000X249	Steel	Steel I/Wide Flange	Gray8Dark	49.29	14.01	11604.12	283.5	
W1000X272	Steel	Steel I/Wide Flange	Blue	53.32	19.29	13189.78	331.55	
W1000X296	Steel	Steel I/Wide Flange	Green	58.75	18.31	14895.56	694.33	
W1000X314	Steel	Steel I/Wide Flange	Cyan	62.16	30.51	15496.19	389.21	
W1000X321	Steel	Steel I/Wide Flange	Red	63.55	24.75	16697.44	802.44	
W1000X350	Steel	Steel I/Wide Flange	Magenta	69.13	41.32	17394.17	444.46	
W1000X371	Steel	Steel I/Wide Flange	Yellow	73.47	38.2	19604.48	924.97	
W1000X393	Steel	Steel I/Wide Flange	Gray8Dark	77.35	56.22	19388.25	492.51	
W1000X412	Steel	Steel I/Wide Flange	Blue	81.53	51.41	21910.89	1040.29	
W1000X415	Steel	Steel I/Wide Flange	Green	82.31	65.11	20493.41	521.34	
W1000X438	Steel	Steel I/Wide Flange	Cyan	86.18	76.64	21910.89	562.19	
W1000X443	Steel	Steel I/Wide Flange	Magenta	87.27	61.26	23208.24	1090.74	
W1000X483	Steel	Steel I/Wide Flange	Yellow	95.33	79.28	25706.85	1220.47	
W1000X486	Steel	Steel I/Wide Flange	Gray8Dark	95.95	103.07	24505.6	639.07	
W1000X494	Steel	Steel I/Wide Flange	Blue	97.65	104.99	24745.85	643.87	
W1000X539	Steel	Steel I/Wide Flange	Green	106.02	109.07	28830.12	1379.04	
W1000X554	Steel	Steel I/Wide Flange	Cyan	110.05	116.04	29550.87	1419.88	
W1000X584	Steel	Steel I/Wide Flange	Red	115.94	172.02	29791.12	802.44	
W1000X591	Steel	Steel I/Wide Flange	Magenta	117.03	141.99	31953.38	1540.01	
W1000X642	Steel	Steel I/Wide Flange	Yellow	126.95	177.06	34836.39	1688.96	
W1000X748	Steel	Steel I/Wide Flange	Gray8Dark	148.03	276.29	41563.42	2039.73	
W1000X883	Steel	Steel I/Wide Flange	Blue	173.6	444.46	50452.7	2522.64	
W100X19.3	Steel	Steel I/Wide Flange	Green	3.83	0.15	11.29	3.87	
W1100X343	Steel	Steel I/Wide Flange	Green	67.74	24.99	20805.73	795.23	
W1100X390	Steel	Steel I/Wide Flange	Cyan	77.19	37.24	24025.1	922.56	
W1100X433	Steel	Steel I/Wide Flange	Red	85.41	50.93	26908.11	1040.29	
W1100X499	Steel	Steel I/Wide Flange	Magenta	98.43	74.72	30992.37	1198.85	
W130X23.8	Steel	Steel I/Wide Flange	Green	4.71	0.19	21.41	7.52	
W130X28.1	Steel	Steel I/Wide Flange	Cyan	5.56	0.32	26.19	9.13	
W150X13	Steel	Steel I/Wide Flange	Red	2.53	0.03	14.9	1.99	
W150X13.5	Steel	Steel I/Wide Flange	Magenta	2.68	0.04	16.41	2.2	
W150X18	Steel	Steel I/Wide Flange	Yellow	3.55	0.09	22.1	2.98	
W150X22.5	Steel	Steel I/Wide Flange	Gray8Dark	4.43	0.1	29.07	9.32	
W150X24	Steel	Steel I/Wide Flange	Blue	4.74	0.22	32.19	4.42	
W150X29.8	Steel	Steel I/Wide Flange	Green	5.87	0.24	41.32	13.31	
W150X37.1	Steel	Steel I/Wide Flange	Cyan	7.35	0.46	53.34	17.11	
W200X100	Steel	Steel I/Wide Flange	Blue	19.69	5.05	271.48	88.65	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
W200X15	Steel	Steel I/Wide Flange	Red	2.96	0.04	30.75	2.09	
W200X19.3	Steel	Steel I/Wide Flange	Magenta	3.84	0.09	39.64	2.74	
W200X22.5	Steel	Steel I/Wide Flange	Yellow	4.43	0.14	48.05	3.41	
W200X26.6	Steel	Steel I/Wide Flange	Gray8Dark	5.25	0.17	61.98	7.98	
W200X31.3	Steel	Steel I/Wide Flange	Blue	6.15	0.28	75.2	9.78	
W200X35.9	Steel	Steel I/Wide Flange	Green	7.08	0.35	82.65	18.31	
W200X41.7	Steel	Steel I/Wide Flange	Cyan	8.25	0.54	98.02	21.69	
W200X46.1	Steel	Steel I/Wide Flange	Red	9.13	0.54	110.03	37	
W200X52	Steel	Steel I/Wide Flange	Magenta	10.31	0.77	127.09	42.52	
W200X59	Steel	Steel I/Wide Flange	Yellow	11.7	1.12	146.07	49.01	
W200X71	Steel	Steel I/Wide Flange	Gray8Dark	14.11	1.96	184.03	60.78	
W200X86	Steel	Steel I/Wide Flange	Gray8Dark	17.05	3.34	228	75.2	
W250X101	Steel	Steel I/Wide Flange	Gray8Dark	19.84	3.56	394.01	134.06	
W250X115	Steel	Steel I/Wide Flange	Blue	22.63	5.12	454.07	154	
W250X131	Steel	Steel I/Wide Flange	Green	26.04	7.52	533.36	178.99	
W250X149	Steel	Steel I/Wide Flange	Cyan	29.3	10.91	622.25	207.1	
W250X167	Steel	Steel I/Wide Flange	Red	32.86	15.11	715.95	235.93	
W250X17.9	Steel	Steel I/Wide Flange	Blue	3.53	0.05	53.82	2.18	
W250X22.3-1	Steel	Steel I/Wide Flange	Green	4.42	0.1	68.95	2.88	
W250X25.3-1	Steel	Steel I/Wide Flange	Cyan	4.99	0.16	81.93	3.56	
W250X28.4-1	Steel	Steel I/Wide Flange	Red	5.63	0.23	96.34	4.3	
W250X32.7	Steel	Steel I/Wide Flange	Magenta	6.49	0.24	117.96	11.41	
W250X38.5	Steel	Steel I/Wide Flange	Yellow	7.61	0.4	143.91	14.1	
W250X44.8	Steel	Steel I/Wide Flange	Gray8Dark	8.84	0.62	170.1	16.7	
W250X49.1-1	Steel	Steel I/Wide Flange	Blue	9.7	0.58	171.06	36.52	
W250X58-1	Steel	Steel I/Wide Flange	Green	11.5	0.98	209.02	44.93	
W250X67-1	Steel	Steel I/Wide Flange	Cyan	13.3	1.51	247.46	53.34	
W250X73	Steel	Steel I/Wide Flange	Red	14.4	1.39	271.48	93.46	
W250X80	Steel	Steel I/Wide Flange	Magenta	15.81	1.82	302.72	103.07	
W250X89-1	Steel	Steel I/Wide Flange	Yellow	17.67	2.47	341.16	116.04	
W310X107-1	Steel	Steel I/Wide Flange	Cyan	21.08	2.93	595.82	195.08	
W310X117-1	Steel	Steel I/Wide Flange	Red	23.25	3.84	663.09	215.99	
W310X129-1	Steel	Steel I/Wide Flange	Magenta	25.58	5.09	739.97	240.25	
W310X143-1	Steel	Steel I/Wide Flange	Yellow	28.21	6.85	833.67	269.08	
W310X158-1	Steel	Steel I/Wide Flange	Gray8Dark	31.16	9.13	932.17	300.31	
W310X179-1	Steel	Steel I/Wide Flange	Blue	35.19	12.9	1069.12	345.96	
W310X202-1	Steel	Steel I/Wide Flange	Green	39.84	18.5	1239.69	398.82	
W310X21-1	Steel	Steel I/Wide Flange	Magenta	4.15	0.07	88.65	2.36	
W310X226-1	Steel	Steel I/Wide Flange	Cyan	44.64	25.71	1429.49	454.07	
W310X23.8-1	Steel	Steel I/Wide Flange	Magenta	4.71	0.1	103.07	2.81	
W310X253-1	Steel	Steel I/Wide Flange	Cyan	50.07	35.56	1650.52	516.54	
W310X28.3-1	Steel	Steel I/Wide Flange	Yellow	5.56	0.18	129.98	3.77	
W310X283-1	Steel	Steel I/Wide Flange	Red	55.96	48.77	1890.78	588.61	
W310X313-1	Steel	Steel I/Wide Flange	Magenta	61.85	64.63	2140.64	663.09	
W310X32.7-1	Steel	Steel I/Wide Flange	Gray8Dark	6.48	0.29	155.92	4.66	
W310X342	Steel	Steel I/Wide Flange	Yellow	67.74	83.85	2426.53	742.38	
W310X375	Steel	Steel I/Wide Flange	Gray8Dark	74.09	108.11	2714.84	828.87	
W310X38.7-1	Steel	Steel I/Wide Flange	Blue	7.66	0.3	203.97	17.3	
W310X415	Steel	Steel I/Wide Flange	Blue	81.84	142.95	3099.24	936.98	
W310X44.5-1	Steel	Steel I/Wide Flange	Green	8.79	0.46	238.09	20.3	
W310X454	Steel	Steel I/Wide Flange	Green	89.44	184.99	3555.71	1049.9	
W310X500	Steel	Steel I/Wide Flange	Cyan	98.89	242.65	4060.24	1189.24	
W310X52	Steel	Steel I/Wide Flange	Cyan	10.31	0.74	285.9	24.51	
W310X60	Steel	Steel I/Wide Flange	Red	11.7	0.91	307.52	44.21	
W310X67	Steel	Steel I/Wide Flange	Magenta	13.1	1.26	348.36	49.97	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
W310X74	Steel	Steel I/Wide Flange	Yellow	14.6	1.71	391.61	56.22	
W310X79-1	Steel	Steel I/Wide Flange	Gray8Dark	15.66	1.58	425.24	95.86	
W310X86-1	Steel	Steel I/Wide Flange	Blue	17.05	2.1	475.7	106.91	
W310X97	Steel	Steel I/Wide Flange	Green	19.07	2.18	533.36	173.94	
W360X101-1	Steel	Steel I/Wide Flange	Yellow	20	3	723.16	121.09	
W360X1086	Steel	Steel I/Wide Flange	Green	215.45	1451.12	14294.93	4708.92	
W360X110-1	Steel	Steel I/Wide Flange	Gray8Dark	21.86	3.87	795.23	134.06	
W360X122-1	Steel	Steel I/Wide Flange	Blue	24.03	5.07	881.72	147.99	
W360X134	Steel	Steel I/Wide Flange	Blue	26.51	4.06	999.44	362.78	
W360X147	Steel	Steel I/Wide Flange	Green	29.14	5.38	1109.96	401.22	
W360X162-1	Steel	Steel I/Wide Flange	Cyan	31.93	7.11	1239.69	446.87	
W360X179-1	Steel	Steel I/Wide Flange	Red	35.34	9.37	1379.04	494.92	
W360X196-1	Steel	Steel I/Wide Flange	Magenta	38.75	12.3	1530.4	547.77	
W360X216-1	Steel	Steel I/Wide Flange	Yellow	42.63	15.21	1710.59	677.51	
W360X237-1	Steel	Steel I/Wide Flange	Gray8Dark	46.66	19.7	1900.39	747.18	
W360X262-1	Steel	Steel I/Wide Flange	Blue	51.77	26.43	2140.64	838.48	
W360X287-1	Steel	Steel I/Wide Flange	Green	56.73	34.84	2400.11	932.17	
W360X314-1	Steel	Steel I/Wide Flange	Cyan	62	44.69	2666.79	1030.68	
W360X32.9-1	Steel	Steel I/Wide Flange	Red	6.49	0.21	198.93	6.99	
W360X347-1	Steel	Steel I/Wide Flange	Red	68.51	59.58	3003.14	1150.8	
W360X382-1	Steel	Steel I/Wide Flange	Magenta	75.64	79.04	3411.56	1290.15	
W360X39-1	Steel	Steel I/Wide Flange	Magenta	7.69	0.36	245.06	8.91	
W360X421-1	Steel	Steel I/Wide Flange	Yellow	83.24	104.03	3844.02	1439.1	
W360X44-1	Steel	Steel I/Wide Flange	Yellow	8.85	0.38	290.7	19.6	
W360X463-1	Steel	Steel I/Wide Flange	Gray8Dark	91.45	135.98	4324.52	1609.68	
W360X509-1	Steel	Steel I/Wide Flange	Blue	101.06	178.03	4901.12	1809.09	
W360X51-1	Steel	Steel I/Wide Flange	Gray8Dark	10	0.57	341.16	23.3	
W360X551-1	Steel	Steel I/Wide Flange	Green	108.97	221.99	5429.67	1989.28	
W360X57.8-1	Steel	Steel I/Wide Flange	Blue	11.21	0.8	384.4	26.67	
W360X592-1	Steel	Steel I/Wide Flange	Cyan	117.03	273.89	6006.27	2169.47	
W360X634-1	Steel	Steel I/Wide Flange	Red	124.93	331.55	6606.9	2359.26	
W360X64	Steel	Steel I/Wide Flange	Green	12.6	1.05	427.65	45.17	
W360X677-1	Steel	Steel I/Wide Flange	Magenta	134.08	394.01	7183.5	2570.69	
W360X72	Steel	Steel I/Wide Flange	Cyan	14.11	1.45	482.9	51.41	
W360X744-1	Steel	Steel I/Wide Flange	Yellow	146.94	514.14	8216.58	2883.01	
W360X79	Steel	Steel I/Wide Flange	Red	15.66	1.94	540.56	57.66	
W360X818	Steel	Steel I/Wide Flange	Yellow	162.75	667.9	9441.86	3243.39	
W360X900	Steel	Steel I/Wide Flange	Gray8Dark	178.25	869.71	10811.29	3675.84	
W360X91-1	Steel	Steel I/Wide Flange	Magenta	17.83	2.19	639.07	106.91	
W360X990	Steel	Steel I/Wide Flange	Blue	195.3	1119.57	12396.95	4180.37	
W410X100	Steel	Steel I/Wide Flange	Cyan	19.53	2.39	953.8	118.92	
W410X114	Steel	Steel I/Wide Flange	Red	22.63	3.58	1109.96	137.9	
W410X132	Steel	Steel I/Wide Flange	Magenta	26.2	5.45	1299.76	162.89	
W410X149	Steel	Steel I/Wide Flange	Yellow	29.45	7.74	1489.56	185.95	
W410X38.8-1	Steel	Steel I/Wide Flange	Cyan	7.67	0.26	300.31	9.59	
W410X46.1-1	Steel	Steel I/Wide Flange	Red	9.13	0.46	374.79	12.4	
W410X53-1	Steel	Steel I/Wide Flange	Magenta	10.6	0.55	446.87	24.51	
W410X60	Steel	Steel I/Wide Flange	Yellow	11.8	0.79	518.94	28.83	
W410X67	Steel	Steel I/Wide Flange	Gray8Dark	13.3	1.11	586.21	32.91	
W410X75	Steel	Steel I/Wide Flange	Blue	14.69	1.52	658.29	37.24	
W410X85	Steel	Steel I/Wide Flange	Green	16.74	2.22	759.19	43	
W460X106	Steel	Steel I/Wide Flange	Blue	20.93	3.48	1170.02	60.3	
W460X113	Steel	Steel I/Wide Flange	Green	22.32	2.83	1330.99	152.08	
W460X128	Steel	Steel I/Wide Flange	Cyan	25.27	4.11	1530.4	174.9	
W460X144	Steel	Steel I/Wide Flange	Red	28.52	5.86	1749.03	201.09	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
W460X158	Steel	Steel I/Wide Flange	Magenta	31.16	7.47	1910	220.07	
W460X177	Steel	Steel I/Wide Flange	Yellow	35.03	10.6	2191.09	252.26	
W460X193	Steel	Steel I/Wide Flange	Gray8Dark	38.29	14.51	2450.56	278.69	
W460X213	Steel	Steel I/Wide Flange	Blue	42.01	19.2	2738.86	309.92	
W460X235	Steel	Steel I/Wide Flange	Green	46.35	25.23	3051.19	345.96	
W460X260	Steel	Steel I/Wide Flange	Cyan	51.46	33.88	3459.61	391.61	
W460X286	Steel	Steel I/Wide Flange	Red	56.27	44.69	3868.04	439.66	
W460X315	Steel	Steel I/Wide Flange	Magenta	62.31	58.62	4324.52	492.51	
W460X349	Steel	Steel I/Wide Flange	Yellow	68.67	78.8	4901.12	557.38	
W460X384	Steel	Steel I/Wide Flange	Gray8Dark	75.95	103.07	5501.75	627.06	
W460X421	Steel	Steel I/Wide Flange	Blue	83.24	134.06	6174.45	703.94	
W460X464	Steel	Steel I/Wide Flange	Green	91.61	176.1	6967.28	795.23	
W460X52-1	Steel	Steel I/Wide Flange	Gray8Dark	10.31	0.51	509.33	15.3	
W460X60-1	Steel	Steel I/Wide Flange	Blue	11.8	0.81	612.64	19.1	
W460X68-1	Steel	Steel I/Wide Flange	Green	13.5	1.22	711.14	22.51	
W460X74-1	Steel	Steel I/Wide Flange	Cyan	14.69	1.24	800.04	40.12	
W460X82-1	Steel	Steel I/Wide Flange	Red	16.28	1.66	888.93	44.93	
W460X89-1	Steel	Steel I/Wide Flange	Yellow	17.67	2.17	985.03	50.21	
W460X97	Steel	Steel I/Wide Flange	Gray8Dark	19.07	2.74	1069.12	54.78	
W530X101-1	Steel	Steel I/Wide Flange	Magenta	20	2.45	1479.95	64.63	
W530X109	Steel	Steel I/Wide Flange	Yellow	21.55	3.03	1600.07	70.63	
W530X123	Steel	Steel I/Wide Flange	Gray8Dark	24.34	4.35	1830.71	81.45	
W530X138	Steel	Steel I/Wide Flange	Blue	27.28	6.03	2070.96	92.98	
W530X150	Steel	Steel I/Wide Flange	Green	29.76	5.21	2426.53	247.46	
W530X165	Steel	Steel I/Wide Flange	Cyan	32.55	6.82	2666.79	273.89	
W530X182	Steel	Steel I/Wide Flange	Red	35.96	8.99	2955.09	305.12	
W530X196	Steel	Steel I/Wide Flange	Magenta	38.75	11.29	3219.36	333.95	
W530X219	Steel	Steel I/Wide Flange	Yellow	43.25	15.4	3627.79	377.19	
W530X248	Steel	Steel I/Wide Flange	Gray8Dark	48.83	23.59	4276.47	434.85	
W530X272	Steel	Steel I/Wide Flange	Blue	53.63	30.75	4732.94	482.9	
W530X300	Steel	Steel I/Wide Flange	Green	59.37	40.84	5309.55	542.97	
W530X66-1	Steel	Steel I/Wide Flange	Cyan	13	0.77	843.28	20.71	
W530X72	Steel	Steel I/Wide Flange	Red	14.11	0.8	958.6	38.68	
W530X74-1	Steel	Steel I/Wide Flange	Magenta	14.69	1.14	985.03	24.99	
W530X82-1	Steel	Steel I/Wide Flange	Yellow	16.28	1.24	1141.19	48.29	
W530X85-1	Steel	Steel I/Wide Flange	Cyan	16.74	1.77	1170.02	30.51	
W530X92-1	Steel	Steel I/Wide Flange	Red	18.29	1.83	1330.99	57.42	
W610X101-1	Steel	Steel I/Wide Flange	Magenta	20.15	1.87	1830.71	70.39	
W610X113-1	Steel	Steel I/Wide Flange	Yellow	22.48	2.69	2099.79	82.41	
W610X125-1	Steel	Steel I/Wide Flange	Gray8Dark	24.65	3.7	2368.87	94.42	
W610X140-1	Steel	Steel I/Wide Flange	Blue	27.75	5.26	2690.81	109.07	
W610X153	Steel	Steel I/Wide Flange	Green	30.23	7.06	3003.14	118.92	
W610X155	Steel	Steel I/Wide Flange	Cyan	30.69	4.71	3099.24	259.47	
W610X174	Steel	Steel I/Wide Flange	Red	34.41	6.73	3531.69	297.91	
W610X195	Steel	Steel I/Wide Flange	Magenta	38.6	9.49	4012.19	341.16	
W610X217	Steel	Steel I/Wide Flange	Magenta	42.94	13.41	4588.79	391.61	
W610X241	Steel	Steel I/Wide Flange	Yellow	47.74	18.5	5165.4	442.06	
W610X262	Steel	Steel I/Wide Flange	Gray8Dark	51.77	23.9	5669.92	478.1	
W610X285	Steel	Steel I/Wide Flange	Blue	56.58	30.75	6270.55	530.95	
W610X307	Steel	Steel I/Wide Flange	Green	60.76	38.2	6823.13	579	
W610X341	Steel	Steel I/Wide Flange	Cyan	67.27	51.41	7639.98	651.08	
W610X372	Steel	Steel I/Wide Flange	Red	73.47	66.55	8480.86	723.16	
W610X415	Steel	Steel I/Wide Flange	Magenta	81.84	90.57	9610.04	824.06	
W610X455	Steel	Steel I/Wide Flange	Yellow	89.75	117	10691.17	920.16	
W610X498	Steel	Steel I/Wide Flange	Gray8Dark	98.27	152.08	11892.42	1030.68	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
W610X551	Steel	Steel I/Wide Flange	Blue	108.97	201.09	13406	1160.41	
W610X82-1	Steel	Steel I/Wide Flange	Cyan	16.28	1.18	1350.21	29.07	
W610X92-1	Steel	Steel I/Wide Flange	Red	18.14	1.71	1549.62	34.6	
W690X125-1	Steel	Steel I/Wide Flange	Green	24.65	2.81	2858.99	105.95	
W690X140-1	Steel	Steel I/Wide Flange	Cyan	27.59	4.04	3267.41	123.97	
W690X152-1	Steel	Steel I/Wide Flange	Red	30.07	5.29	3627.79	139.11	
W690X170-1	Steel	Steel I/Wide Flange	Magenta	33.64	7.33	4084.27	159.05	
W690X192-1	Steel	Steel I/Wide Flange	Yellow	37.82	11.1	4756.97	184.03	
W690X217	Steel	Steel I/Wide Flange	Gray8Dark	43.25	11.29	5669.92	442.06	
W690X240	Steel	Steel I/Wide Flange	Blue	47.59	15.11	6318.6	497.32	
W690X265	Steel	Steel I/Wide Flange	Green	52.55	20.11	7015.33	554.98	
W690X289	Steel	Steel I/Wide Flange	Cyan	57.04	27.15	7856.21	619.85	
W690X323	Steel	Steel I/Wide Flange	Red	63.86	37.72	8913.31	703.94	
W690X350	Steel	Steel I/Wide Flange	Magenta	69.44	47.09	9706.14	768.8	
W690X384	Steel	Steel I/Wide Flange	Yellow	76.11	61.5	10811.29	860.1	
W690X419	Steel	Steel I/Wide Flange	Gray8Dark	83.08	79.52	11892.42	953.8	
W690X457	Steel	Steel I/Wide Flange	Blue	90.21	100.91	13093.68	1049.9	
W690X500	Steel	Steel I/Wide Flange	Green	99.2	130.94	14607.26	1179.63	
W690X548	Steel	Steel I/Wide Flange	Cyan	108.97	170.1	16192.91	1309.37	
W690X802	Steel	Steel I/Wide Flange	Yellow	159.65	494.92	25706.85	2109.4	
W760X134-1	Steel	Steel I/Wide Flange	Gray8Dark	26.35	2.83	3603.76	115.08	
W760X147-1	Steel	Steel I/Wide Flange	Blue	28.99	3.77	3988.17	128.05	
W760X161-1	Steel	Steel I/Wide Flange	Green	31.78	5	4468.67	146.07	
W760X173-1	Steel	Steel I/Wide Flange	Cyan	34.26	6.44	4925.14	164.09	
W760X185-1	Steel	Steel I/Wide Flange	Red	36.43	8	5357.6	180.91	
W760X196-1	Steel	Steel I/Wide Flange	Magenta	38.75	9.73	5766.02	196.04	
W760X220	Steel	Steel I/Wide Flange	Yellow	43.56	14.51	6678.98	227.04	
W760X257	Steel	Steel I/Wide Flange	Gray8Dark	50.84	15.59	8240.61	598.22	
W760X284	Steel	Steel I/Wide Flange	Blue	56.11	21	9201.61	672.7	
W760X314	Steel	Steel I/Wide Flange	Green	62.31	28.35	10306.77	756.79	
W760X350	Steel	Steel I/Wide Flange	Cyan	69.29	40.36	11700.22	855.29	
W760X389	Steel	Steel I/Wide Flange	Red	77.04	54.06	13093.68	958.6	
W760X434	Steel	Steel I/Wide Flange	Magenta	86.03	75.2	14895.56	1100.35	
W760X484	Steel	Steel I/Wide Flange	Yellow	95.95	103.07	16793.54	1239.69	
W760X531	Steel	Steel I/Wide Flange	Gray8Dark	104.94	134.06	18691.52	1391.05	
W760X582	Steel	Steel I/Wide Flange	Blue	115.01	172.98	20709.63	1549.62	
W840X176-1	Steel	Steel I/Wide Flange	Green	34.72	5.31	5910.17	186.92	
W840X193-1	Steel	Steel I/Wide Flange	Cyan	38.29	7.38	6703	217.91	
W840X210-1	Steel	Steel I/Wide Flange	Red	41.54	9.71	7447.78	245.06	
W840X226-1	Steel	Steel I/Wide Flange	Magenta	44.95	12.4	8168.53	273.89	
W840X251	Steel	Steel I/Wide Flange	Yellow	49.45	17.71	9297.71	309.92	
W840X299	Steel	Steel I/Wide Flange	Gray8Dark	59.06	20.81	11604.12	749.58	
W840X329	Steel	Steel I/Wide Flange	Blue	65.26	27.87	12901.48	840.88	
W840X359	Steel	Steel I/Wide Flange	Green	71.15	36.28	14198.83	932.17	
W840X392	Steel	Steel I/Wide Flange	Cyan	77.35	48.77	15904.61	1040.29	
W840X433	Steel	Steel I/Wide Flange	Red	85.56	65.11	17706.5	1160.41	
W840X473	Steel	Steel I/Wide Flange	Magenta	93.78	84.33	19508.38	1290.15	
W840X527	Steel	Steel I/Wide Flange	Yellow	104.01	115.08	22006.99	1460.73	
W840X576	Steel	Steel I/Wide Flange	Gray8Dark	113.93	147.99	24265.35	1619.29	
W920X201	Steel	Steel I/Wide Flange	Yellow	39.84	6.99	7808.16	225.12	
W920X223-1	Steel	Steel I/Wide Flange	Gray8Dark	44.33	10.09	9033.44	269.08	
W920X238-1	Steel	Steel I/Wide Flange	Blue	46.97	12.4	9754.19	295.51	
W920X253-1	Steel	Steel I/Wide Flange	Green	50.07	15.11	10498.97	319.53	
W920X271-1	Steel	Steel I/Wide Flange	Cyan	53.63	18.5	11291.8	345.96	
W920X289	Steel	Steel I/Wide Flange	Red	57.04	22.2	12108.65	374.79	

Name	Material	Shape	Color	Area in2	J in4	I33 in4	I22 in4	I23 in4
W920X313	Steel	Steel I/Wide Flange	Magenta	61.85	28.11	13189.78	410.83	
W920X344	Steel	Steel I/Wide Flange	Yellow	68.2	28.59	15592.29	939.38	
W920X345	Steel	Steel I/Wide Flange	Gray8Dark	68.05	39.64	14991.66	468.49	
W920X368	Steel	Steel I/Wide Flange	Blue	72.54	34.6	16697.44	1009.05	
W920X381	Steel	Steel I/Wide Flange	Green	75.33	52.86	16793.54	528.55	
W920X390	Steel	Steel I/Wide Flange	Cyan	77.19	41.56	17898.7	1090.74	
W920X420	Steel	Steel I/Wide Flange	Red	82.93	52.61	19604.48	1198.85	
W920X449	Steel	Steel I/Wide Flange	Magenta	88.97	64.39	21094.03	1299.76	
W920X491	Steel	Steel I/Wide Flange	Yellow	96.88	84.33	23304.34	1419.88	
W920X537	Steel	Steel I/Wide Flange	Gray8Dark	106.02	109.07	25706.85	1568.84	
W920X588	Steel	Steel I/Wide Flange	Blue	115.94	141.99	28589.86	1749.03	
W920X656	Steel	Steel I/Wide Flange	Green	130.05	193.88	32193.63	1989.28	
W920X725	Steel	Steel I/Wide Flange	Cyan	143.07	257.07	36037.64	2251.15	
W920X787	Steel	Steel I/Wide Flange	Red	156.55	326.74	39641.41	2498.61	
W920X970	Steel	Steel I/Wide Flange	Magenta	192.2	593.42	50692.95	3219.36	

Table 2.2 - Frame Section Property Definitions - Summary (Part 2 of 4)

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
Column									
HSS101.6X101.6X12.7				3.73	3.73	5.92	5.92	5.92	5.92
HSS101.6X101.6X3.2				0.93	0.93	2.19	2.19	2.19	2.19
HSS101.6X101.6X4.8				1.4	1.4	3.09	3.09	3.09	3.09
HSS101.6X101.6X6.4				1.87	1.87	3.89	3.89	3.89	3.89
HSS101.6X101.6X7.9				2.34	2.34	4.55	4.55	4.55	4.55
HSS101.6X101.6X9.5				2.8	2.8	5.13	5.13	5.13	5.13
HSS101.6X50.8X3.2				0.93	0.46	1.32	1.32	0.9	0.9
HSS101.6X50.8X4.8				1.4	0.7	1.82	1.82	1.22	1.22
HSS101.6X50.8X6.4				1.87	0.93	2.24	2.24	1.48	1.48
HSS101.6X50.8X7.9				2.34	1.16	2.56	2.56	1.67	1.67
HSS101.6X50.8X9.5				2.8	1.4	2.79	2.79	1.8	1.8
HSS101.6X63.5X3.2				0.93	0.58	1.54	1.54	1.19	1.19
HSS101.6X63.5X4.8				1.4	0.87	2.14	2.14	1.65	1.65
HSS101.6X63.5X6.4				1.87	1.17	2.64	2.64	2.02	2.02
HSS101.6X63.5X7.9				2.34	1.45	3.05	3.05	2.31	2.31
HSS101.6X63.5X9.5				2.8	1.74	3.37	3.37	2.54	2.54
HSS101.6X76.2X3.2				0.93	0.7	1.76	1.76	1.51	1.51
HSS101.6X76.2X4.8				1.4	1.04	2.45	2.45	2.11	2.11
HSS101.6X76.2X6.4				1.87	1.4	3.06	3.06	2.61	2.61
HSS101.6X76.2X7.9				2.34	1.75	3.55	3.55	3.01	3.01
HSS101.6X76.2X9.5				2.8	2.09	3.95	3.95	3.35	3.35
HSS114.3X114.3X12.7				4.17	4.17	8.06	8.06	8.06	8.06
HSS114.3X114.3X3.2				1.04	1.04	2.83	2.83	2.83	2.83
HSS114.3X114.3X4.8				1.56	1.56	4.01	4.01	4.01	4.01
HSS114.3X114.3X6.4				2.09	2.09	5.09	5.09	5.09	5.09
HSS114.3X114.3X7.9				2.61	2.61	6.02	6.02	6.02	6.02
HSS114.3X114.3X9.5				3.13	3.13	6.82	6.82	6.82	6.82
HSS127X101.6X12.7				4.65	3.73	8.48	8.48	7.42	7.42
HSS127X101.6X3.2				1.16	0.93	2.97	2.97	2.62	2.62
HSS127X101.6X4.8				1.74	1.4	4.24	4.24	3.72	3.72
HSS127X101.6X6.4				2.33	1.87	5.36	5.36	4.71	4.71
HSS127X101.6X7.9				2.91	2.34	6.32	6.32	5.53	5.53
HSS127X101.6X9.5				3.49	2.8	7.16	7.16	6.27	6.27
HSS127X127X12.7				4.65	4.65	10.38	10.38	10.38	10.38
HSS127X127X3.2				1.16	1.16	3.52	3.52	3.52	3.52

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS127X127X4.8				1.74	1.74	5.04	5.04	5.04	5.04
HSS127X127X6.4				2.33	2.33	6.4	6.4	6.4	6.4
HSS127X127X7.9				2.91	2.91	7.6	7.6	7.6	7.6
HSS127X127X9.5				3.49	3.49	8.68	8.68	8.68	8.68
HSS127X50.8X3.2				1.16	0.46	1.86	1.86	1.1	1.1
HSS127X50.8X4.8				1.74	0.7	2.6	2.6	1.51	1.51
HSS127X50.8X6.4				2.33	0.93	3.23	3.23	1.84	1.84
HSS127X50.8X7.9				2.91	1.16	3.74	3.74	2.1	2.1
HSS127X50.8X9.5				3.49	1.4	4.16	4.16	2.28	2.28
HSS127X63.5X3.2				1.16	0.58	2.13	2.13	1.46	1.46
HSS127X63.5X4.8				1.74	0.87	3.01	3.01	2.02	2.02
HSS127X63.5X6.4				2.33	1.17	3.76	3.76	2.5	2.5
HSS127X76.2X12.7				4.65	2.79	6.56	6.56	4.79	4.79
HSS127X76.2X3.2				1.16	0.7	2.41	2.41	1.83	1.83
HSS127X76.2X4.8				1.74	1.04	3.41	3.41	2.56	2.56
HSS127X76.2X6.4				2.33	1.4	4.28	4.28	3.2	3.2
HSS127X76.2X7.9				2.91	1.75	5.04	5.04	3.73	3.73
HSS127X76.2X9.5				3.49	2.09	5.64	5.64	4.16	4.16
HSS139.7X139.7X3.2				1.28	1.28	4.28	4.28	4.28	4.28
HSS139.7X139.7X4.8				1.92	1.92	6.17	6.17	6.17	6.17
HSS139.7X139.7X6.4				2.57	2.57	7.87	7.87	7.87	7.87
HSS139.7X139.7X7.9				3.21	3.21	9.42	9.42	9.42	9.42
HSS139.7X139.7X9.5				3.85	3.85	10.81	10.81	10.81	10.81
HSS152.4X101.6X12.7				5.56	3.73	11.4	11.4	8.87	8.87
HSS152.4X101.6X3.2				1.39	0.93	3.81	3.81	3.06	3.06
HSS152.4X101.6X4.8				2.08	1.4	5.48	5.48	4.37	4.37
HSS152.4X101.6X6.4				2.79	1.87	6.99	6.99	5.53	5.53
HSS152.4X101.6X7.9				3.48	2.34	8.27	8.27	6.57	6.57
HSS152.4X101.6X9.5				4.17	2.8	9.47	9.47	7.42	7.42
HSS152.4X127X12.7				5.56	4.65	13.73	13.73	12.3	12.3
HSS152.4X127X3.2				1.39	1.16	4.48	4.48	4.08	4.08
HSS152.4X127X4.8				2.08	1.74	6.45	6.45	5.84	5.84
HSS152.4X127X6.4				2.79	2.33	8.27	8.27	7.48	7.48
HSS152.4X127X7.9				3.48	2.91	9.88	9.88	8.92	8.92
HSS152.4X127X9.5				4.17	3.49	11.32	11.32	10.19	10.19
HSS152.4X152.4X12.7				5.56	5.56	16.14	16.14	16.14	16.14
HSS152.4X152.4X15.9				6.97	6.97	18.47	18.47	18.47	18.47
HSS152.4X152.4X3.2				1.39	1.39	5.18	5.18	5.18	5.18
HSS152.4X152.4X4.8				2.08	2.08	7.45	7.45	7.45	7.45
HSS152.4X152.4X6.4				2.79	2.79	9.56	9.56	9.56	9.56
HSS152.4X152.4X7.9				3.48	3.48	11.48	11.48	11.48	11.48
HSS152.4X152.4X9.5				4.17	4.17	13.17	13.17	13.17	13.17
HSS152.4X50.8X3.2				1.39	0.46	2.48	2.48	1.31	1.31
HSS152.4X50.8X4.8				2.08	0.7	3.51	3.51	1.8	1.8
HSS152.4X50.8X6.4				2.79	0.93	4.38	4.38	2.21	2.21
HSS152.4X50.8X7.9				3.48	1.16	5.11	5.11	2.52	2.52
HSS152.4X50.8X9.5				4.17	1.4	5.72	5.72	2.76	2.76
HSS152.4X76.2X12.7				5.56	2.79	8.99	8.99	5.8	5.8
HSS152.4X76.2X3.2				1.39	0.7	3.16	3.16	2.15	2.15
HSS152.4X76.2X4.8				2.08	1.04	4.48	4.48	3.03	3.03
HSS152.4X76.2X6.4				2.79	1.4	5.68	5.68	3.8	3.8
HSS152.4X76.2X7.9				3.48	1.75	6.72	6.72	4.45	4.45
HSS152.4X76.2X9.5				4.17	2.09	7.59	7.59	4.98	4.98
HSS177.8X101.6X12.7				6.51	3.73	14.47	14.47	10.31	10.31
HSS177.8X101.6X3.2				1.63	0.93	4.74	4.74	3.51	3.51

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS177.8X101.6X4.8				2.44	1.4	6.79	6.79	4.98	4.98
HSS177.8X101.6X6.4				3.27	1.87	8.71	8.71	6.38	6.38
HSS177.8X101.6X7.9				4.08	2.34	10.42	10.42	7.57	7.57
HSS177.8X101.6X9.5				4.89	2.8	11.93	11.93	8.62	8.62
HSS177.8X127X12.7				6.51	4.65	17.28	17.28	14.22	14.22
HSS177.8X127X3.2				1.63	1.16	5.51	5.51	4.64	4.64
HSS177.8X127X4.8				2.44	1.74	7.95	7.95	6.64	6.64
HSS177.8X127X6.4				3.27	2.33	10.22	10.22	8.52	8.52
HSS177.8X127X7.9				4.08	2.91	12.27	12.27	10.19	10.19
HSS177.8X127X9.5				4.89	3.49	14.12	14.12	11.72	11.72
HSS177.8X177.8X12.7				6.51	6.51	22.97	22.97	22.97	22.97
HSS177.8X177.8X15.9				8.17	8.17	26.67	26.67	26.67	26.67
HSS177.8X177.8X3.2				1.63	1.63	7.06	7.06	7.06	7.06
HSS177.8X177.8X4.8				2.44	2.44	10.28	10.28	10.28	10.28
HSS177.8X177.8X6.4				3.27	3.27	13.3	13.3	13.3	13.3
HSS177.8X177.8X7.9				4.08	4.08	16.04	16.04	16.04	16.04
HSS177.8X177.8X9.5				4.89	4.89	18.58	18.58	18.58	18.58
HSS177.8X50.8X3.2				1.63	0.46	3.17	3.17	1.52	1.52
HSS177.8X50.8X4.8				2.44	0.7	4.48	4.48	2.1	2.1
HSS177.8X50.8X6.4				3.27	0.93	5.65	5.65	2.57	2.57
HSS177.8X76.2X12.7				6.51	2.79	11.59	11.59	6.81	6.81
HSS177.8X76.2X3.2				1.63	0.7	3.94	3.94	2.47	2.47
HSS177.8X76.2X4.8				2.44	1.04	5.65	5.65	3.49	3.49
HSS177.8X76.2X6.4				3.27	1.4	7.2	7.2	4.4	4.4
HSS177.8X76.2X7.9				4.08	1.75	8.5	8.5	5.16	5.16
HSS177.8X76.2X9.5				4.89	2.09	9.74	9.74	5.81	5.81
HSS203.2X101.6X12.7				7.43	3.73	17.98	17.98	11.75	11.75
HSS203.2X101.6X15.9				9.31	4.68	20.5	20.5	13.28	13.28
HSS203.2X101.6X3.2				1.86	0.93	5.73	5.73	3.94	3.94
HSS203.2X101.6X4.8				2.78	1.4	8.3	8.3	5.62	5.62
HSS203.2X101.6X6.4				3.73	1.87	10.64	10.64	7.17	7.17
HSS203.2X101.6X7.9				4.65	2.34	12.75	12.75	8.57	8.57
HSS203.2X101.6X9.5				5.58	2.8	14.67	14.67	9.76	9.76
HSS203.2X152.4X12.7				7.43	5.56	24.59	24.59	20.88	20.88
HSS203.2X152.4X15.9				9.31	6.97	28.56	28.56	24.17	24.17
HSS203.2X152.4X4.8				2.78	2.08	10.94	10.94	9.39	9.39
HSS203.2X152.4X6.4				3.73	2.79	14.19	14.19	12.2	12.2
HSS203.2X152.4X7.9				4.65	3.48	17.07	17.07	14.61	14.61
HSS203.2X152.4X9.5				5.58	4.17	19.78	19.78	16.94	16.94
HSS203.2X203.2X12.7				7.43	7.43	31.26	31.26	31.26	31.26
HSS203.2X203.2X15.9				9.31	9.31	36.55	36.55	36.55	36.55
HSS203.2X203.2X3.2				1.86	1.86	9.38	9.38	9.38	9.38
HSS203.2X203.2X4.8				2.78	2.78	13.59	13.59	13.59	13.59
HSS203.2X203.2X6.4				3.73	3.73	17.68	17.68	17.68	17.68
HSS203.2X203.2X7.9				4.65	4.65	21.4	21.4	21.4	21.4
HSS203.2X203.2X9.5				5.58	5.58	25.01	25.01	25.01	25.01
HSS203.2X50.8X3.2				1.86	0.46	3.93	3.93	1.72	1.72
HSS203.2X50.8X4.8				2.78	0.7	5.6	5.6	2.39	2.39
HSS203.2X50.8X6.4				3.73	0.93	7.15	7.15	2.93	2.93
HSS203.2X50.8X7.9				4.65	1.16	8.42	8.42	3.39	3.39
HSS203.2X50.8X9.5				5.58	1.4	9.56	9.56	3.72	3.72
HSS203.2X76.2X12.7				7.43	2.79	14.67	14.67	7.8	7.8
HSS203.2X76.2X3.2				1.86	0.7	4.83	4.83	2.8	2.8
HSS203.2X76.2X4.8				2.78	1.04	6.97	6.97	3.96	3.96
HSS203.2X76.2X6.4				3.73	1.4	8.9	8.9	5	5

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS203.2X76.2X7.9				4.65	1.75	10.58	10.58	5.88	5.88
HSS203.2X76.2X9.5				5.58	2.09	12.14	12.14	6.66	6.66
HSS228.6X127X12.7				8.38	4.65	25.53	25.53	18.07	18.07
HSS228.6X127X15.9				10.51	5.83	29.53	29.53	20.76	20.76
HSS228.6X127X4.8				3.14	1.74	11.35	11.35	8.28	8.28
HSS228.6X127X6.4				4.2	2.33	14.66	14.66	10.67	10.67
HSS228.6X127X7.9				5.25	2.91	17.69	17.69	12.78	12.78
HSS228.6X127X9.5				6.29	3.49	20.52	20.52	14.7	14.7
HSS228.6X177.8X12.7				8.38	6.51	33.04	33.04	28.52	28.52
HSS228.6X177.8X15.9				10.51	8.17	38.59	38.59	33.39	33.39
HSS228.6X177.8X4.8				3.14	2.44	14.34	14.34	12.62	12.62
HSS228.6X177.8X6.4				4.2	3.27	18.65	18.65	16.32	16.32
HSS228.6X177.8X7.9				5.25	4.08	22.65	22.65	19.75	19.75
HSS228.6X177.8X9.5				6.29	4.89	26.38	26.38	22.97	22.97
HSS228.6X228.6X12.7				8.38	8.38	40.61	40.61	40.61	40.61
HSS228.6X228.6X15.9				10.51	10.51	47.91	47.91	47.91	47.91
HSS228.6X228.6X3.2				2.09	2.09	11.88	11.88	11.88	11.88
HSS228.6X228.6X6.4				4.2	4.2	22.65	22.65	22.65	22.65
HSS228.6X228.6X7.9				5.25	5.25	27.5	27.5	27.5	27.5
HSS228.6X228.6X74.8				3.14	3.14	17.32	17.32	17.32	17.32
HSS228.6X228.6X9.5				6.29	6.29	32.19	32.19	32.19	32.19
HSS228.6X76.2X12.7				8.38	2.79	17.91	17.91	8.79	8.79
HSS228.6X76.2X4.8				3.14	1.04	8.37	8.37	4.42	4.42
HSS228.6X76.2X6.4				4.2	1.4	10.71	10.71	5.59	5.59
HSS228.6X76.2X7.9				5.25	1.75	12.79	12.79	6.58	6.58
HSS228.6X76.2X9.5				6.29	2.09	14.71	14.71	7.46	7.46
HSS254X101.6X12.7				9.29	3.73	25.8	25.8	14.72	14.72
HSS254X101.6X15.9				11.65	4.68	29.79	29.79	16.63	16.63
HSS254X101.6X3.2				2.32	0.93	7.98	7.98	4.81	4.81
HSS254X101.6X4.8				3.48	1.4	11.58	11.58	6.93	6.93
HSS254X101.6X6.4				4.66	1.87	14.94	14.94	8.82	8.82
HSS254X101.6X7.9				5.82	2.34	18.02	18.02	10.55	10.55
HSS254X101.6X9.5				6.98	2.8	20.81	20.81	12.09	12.09
HSS254X127X4.8				3.48	1.74	13.26	13.26	9.08	9.08
HSS254X127X6.4				4.66	2.33	17.15	17.15	11.72	11.72
HSS254X127X7.9				5.82	2.91	20.81	20.81	14.13	14.13
HSS254X127X9.5				6.98	3.49	23.98	23.98	16.24	16.24
HSS254X152.4X12.7				9.29	5.56	34.21	34.21	25.69	25.69
HSS254X152.4X15.9				11.65	6.97	40.22	40.22	29.87	29.87
HSS254X152.4X4.8				3.48	2.08	14.94	14.94	11.4	11.4
HSS254X152.4X6.4				4.66	2.79	19.36	19.36	14.77	14.77
HSS254X152.4X7.9				5.82	3.48	23.59	23.59	17.83	17.83
HSS254X152.4X9.5				6.98	4.17	27.39	27.39	20.64	20.64
HSS254X203.2X12.7				9.29	7.43	42.81	42.81	37.82	37.82
HSS254X203.2X15.9				11.65	9.31	50.45	50.45	44.55	44.55
HSS254X203.2X4.8				3.48	2.78	18.26	18.26	16.29	16.29
HSS254X203.2X6.4				4.66	3.73	23.78	23.78	21.22	21.22
HSS254X203.2X7.9				5.82	4.65	29.02	29.02	25.79	25.79
HSS254X203.2X9.5				6.98	5.58	33.78	33.78	30	30
HSS254X254X12.7				9.29	9.29	51.41	51.41	51.41	51.41
HSS254X254X15.9				11.65	11.65	61.02	61.02	61.02	61.02
HSS254X254X4.8				3.48	3.48	21.62	21.62	21.62	21.62
HSS254X254X6.4				4.66	4.66	28.21	28.21	28.21	28.21
HSS254X254X7.9				5.82	5.82	34.4	34.4	34.4	34.4
HSS254X254X9.5				6.98	6.98	40.41	40.41	40.41	40.41

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS254X50.8X3.2				2.32	0.46	5.72	5.72	2.14	2.14
HSS254X50.8X4.8				3.48	0.7	8.22	8.22	2.98	2.98
HSS254X50.8X6.4				4.66	0.93	10.52	10.52	3.68	3.68
HSS254X50.8X7.9				5.82	1.16	12.54	12.54	4.23	4.23
HSS254X50.8X9.5				6.98	1.4	14.32	14.32	4.71	4.71
HSS254X76.2X3.2				2.32	0.7	6.82	6.82	3.44	3.44
HSS254X76.2X4.8				3.48	1.04	9.9	9.9	4.89	4.89
HSS254X76.2X6.4				4.66	1.4	12.73	12.73	6.18	6.18
HSS254X76.2X7.9				5.82	1.75	15.28	15.28	7.34	7.34
HSS254X76.2X9.5				6.98	2.09	17.59	17.59	8.26	8.26
HSS254X88.9X12.7				9.29	3.25	23.59	23.59	12.23	12.23
HSS254X88.9X3.2				2.32	0.81	7.4	7.4	4.13	4.13
HSS254X88.9X4.8				3.48	1.22	10.72	10.72	5.89	5.89
HSS254X88.9X6.4				4.66	1.63	13.84	13.84	7.48	7.48
HSS254X88.9X7.9				5.82	2.04	16.63	16.63	8.91	8.91
HSS254X88.9X9.5				6.98	2.44	19.22	19.22	10.17	10.17
HSS304.8X101.6X12.7				11.16	3.73	34.97	34.97	17.59	17.59
HSS304.8X101.6X15.9				13.99	4.68	40.82	40.82	20.1	20.1
HSS304.8X101.6X4.8				4.18	1.4	15.29	15.29	8.17	8.17
HSS304.8X101.6X6.4				5.6	1.87	19.81	19.81	10.46	10.46
HSS304.8X101.6X7.9				6.99	2.34	23.97	23.97	12.56	12.56
HSS304.8X101.6X9.5				8.38	2.8	27.97	27.97	14.36	14.36
HSS304.8X152.4X12.7				11.16	5.56	45.22	45.22	30.43	30.43
HSS304.8X152.4X15.9				13.99	6.97	53.62	53.62	35.73	35.73
HSS304.8X152.4X4.8				4.18	2.08	19.33	19.33	13.33	13.33
HSS304.8X152.4X6.4				5.6	2.79	25.17	25.17	17.34	17.34
HSS304.8X152.4X7.9				6.99	3.48	30.65	30.65	20.96	20.96
HSS304.8X152.4X9.5				8.38	4.17	35.81	35.81	24.33	24.33
HSS304.8X203.2X12.7				11.16	7.43	55.62	55.62	44.55	44.55
HSS304.8X203.2X15.9				13.99	9.31	66.03	66.03	52.55	52.55
HSS304.8X203.2X4.8				4.18	2.78	23.33	23.33	18.94	18.94
HSS304.8X203.2X6.4				5.6	3.73	30.65	30.65	24.71	24.71
HSS304.8X203.2X7.9				6.99	4.65	37.29	37.29	30	30
HSS304.8X203.2X9.5				8.38	5.58	43.62	43.62	35.05	35.05
HSS304.8X254X12.7				11.16	9.29	65.63	65.63	59.58	59.58
HSS304.8X254X6.4				5.6	4.66	35.97	35.97	32.82	32.82
HSS304.8X254X7.9				6.99	5.82	44.02	44.02	39.98	39.98
HSS304.8X254X9.5				8.38	6.98	51.62	51.62	46.8	46.8
HSS304.8X304.8X12.7				11.16	11.16	76.03	76.03	76.03	76.03
HSS304.8X304.8X15.9				13.99	13.99	91.24	91.24	91.24	91.24
HSS304.8X304.8X4.8				4.18	4.18	31.49	31.49	31.49	31.49
HSS304.8X304.8X6.4				5.6	5.6	41.22	41.22	41.22	41.22
HSS304.8X304.8X7.9				6.99	6.99	50.82	50.82	50.82	50.82
HSS304.8X304.8X9.5				8.38	8.38	59.62	59.62	59.62	59.62
HSS304.8X50.8X4.8				4.18	0.7	11.24	11.24	3.56	3.56
HSS304.8X50.8X6.4				5.6	0.93	14.49	14.49	4.42	4.42
HSS304.8X50.8X7.9				6.99	1.16	17.33	17.33	5.09	5.09
HSS304.8X76.2X4.8				4.18	1.04	13.25	13.25	5.81	5.81
HSS304.8X76.2X6.4				5.6	1.4	17.17	17.17	7.4	7.4
HSS304.8X76.2X7.9				6.99	1.75	20.65	20.65	8.73	8.73
HSS304.8X88.9X7.9				6.99	2.04	22.33	22.33	10.63	10.63
HSS304.8X88.9X9.5				8.38	2.44	25.97	25.97	12.18	12.18
HSS355.6X101.6X12.7				13.02	3.73	45.25	45.25	20.46	20.46
HSS355.6X101.6X15.9				16.33	4.68	53.14	53.14	23.45	23.45
HSS355.6X101.6X4.8				4.88	1.4	19.54	19.54	9.46	9.46

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS355.6X101.6X6.4				6.53	1.87	25.4	25.4	12.2	12.2
HSS355.6X101.6X7.9				8.16	2.34	30.82	30.82	14.6	14.6
HSS355.6X101.6X9.5				9.78	2.8	36	36	16.75	16.75
HSS355.6X152.4X12.7				13.02	5.56	57.25	57.25	35.09	35.09
HSS355.6X152.4X15.9				16.33	6.97	68.22	68.22	41.43	41.43
HSS355.6X152.4X4.8				4.88	2.08	24.27	24.27	15.34	15.34
HSS355.6X152.4X6.4				6.53	2.79	31.68	31.68	19.91	19.91
HSS355.6X152.4X7.9				8.16	3.48	38.74	38.74	24.17	24.17
HSS355.6X152.4X9.5				9.78	4.17	45.25	45.25	28.1	28.1
HSS355.6X254X12.7				13.02	9.29	81.94	81.94	68.23	68.23
HSS355.6X254X15.9				16.33	11.65	98.05	98.05	81.2	81.2
HSS355.6X254X6.4				6.53	4.66	44.23	44.23	37.19	37.19
HSS355.6X254X7.9				8.16	5.82	54.17	54.17	45.41	45.41
HSS355.6X254X9.5				9.78	6.98	63.77	63.77	53.34	53.34
HSS355.6X355.6X12.7				13.02	13.02	105.93	105.93	105.93	105.93
HSS355.6X355.6X15.9				16.33	16.33	127.88	127.88	127.88	127.88
HSS355.6X355.6X7.9				8.16	8.16	69.94	69.94	69.94	69.94
HSS355.6X355.6X9.5				9.78	9.78	82.28	82.28	82.28	82.28
HSS406.4X101.6X12.7				14.85	3.73	56.82	56.82	23.45	23.45
HSS406.4X101.6X15.9				18.63	4.68	67.34	67.34	26.92	26.92
HSS406.4X101.6X4.8				5.56	1.4	24.14	24.14	10.71	10.71
HSS406.4X101.6X6.4				7.45	1.87	31.56	31.56	13.76	13.76
HSS406.4X101.6X7.9				9.3	2.34	38.48	38.48	16.51	16.51
HSS406.4X101.6X9.5				11.15	2.8	45.09	45.09	19.03	19.03
HSS406.4X203.2X12.7				14.85	7.43	85.07	85.07	57.54	57.54
HSS406.4X203.2X15.9				18.63	9.31	101.91	101.91	68.54	68.54
HSS406.4X203.2X6.4				7.45	3.73	45.99	45.99	31.8	31.8
HSS406.4X203.2X7.9				9.3	4.65	56.51	56.51	38.78	38.78
HSS406.4X203.2X9.5				11.15	5.58	66.43	66.43	45.27	45.27
HSS406.4X304.8X12.7				14.85	11.16	113.03	113.03	96.84	96.84
HSS406.4X304.8X15.9				18.63	13.99	136.48	136.48	116.45	116.45
HSS406.4X304.8X7.9				9.3	6.99	74.55	74.55	64.02	64.02
HSS406.4X304.8X9.5				11.15	8.38	87.78	87.78	75.23	75.23
HSS406.4X406.4X12.7				14.85	14.85	141.29	141.29	141.29	141.29
HSS406.4X406.4X15.9				18.63	18.63	171.35	171.35	171.35	171.35
HSS406.4X406.4X7.9				9.3	9.3	92.59	92.59	92.59	92.59
HSS406.4X406.4X9.5				11.15	11.15	109.12	109.12	109.12	109.12
HSS457.2X152.4X12.7				16.72	5.56	85.46	85.46	44.8	44.8
HSS457.2X152.4X15.9				20.97	6.97	102.55	102.55	52.83	52.83
HSS457.2X152.4X6.4				8.39	2.79	46.47	46.47	25.13	25.13
HSS457.2X152.4X7.9				10.47	3.48	57.15	57.15	30.51	30.51
HSS457.2X152.4X9.5				12.55	4.17	67.03	67.03	35.41	35.41
HSS50.8X25.4X3.2				0.46	0.23	0.28	0.28	0.18	0.18
HSS50.8X25.4X4.8				0.7	0.35	0.35	0.35	0.22	0.22
HSS50.8X38.1X3.2				0.46	0.35	0.38	0.38	0.33	0.33
HSS50.8X38.1X4.8				0.7	0.52	0.49	0.49	0.42	0.42
HSS50.8X50.8X3.2				0.46	0.46	0.49	0.49	0.49	0.49
HSS50.8X50.8X4.8				0.7	0.7	0.64	0.64	0.64	0.64
HSS50.8X50.8X6.4				0.93	0.93	0.75	0.75	0.75	0.75
HSS508X101.6X12.7				18.58	3.73	83.85	83.85	29.2	29.2
HSS508X101.6X6.4				9.32	1.87	45.89	45.89	17.11	17.11
HSS508X101.6X7.9				11.64	2.34	55.98	55.98	20.46	20.46
HSS508X101.6X9.5				13.95	2.8	65.59	65.59	23.69	23.69
HSS508X203.2X12.7				18.58	7.43	118.92	118.92	70.94	70.94
HSS508X203.2X15.9				23.31	9.31	143.91	143.91	84.77	84.77

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS508X203.2X7.9				11.64	4.65	78.56	78.56	47.32	47.32
HSS508X203.2X9.5				13.95	5.58	92.5	92.5	55.55	55.55
HSS508X304.8X12.7				18.58	11.16	154.96	154.96	117.25	117.25
HSS508X304.8X15.9				23.31	13.99	188.12	188.12	141.66	141.66
HSS508X304.8X7.9				11.64	6.99	100.91	100.91	77.23	77.23
HSS508X304.8X9.5				13.95	8.38	119.89	119.89	91.24	91.24
HSS57.2X50.8X3.2				0.52	0.46	0.57	0.57	0.54	0.54
HSS57.2X50.8X4.8				0.78	0.7	0.76	0.76	0.71	0.71
HSS57.2X57.2X3.2				0.52	0.52	0.63	0.63	0.63	0.63
HSS57.2X57.2X4.8				0.78	0.78	0.85	0.85	0.85	0.85
HSS57.2X57.2X6.4				1.05	1.05	1	1	1	1
HSS63.5X25.4X3.2				0.58	0.23	0.4	0.4	0.23	0.23
HSS63.5X25.4X4.8				0.87	0.35	0.52	0.52	0.29	0.29
HSS63.5X38.1X3.2				0.58	0.35	0.53	0.53	0.4	0.4
HSS63.5X38.1X4.8				0.87	0.52	0.71	0.71	0.52	0.52
HSS63.5X38.1X6.4				1.17	0.7	0.82	0.82	0.6	0.6
HSS63.5X50.8X3.2				0.58	0.46	0.67	0.67	0.59	0.59
HSS63.5X50.8X4.8				0.87	0.7	0.9	0.9	0.79	0.79
HSS63.5X50.8X6.4				1.17	0.93	1.06	1.06	0.93	0.93
HSS63.5X63.5X3.2				0.58	0.58	0.8	0.8	0.8	0.8
HSS63.5X63.5X4.8				0.87	0.87	1.08	1.08	1.08	1.08
HSS63.5X63.5X6.4				1.17	1.17	1.3	1.3	1.3	1.3
HSS63.5X63.5X7.9				1.45	1.45	1.46	1.46	1.46	1.46
HSS76.2X25.4X3.2				0.7	0.23	0.54	0.54	0.28	0.28
HSS76.2X25.4X4.8				1.04	0.35	0.71	0.71	0.35	0.35
HSS76.2X38.1X3.2				0.7	0.35	0.71	0.71	0.47	0.47
HSS76.2X38.1X4.8				1.04	0.52	0.95	0.95	0.62	0.62
HSS76.2X38.1X6.4				1.4	0.7	1.12	1.12	0.72	0.72
HSS76.2X50.8X3.2				0.7	0.46	0.87	0.87	0.69	0.69
HSS76.2X50.8X4.8				1.04	0.7	1.18	1.18	0.93	0.93
HSS76.2X50.8X6.4				1.4	0.93	1.42	1.42	1.11	1.11
HSS76.2X50.8X7.9				1.75	1.16	1.59	1.59	1.24	1.24
HSS76.2X63.5X3.2				0.7	0.58	1.03	1.03	0.93	0.93
HSS76.2X63.5X4.8				1.04	0.87	1.41	1.41	1.27	1.27
HSS76.2X63.5X6.4				1.4	1.17	1.71	1.71	1.54	1.54
HSS76.2X63.5X7.9				1.75	1.45	1.95	1.95	1.74	1.74
HSS76.2X76.2X3.2				0.7	0.7	1.19	1.19	1.19	1.19
HSS76.2X76.2X4.8				1.04	1.04	1.63	1.63	1.63	1.63
HSS76.2X76.2X6.4				1.4	1.4	2.02	2.02	2.02	2.02
HSS76.2X76.2X7.9				1.75	1.75	2.31	2.31	2.31	2.31
HSS76.2X76.2X9.5				2.09	2.09	2.51	2.51	2.51	2.51
HSS88.9X38.1X3.2				0.81	0.35	0.9	0.9	0.55	0.55
HSS88.9X38.1X4.8				1.22	0.52	1.21	1.21	0.72	0.72
HSS88.9X38.1X6.4				1.63	0.7	1.46	1.46	0.85	0.85
HSS88.9X50.8X3.2				0.81	0.46	1.09	1.09	0.8	0.8
HSS88.9X50.8X4.8				1.22	0.7	1.5	1.5	1.08	1.08
HSS88.9X50.8X6.4				1.63	0.93	1.81	1.81	1.3	1.3
HSS88.9X63.5X3.2				0.81	0.58	1.28	1.28	1.06	1.06
HSS88.9X63.5X4.8				1.22	0.87	1.76	1.76	1.46	1.46
HSS88.9X63.5X6.4				1.63	1.17	2.17	2.17	1.78	1.78
HSS88.9X63.5X7.9				2.04	1.46	2.48	2.48	2.04	2.04
HSS88.9X63.5X9.5				2.44	1.74	2.72	2.72	2.21	2.21
HSS88.9X88.9X3.2				0.81	0.81	1.66	1.66	1.66	1.66
HSS88.9X88.9X4.8				1.22	1.22	2.32	2.32	2.32	2.32
HSS88.9X88.9X6.4				1.63	1.63	2.88	2.88	2.88	2.88

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
HSS88.9X88.9X7.9				2.04	2.04	3.34	3.34	3.34	3.34
HSS88.9X88.9X9.5				2.44	2.44	3.71	3.71	3.71	3.71
L102X102X11.1	8	2.09	-45	1.75	1.75	1.72	4.28	4.28	1.72
L102X102X12.7	8.94	2.35	-45	2.01	2.01	1.95	4.68	4.68	1.95
L102X102X15.9	10.67	2.86	-45	2.51	2.51	2.37	5.43	5.43	2.37
L102X102X19	12.22	3.36	-45	3.02	3.02	2.78	5.99	5.99	2.78
L102X102X6.4	4.91	1.28	-45	1	1	1.02	2.78	2.78	1.02
L102X102X7.9	6	1.56	-45	1.26	1.26	1.27	3.31	3.31	1.27
L102X102X9.5	7.03	1.83	-45	1.51	1.51	1.5	3.83	3.83	1.5
L102X76X12.7	6.21	1.34	-61.821	2.01	1.5	1.86	3.81	2.92	1.1
L102X76X15.9	7.38	1.63	-62.181	2.51	1.88	2.27	4.38	3.3	1.34
L102X76X6.4	3.45	0.74	-61.361	1	0.75	0.98	2.24	1.84	0.58
L102X76X7.9	4.2	0.9	-61.432	1.26	0.94	1.22	2.69	2.15	0.72
L102X76X9.5	4.91	1.05	-61.539	1.51	1.13	1.44	3.1	2.44	0.85
L102X89X12.7	7.35	1.85	-53.433	2.01	1.75	1.91	4.28	3.8	1.5
L102X89X6.4	4.06	1.01	-53.206	1	0.88	1	2.53	2.31	0.8
L102X89X7.9	4.96	1.23	-53.238	1.26	1.09	1.24	3.02	2.74	0.98
L102X89X9.5	5.8	1.44	-53.291	1.51	1.31	1.48	3.46	3.11	1.16
L127X127X11.1	15.96	4.15	-45	2.19	2.19	2.78	7.13	7.13	2.78
L127X127X12.7	17.91	4.67	-45	2.5	2.5	3.16	7.94	7.94	3.16
L127X127X15.9	21.58	5.68	-45	3.13	3.13	3.85	9.26	9.26	3.85
L127X127X19	24.95	6.66	-45	3.76	3.76	4.51	10.32	10.32	4.51
L127X127X22.2	27.93	7.59	-45	4.37	4.37	5.17	11.42	11.42	5.17
L127X127X7.9	11.85	3.1	-45	1.56	1.56	2.04	5.52	5.52	2.04
L127X127X9.5	13.96	3.64	-45	1.88	1.88	2.42	6.4	6.4	2.42
L127X76X11.1	9.39	1.42	-70.366	2.19	1.31	2.56	4.89	3.18	1
L127X76X12.7	10.51	1.59	-70.516	2.5	1.5	2.9	5.43	3.42	1.13
L127X76X6.4	5.76	0.88	-70.09	1.25	0.75	1.52	3.1	2.17	0.6
L127X76X7.9	7.03	1.06	-70.144	1.56	0.94	1.88	3.74	2.56	0.74
L127X76X9.5	8.24	1.24	-70.242	1.88	1.13	2.22	4.35	2.89	0.87
L127X89X12.7	11.79	2.3	-64.571	2.5	1.75	2.98	6.06	4.45	1.54
L127X89X15.9	14.14	2.79	-64.854	3.13	2.19	3.62	7.1	5.06	1.88
L127X89X19	16.27	3.27	-65.19	3.76	2.63	4.27	7.99	5.57	2.2
L127X89X6.4	6.43	1.27	-64.25	1.25	0.88	1.55	3.45	2.74	0.82
L127X89X7.9	7.86	1.54	-64.283	1.56	1.09	1.92	4.19	3.24	1.01
L127X89X9.5	9.23	1.8	-64.356	1.88	1.31	2.28	4.85	3.68	1.19
L152X102X11.1	17.88	3.27	-66.086	2.62	1.75	3.82	7.9	5.8	1.81
L152X102X12.7	20.08	3.67	-66.176	2.99	2.01	4.32	8.74	6.35	2.05
L152X102X14.3	22.21	4.06	-66.278	3.37	2.26	4.82	9.6	6.87	2.28
L152X102X15.9	24.25	4.45	-66.399	3.75	2.51	5.31	10.34	7.24	2.5
L152X102X19	28.1	5.2	-66.662	4.5	3.02	6.26	11.83	8.05	2.93
L152X102X22.2	31.55	5.93	-66.953	5.23	3.51	7.15	13.04	8.68	3.35
L152X102X7.9	13.27	2.44	-65.976	1.87	1.26	2.8	6	4.54	1.33
L152X102X9.5	15.63	2.86	-66.015	2.25	1.51	3.31	6.95	5.2	1.57
L152X152X11.1	27.96	7.23	-45	2.62	2.62	4.06	10.68	10.68	4.06
L152X152X12.7	31.49	8.14	-45	2.99	2.99	4.61	11.92	11.92	4.61
L152X152X14.3	34.89	9.04	-45	3.37	3.37	5.14	12.94	12.94	5.14
L152X152X15.9	38.18	9.93	-45	3.75	3.75	5.63	13.96	13.96	5.63
L152X152X19	44.41	11.65	-45	4.5	4.5	6.67	15.87	15.87	6.67
L152X152X22.2	50.04	13.29	-45	5.23	5.23	7.66	17.64	17.64	7.66
L152X152X25.4	55.44	14.95	-45	5.98	5.98	8.56	19.01	19.01	8.56
L152X152X7.9	20.64	5.37	-45	1.87	1.87	2.96	8.13	8.13	2.96
L152X152X9.5	24.39	6.31	-45	2.25	2.25	3.53	9.52	9.52	3.53
L152X89X12.7	18.24	2.62	-71.145	2.99	1.75	4.24	8.02	5.09	1.58
L152X89X7.9	12.08	1.75	-70.892	1.87	1.09	2.74	5.45	3.75	1.03

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
L152X89X9.5	14.21	2.05	-70.947	2.25	1.31	3.25	6.39	4.28	1.23
L178X102X11.1	26.21	3.63	-71.513	3.06	1.75	5.1	9.91	6.54	1.85
L178X102X12.7	29.5	4.08	-71.601	3.5	2.01	5.79	11.1	7.13	2.09
L178X102X15.9	35.77	4.94	-71.817	4.39	2.51	7.11	13.24	8.14	2.54
L178X102X19	41.63	5.78	-72.066	5.27	3.02	8.37	15.09	9.01	2.99
L178X102X9.5	22.87	3.18	-71.441	2.63	1.51	4.4	8.72	5.88	1.61
L203X102X11.1	36.53	3.9	-75.007	3.49	1.75	6.58	12.14	7.26	1.89
L203X102X12.7	41.18	4.38	-75.09	4	2.01	7.51	13.63	7.9	2.14
L203X102X14.3	45.7	4.85	-75.182	4.5	2.26	8.38	15.05	8.48	2.37
L203X102X15.9	50.09	5.32	-75.289	5	2.51	9.23	16.3	9.01	2.61
L203X102X19	58.48	6.22	-75.515	6.01	3.02	10.89	18.71	9.88	3.06
L203X102X22.2	66.15	7.08	-75.76	6.99	3.51	12.53	20.98	10.54	3.48
L203X102X25.4	73.61	7.95	-76.032	7.99	4.02	14.05	22.98	11.16	3.9
L203X152X11.1	48.17	10.24	-61.034	3.49	2.62	7.08	16.22	13.39	4.25
L203X152X12.7	54.38	11.54	-61.08	4	2.99	8.04	18.06	14.85	4.8
L203X152X14.3	60.42	12.82	-61.132	4.5	3.37	8.98	19.95	16.14	5.34
L203X152X15.9	66.3	14.08	-61.197	5	3.75	9.89	21.72	17.48	5.91
L203X152X19	77.56	16.54	-61.334	6.01	4.5	11.66	24.86	19.72	6.95
L203X152X22.2	87.87	18.85	-61.486	6.99	5.23	13.41	27.83	21.79	7.94
L203X152X25.4	97.92	21.19	-61.655	7.99	5.98	15.15	30.56	23.45	8.92
L203X203X12.7	77.43	20	-45	4	4	8.38	22.48	22.48	8.38
L203X203X14.3	86.15	22.23	-45	4.5	4.5	9.36	24.8	24.8	9.36
L203X203X15.9	94.65	24.42	-45	5	5	10.3	26.98	26.98	10.3
L203X203X19	111.01	28.73	-45	6.01	6.01	12.2	30.94	30.94	12.2
L203X203X22.2	126.06	32.79	-45	6.99	6.99	14.04	34.51	34.51	14.04
L203X203X25.4	140.81	36.89	-45	7.99	7.99	15.82	37.8	37.8	15.82
L203X203X28.6	154.78	40.93	-45	9	9	17.53	40.82	40.82	17.53
L51X51X3.2	0.3	0.08	-45	0.25	0.25	0.13	0.35	0.35	0.13
L51X51X4.8	0.43	0.11	-45	0.37	0.37	0.19	0.49	0.49	0.19
L51X51X6.4	0.55	0.15	-45	0.5	0.5	0.24	0.59	0.59	0.24
L51X51X7.9	0.66	0.18	-45	0.63	0.63	0.3	0.68	0.68	0.3
L51X51X9.5	0.75	0.21	-45	0.75	0.75	0.35	0.75	0.75	0.35
L64X38X4.8	0.52	0.08	-70.301	0.47	0.28	0.28	0.55	0.36	0.11
L64X38X6.4	0.66	0.1	-70.554	0.63	0.38	0.36	0.69	0.43	0.14
L64X51X4.8	0.65	0.15	-57.995	0.47	0.37	0.29	0.68	0.58	0.2
L64X51X6.4	0.84	0.19	-58.125	0.63	0.5	0.38	0.84	0.7	0.25
L64X51X7.9	1	0.24	-58.309	0.78	0.63	0.47	0.98	0.8	0.31
L64X51X9.5	1.15	0.28	-58.532	0.94	0.75	0.55	1.1	0.89	0.36
L64X64X12.7	1.92	0.53	-45	1.25	1.25	0.72	1.52	1.52	0.72
L64X64X4.8	0.87	0.23	-45	0.47	0.47	0.3	0.78	0.78	0.3
L64X64X6.4	1.12	0.29	-45	0.63	0.63	0.39	0.97	0.97	0.39
L64X64X7.9	1.35	0.35	-45	0.78	0.78	0.47	1.14	1.14	0.47
L64X64X9.5	1.56	0.42	-45	0.94	0.94	0.56	1.28	1.28	0.56
L76X51X12.7	2.19	0.41	-67.572	1.5	1	1	1.78	1.15	0.47
L76X51X4.8	0.98	0.18	-66.418	0.56	0.37	0.41	0.89	0.66	0.2
L76X51X6.4	1.26	0.23	-66.532	0.75	0.5	0.54	1.11	0.8	0.26
L76X51X7.9	1.52	0.28	-66.725	0.94	0.63	0.66	1.3	0.91	0.31
L76X51X9.5	1.76	0.32	-66.968	1.13	0.75	0.78	1.49	1.01	0.37
L76X64X11.1	2.45	0.61	-56.234	1.31	1.09	0.92	1.92	1.62	0.66
L76X64X12.7	2.71	0.68	-56.397	1.5	1.25	1.03	2.08	1.73	0.73
L76X64X4.8	1.21	0.3	-55.891	0.56	0.47	0.42	1.03	0.91	0.3
L76X64X6.4	1.55	0.38	-55.906	0.75	0.63	0.55	1.29	1.12	0.4
L76X64X7.9	1.88	0.46	-55.982	0.94	0.78	0.68	1.52	1.31	0.49
L76X64X9.5	2.18	0.53	-56.095	1.13	0.94	0.8	1.74	1.47	0.57
L76X76X11.1	3.15	0.84	-45	1.31	1.31	0.95	2.19	2.19	0.95

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
L76X76X12.7	3.5	0.95	-45	1.5	1.5	1.06	2.37	2.37	1.06
L76X76X4.8	1.54	0.41	-45	0.56	0.56	0.43	1.17	1.17	0.43
L76X76X6.4	1.98	0.52	-45	0.75	0.75	0.57	1.47	1.47	0.57
L76X76X7.9	2.4	0.63	-45	0.94	0.94	0.7	1.75	1.75	0.7
L76X76X9.5	2.79	0.74	-45	1.13	1.13	0.83	1.97	1.97	0.83
L89X64X12.7	3.83	0.79	-64.198	1.75	1.25	1.41	2.7	1.94	0.76
L89X64X6.4	2.16	0.44	-63.553	0.88	0.63	0.75	1.65	1.28	0.41
L89X64X7.9	2.62	0.53	-63.656	1.09	0.78	0.93	1.95	1.48	0.5
L89X64X9.5	3.05	0.62	-63.806	1.31	0.94	1.09	2.24	1.67	0.59
L89X76X11.1	4.16	1.04	-54.439	1.53	1.31	1.29	2.84	2.47	0.97
L89X76X12.7	4.63	1.17	-54.545	1.75	1.5	1.45	3.09	2.67	1.09
L89X76X6.4	2.6	0.64	-54.225	0.88	0.75	0.77	1.88	1.68	0.58
L89X76X7.9	3.16	0.78	-54.273	1.09	0.94	0.95	2.22	1.98	0.72
L89X76X9.5	3.68	0.91	-54.347	1.31	1.13	1.13	2.56	2.24	0.85
L89X89X11.1	5.17	1.36	-45	1.53	1.53	1.31	3.14	3.14	1.31
L89X89X12.7	5.76	1.53	-45	1.75	1.75	1.48	3.45	3.45	1.48
L89X89X6.4	3.21	0.84	-45	0.88	0.88	0.78	2.1	2.1	0.78
L89X89X7.9	3.9	1.02	-45	1.09	1.09	0.97	2.5	2.5	0.97
L89X89X9.5	4.56	1.19	-45	1.31	1.31	1.14	2.86	2.86	1.14
Rufter									
Truss									
W1000X222				24.06	16.35	513.35	513.35	38.77	38.77
W1000X249				25.06	20.31	601.52	601.52	48.01	48.01
W1000X272				25.34	23.64	676.13	676.13	56.14	56.14
W1000X296				25.14	28.18	769.78	769.78	87.96	87.96
W1000X314				29.61	27.98	787.21	787.21	65.91	65.91
W1000X321				25.34	32.11	855.93	855.93	101.66	101.66
W1000X350				33.03	31.28	874.87	874.87	74.76	74.76
W1000X371				29.61	37.4	995.91	995.91	117.18	117.18
W1000X393				38.58	34.25	965.61	965.61	82.85	82.85
W1000X412				33.03	41.54	1102.05	1102.05	131.79	131.79
W1000X415				41.42	36.24	1020.65	1020.65	86.83	86.83
W1000X438				42.95	38.61	1080.65	1080.65	93.64	93.64
W1000X443				36.95	43.41	1167.31	1167.31	138.18	138.18
W1000X483				40.16	48.01	1280.3	1280.3	153.47	153.47
W1000X486				48.36	42.91	1197	1197	105.75	105.75
W1000X494				49.97	43.33	1208.74	1208.74	105.51	105.51
W1000X539				45.34	53.6	1421.91	1421.91	172.55	172.55
W1000X554				47.1	55.05	1457.46	1457.46	176.36	176.36
W1000X584				59.31	52.08	1427.73	1427.73	129.41	129.41
W1000X591				49.97	59.06	1560.8	1560.8	191.28	191.28
W1000X642				55.34	63.6	1685.42	1685.42	208.76	208.76
W1000X748				64.85	75.52	1973.29	1973.29	248.49	248.49
W1000X883				76.87	89.82	2351.37	2351.37	302.24	302.24
W100X19.3				1.17	2.33	5.41	5.41	1.91	1.91
W1100X343				30.41	32.11	969.66	969.66	100.74	100.74
W1100X390				33.93	37.4	1109.52	1109.52	116.87	116.87
W1100X433				37.85	41.54	1231.47	1231.47	131.79	131.79
W1100X499				45.48	46.97	1405.73	1405.73	150.75	150.75
W130X23.8				1.2	3	8.56	8.56	3.01	3.01
W130X28.1				1.39	3.6	10.16	10.16	3.62	3.62
W150X13				0.99	1.28	5.11	5.11	1.01	1.01
W150X13.5				1	1.41	5.56	5.56	1.12	1.12
W150X18				1.38	1.87	7.34	7.34	1.48	1.48
W150X22.5				1.38	2.59	9.72	9.72	3.12	3.12

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
W150X24				1.64	2.71	10.22	10.22	2.2	2.2
W150X29.8				1.61	3.66	13.37	13.37	4.42	4.42
W150X37.1				2.04	4.61	16.73	16.73	5.64	5.64
W200X100				5.15	12.86	60.22	60.22	21.45	21.45
W200X15				1.34	1.35	7.81	7.81	1.06	1.06
W200X19.3				1.84	1.71	9.92	9.92	1.36	1.36
W200X22.5				1.99	2.11	11.85	11.85	1.7	1.7
W200X26.6				1.87	2.88	15.21	15.21	3.05	3.05
W200X31.3				2.07	3.53	18.19	18.19	3.71	3.71
W200X35.9				1.94	4.35	20.89	20.89	5.64	5.64
W200X41.7				2.3	5.06	24.29	24.29	6.64	6.64
W200X46.1				2.28	5.77	27.54	27.54	9.26	9.26
W200X52				2.51	6.64	31.34	31.34	10.59	10.59
W200X59				2.98	7.52	35.34	35.34	12.15	12.15
W200X71				3.41	9.26	43.28	43.28	14.99	14.99
W200X86				4.47	11.12	52.17	52.17	18.28	18.28
W250X101				4.87	13.01	75.82	75.82	26.5	26.5
W250X115				5.63	14.79	85.75	85.75	30.21	30.21
W250X131				6.54	16.99	98.89	98.89	34.7	34.7
W250X149				7.56	19.22	112.09	112.09	40.15	40.15
W250X167				8.63	21.69	125.41	125.41	45.4	45.4
W250X17.9				1.88	1.39	10.89	10.89	1.1	1.1
W250X22.3-1				2.3	1.81	13.79	13.79	1.44	1.44
W250X25.3-1				2.43	2.21	16.19	16.19	1.77	1.77
W250X28.4-1				2.55	2.64	18.9	18.9	2.14	2.14
W250X32.7				2.45	3.45	23.14	23.14	3.97	3.97
W250X38.5				2.68	4.25	27.9	27.9	4.87	4.87
W250X44.8				3.15	4.97	32.36	32.36	5.73	5.73
W250X49.1-1				2.82	5.74	35.18	35.18	9.18	9.18
W250X58-1				3.12	7.08	42.14	42.14	11.24	11.24
W250X67-1				3.54	8.27	48.91	48.91	13.28	13.28
W250X73				3.4	9.32	54.3	54.3	18.69	18.69
W250X80				3.74	10.24	59.84	59.84	20.61	20.61
W250X89-1				4.3	11.49	66.91	66.91	22.94	22.94
W310X107-1				5.27	13.39	97.01	97.01	32.49	32.49
W310X117-1				5.81	14.83	106.94	106.94	35.74	35.74
W310X129-1				6.46	16.34	118.21	118.21	39.75	39.75
W310X143-1				7.01	18.34	131.12	131.12	44.09	44.09
W310X158-1				7.88	20.1	144.37	144.37	49.21	49.21
W310X179-1				9.29	22.73	163.1	163.1	56.33	56.33
W310X202-1				10.59	25.88	185.23	185.23	64.32	64.32
W310X21-1				2.38	1.49	14.91	14.91	1.19	1.19
W310X226-1				11.92	29.25	208.67	208.67	72.54	72.54
W310X23.8-1				2.64	1.76	17.17	17.17	1.41	1.41
W310X253-1				13.46	32.74	235.52	235.52	82	82
W310X28.3-1				2.87	2.34	21.3	21.3	1.88	1.88
W310X283-1				15.26	36.88	262.44	262.44	92.57	92.57
W310X313-1				17.34	40.55	291.54	291.54	103.65	103.65
W310X32.7-1				3.19	2.85	25.39	25.39	2.32	2.32
W310X342				19.52	44.57	321.01	321.01	114.98	114.98
W310X375				21.58	48.76	352.72	352.72	127.6	127.6
W310X38.7-1				2.81	4.11	33.43	33.43	5.33	5.33
W310X415				24.36	53.94	389.71	389.71	142.94	142.94
W310X44.5-1				3.19	4.8	38.77	38.77	6.21	6.21
W310X454				26.57	59.54	436.31	436.31	159.21	159.21

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
W310X500				29.92	66.05	483.05	483.05	177.69	177.69
W310X52				3.76	5.69	45.67	45.67	7.45	7.45
W310X60				3.51	6.87	51.73	51.73	11.06	11.06
W310X67				4.05	7.69	57.64	57.64	12.44	12.44
W310X74				4.52	8.63	64.17	64.17	13.93	13.93
W310X79-1				4.17	9.58	70.37	70.37	19.17	19.17
W310X86-1				4.39	10.7	77.95	77.95	21.38	21.38
W310X97				4.72	12.13	88.26	88.26	28.97	28.97
W360X101-1				5.79	12.01	103.19	103.19	24.22	24.22
W360X1086				68.79	146.93	1276.24	1276.24	525.74	525.74
W360X110-1				6.38	13.21	111.91	111.91	26.5	26.5
W360X122-1				7.31	14.41	123.39	123.39	29.25	29.25
W360X134				6.18	17.11	142.62	142.62	50.08	50.08
W360X147				6.88	18.98	156.19	156.19	54.94	54.94
W360X162-1				7.48	20.89	173.49	173.49	61.19	61.19
W360X179-1				8.56	23.03	190.37	190.37	67.4	67.4
W360X196-1				9.48	25.25	208.43	208.43	74.6	74.6
W360X216-1				10.08	28.19	231.11	231.11	87.35	87.35
W360X237-1				11.16	30.89	253.38	253.38	95.85	95.85
W360X262-1				12.62	34.32	281.72	281.72	106.75	106.75
W360X287-1				13.8	37.73	309.46	309.46	118.68	118.68
W360X314-1				15.4	41.02	339.53	339.53	130.57	130.57
W360X32.9-1				3.15	2.79	29.04	29.04	2.8	2.8
W360X347-1				17.12	45.61	375.76	375.76	144.7	144.7
W360X382-1				19.39	50.34	415.61	415.61	161.43	161.43
W360X39-1				3.55	3.54	35.27	35.27	3.54	3.54
W360X421-1				21.56	55.58	460.56	460.56	178.74	178.74
W360X44-1				3.73	4.32	42.07	42.07	5.82	5.82
W360X463-1				24.08	60.94	506.19	506.19	198.96	198.96
W360X509-1				26.97	67.54	559.5	559.5	220.39	220.39
W360X51-1				4	5.12	48.68	48.68	6.92	6.92
W360X551-1				29.76	73.17	606.21	606.21	241.18	241.18
W360X57.8-1				4.37	5.82	54.55	54.55	7.88	7.88
W360X592-1				32.43	78.93	656.17	656.17	261.16	261.16
W360X634-1				35.19	84.56	706.59	706.59	282.67	282.67
W360X64				4.18	7.08	62.43	62.43	11.3	11.3
W360X677-1				38.41	89.9	755.53	755.53	305.83	305.83
W360X72				4.7	7.96	69.89	69.89	12.8	12.8
W360X744-1				42.92	99.21	838.16	838.16	339.02	339.02
W360X79				5.14	8.9	77.79	77.79	14.29	14.29
W360X818				48.11	109.51	934.98	934.98	377.03	377.03
W360X900				54.32	121.03	1034.3	1034.3	422.47	422.47
W360X91-1				5.21	10.76	91.97	91.97	21.38	21.38
W360X990				61.18	133.69	1147.11	1147.11	471.92	471.92
W410X100				6.42	11.31	117.04	117.04	23.33	23.33
W410X114				7.53	13.06	134.57	134.57	26.74	26.74
W410X132				8.8	15.14	154.63	154.63	31.34	31.34
W410X149				9.98	17.05	175.16	175.16	35.78	35.78
W410X38.8-1				3.93	3.17	38.24	38.24	3.48	3.48
W410X46.1-1				4.38	4.05	47.13	47.13	4.5	4.5
W410X53-1				4.69	5.01	56.19	56.19	6.99	6.99
W410X60				4.88	5.89	64.93	64.93	8.23	8.23
W410X67				5.55	6.66	72.81	72.81	9.34	9.34
W410X75				6.19	7.44	80.78	80.78	10.51	10.51
W410X85				7.05	8.51	92.49	92.49	12.07	12.07

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
W460X106				9.18	10.32	126.46	126.46	15.79	15.79
W460X113				7.73	12.47	146.35	146.35	27.69	27.69
W460X128				8.83	14.28	166.48	166.48	31.51	31.51
W460X144				9.95	16.1	188.24	188.24	36.22	36.22
W460X158				11.04	17.53	204.27	204.27	39.36	39.36
W460X177				12.43	19.94	230.45	230.45	44.65	44.65
W460X193				12.91	22.38	254.06	254.06	49.85	49.85
W460X213				14.19	24.58	281.08	281.08	55.44	55.44
W460X235				15.97	27.14	310	310	61.24	61.24
W460X260				17.8	30.27	345.96	345.96	68.6	68.6
W460X286				19.59	33.57	379.34	379.34	76.49	76.49
W460X315				21.93	36.96	417.65	417.65	84.81	84.81
W460X349				24.51	41.12	464.51	464.51	95.34	95.34
W460X384				27.5	45.26	511.88	511.88	106.18	106.18
W460X421				30.68	49.54	564.14	564.14	118.41	118.41
W460X464				33.86	54.84	625.33	625.33	132.45	132.45
W460X52-1				5.31	4.24	57.5	57.5	5.11	5.11
W460X60-1				5.64	5.26	68.4	68.4	6.34	6.34
W460X68-1				6.52	6.13	78.53	78.53	7.43	7.43
W460X74-1				6.39	7.15	88.93	88.93	10.67	10.67
W460X82-1				7.07	7.89	98.17	98.17	11.95	11.95
W460X89-1				7.52	8.78	108.31	108.31	13.29	13.29
W460X97				8.25	9.52	116.3	116.3	14.42	14.42
W530X101-1				9.06	9.44	140.26	140.26	15.63	15.63
W530X109				9.67	10.25	151.08	151.08	17.01	17.01
W530X123				11.05	11.61	170.96	170.96	19.52	19.52
W530X138				12.51	13.05	191.63	191.63	22.07	22.07
W530X150				10.71	16.36	226.6	226.6	40.29	40.29
W530X165				11.85	17.89	248.12	248.12	44.59	44.59
W530X182				12.98	19.86	272.45	272.45	49.21	49.21
W530X196				14.17	21.48	295.21	295.21	53.86	53.86
W530X219				15.91	23.99	328.51	328.51	60.26	60.26
W530X248				16.93	28.07	379.8	379.8	70.13	70.13
W530X272				18.87	30.89	416.7	416.7	77.14	77.14
W530X300				20.91	34.22	461.86	461.86	86.2	86.2
W530X66-1				7.25	4.86	81.44	81.44	6.38	6.38
W530X72				7.21	5.83	93.11	93.11	9.49	9.49
W530X74-1				7.9	5.83	94.77	94.77	7.65	7.65
W530X82-1				7.8	7.18	109.8	109.8	11.74	11.74
W530X85-1				8.56	7.12	110.89	110.89	9.28	9.28
W530X92-1				8.43	8.42	126.86	126.86	13.96	13.96
W610X101-1				9.8	8.78	154.49	154.49	15.68	15.68
W610X113-1				10.54	10.19	175.73	175.73	18.36	18.36
W610X125-1				11.29	11.6	196.63	196.63	20.95	20.95
W610X140-1				12.53	13.19	221.54	221.54	24.09	24.09
W610X153				13.5	14.73	245.27	245.27	26.38	26.38
W610X155				12.05	16.04	257.26	257.26	40.56	40.56
W610X174				13.39	18.14	290.78	290.78	46.57	46.57
W610X195				14.85	20.67	327.68	327.68	52.84	52.84
W610X217				16.04	23.47	371.79	371.79	60.65	60.65
W610X241				17.62	26.43	413.23	413.23	68.05	68.05
W610X262				18.95	28.81	450.05	450.05	74.05	74.05
W610X285				20.69	31.63	491.58	491.58	81.73	81.73
W610X307				22.37	34.01	530.8	530.8	89.13	89.13
W610X341				24.96	37.77	588.05	588.05	99.32	99.32

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
W610X372				27.33	41.54	644.95	644.95	109.66	109.66
W610X415				31	46.37	720.04	720.04	123.85	123.85
W610X455				34.12	50.86	789.41	789.41	137.48	137.48
W610X498				38.03	55.82	864.28	864.28	152.65	152.65
W610X551				42.54	62.12	957.84	957.84	169.39	169.39
W610X82-1				9.28	5.89	114.51	114.51	8.3	8.3
W610X92-1				10.17	6.94	130.77	130.77	9.82	9.82
W690X125-1				12.3	10.7	214.21	214.21	21.19	21.19
W690X140-1				13.13	12.4	243.02	243.02	24.79	24.79
W690X152-1				13.97	13.85	267.87	267.87	27.82	27.82
W690X170-1				15.58	15.67	299.39	299.39	31.44	31.44
W690X192-1				16.84	18.31	344.73	344.73	36.81	36.81
W690X217				16.61	22.81	413.84	413.84	63.08	63.08
W690X240				18.25	25.2	457.9	457.9	70.97	70.97
W690X265				20.14	27.93	504.79	504.79	78.75	78.75
W690X289				21.14	31.27	558.96	558.96	88.45	88.45
W690X323				23.58	35.24	628.01	628.01	99.89	99.89
W690X350				26.1	38.14	676.37	676.37	108.19	108.19
W690X384				28.44	42.2	745.2	745.2	120.37	120.37
W690X419				31.02	46.33	812.01	812.01	132.38	132.38
W690X457				34.39	50.21	884.52	884.52	145.72	145.72
W690X500				37.8	55.49	973.82	973.82	161.52	161.52
W690X548				42	60.71	1065.54	1065.54	178.33	178.33
W690X802				64.02	90.34	1581	1581	275.47	275.47
W760X134-1				13.82	10.57	244.42	244.42	22.14	22.14
W760X147-1				15.43	11.73	268.7	268.7	24.36	24.36
W760X161-1				16.19	13.31	299.88	299.88	27.79	27.79
W760X173-1				17.01	14.9	328.34	328.34	31.22	31.22
W760X185-1				17.71	16.28	354.84	354.84	34.42	34.42
W760X196-1				18.62	17.52	380.41	380.41	37.3	37.3
W760X220				19.95	20.69	434.99	434.99	43.2	43.2
W760X257				19.86	26.77	542.26	542.26	79.76	79.76
W760X284				21.76	29.72	599.28	599.28	89.69	89.69
W760X314				23.97	33.23	666.99	666.99	100.12	100.12
W760X350				26	37.8	747.64	747.64	113.15	113.15
W760X389				29.37	41.78	828.34	828.34	126.16	126.16
W760X434				32.64	47.23	930.74	930.74	143.7	143.7
W760X484				36.99	52.63	1036.59	1036.59	161.07	161.07
W760X531				40.67	57.91	1139.89	1139.89	179.35	179.35
W760X582				45.08	63.43	1247.98	1247.98	198.79	198.79
W840X176-1				18.14	14.18	359.13	359.13	32.52	32.52
W840X193-1				19.16	16.37	404.89	404.89	37.91	37.91
W840X210-1				20.19	18.41	447.22	447.22	42.63	42.63
W840X226-1				21.24	20.5	487.62	487.62	47.16	47.16
W840X251				22.63	23.38	549.85	549.85	53.92	53.92
W840X299				24.15	30.1	688.66	688.66	95.44	95.44
W840X329				26.29	33.67	761.2	761.2	106.53	106.53
W840X359				28.42	37.15	830.04	830.04	117.21	117.21
W840X392				30.01	41.33	922.32	922.32	131.79	131.79
W840X433				33.43	45.82	1017.52	1017.52	145.91	145.91
W840X473				36.58	50.34	1108.53	1108.53	161.43	161.43
W840X527				41.34	56.1	1236.68	1236.68	181.43	181.43
W840X576				45.33	61.48	1348.67	1348.67	200.15	200.15
W920X201				21.3	15.84	438.78	438.78	37.49	37.49
W920X223-1				22.48	18.83	503.18	503.18	44.82	44.82

Name	IMajor in4	IMinor in4	MajorAngle deg	As2 in2	As3 in2	S33Pos in3	S33Neg in3	S22Pos in3	S22Neg in3
W920X238-1				23.38	20.41	542.14	542.14	49.22	49.22
W920X253-1				24.64	21.98	580.36	580.36	53.22	53.22
W920X271-1				26.3	23.79	622.15	622.15	57.25	57.25
W920X289				27.87	25.38	663.56	663.56	62.02	62.02
W920X313				30.48	27.63	718.93	718.93	67.32	67.32
W920X344				27.73	34.64	854.46	854.46	113.89	113.89
W920X345				32.3	31.64	807.61	807.61	77.52	77.52
W920X368				29.33	37.13	910.12	910.12	122.34	122.34
W920X381				35.97	35.16	897.07	897.07	86.61	86.61
W920X390				30.94	39.9	970.39	970.39	131.3	131.3
W920X420				32.85	43.5	1057.23	1057.23	144.32	144.32
W920X449				35.23	46.77	1131.55	1131.55	155.73	155.73
W920X491				38.46	51.24	1235.76	1235.76	170.92	170.92
W920X537				42.48	55.97	1353.27	1353.27	187.96	187.96
W920X588				46.85	61.66	1489.61	1489.61	208.08	208.08
W920X656				52.83	69.19	1655.3	1655.3	233.92	233.92
W920X725				58.94	76.35	1834.38	1834.38	263.5	263.5
W920X787				64.03	83.43	1993.85	1993.85	290.46	290.46
W920X970				80.6	103.81	2476.16	2476.16	365.87	365.87

Table 2.2 - Frame Section Property Definitions - Summary (Part 3 of 4)

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
Column									
HSS101.6X101.6X12.7	7.69	7.69	1.4062	1.4062			0	0	0
HSS101.6X101.6X3.2	2.56	2.56	1.5774	1.5774			0	0	0
HSS101.6X101.6X4.8	3.67	3.67	1.5521	1.5521			0	0	0
HSS101.6X101.6X6.4	4.69	4.69	1.5236	1.5236			0	0	0
HSS101.6X101.6X7.9	5.59	5.59	1.4909	1.4909			0	0	0
HSS101.6X101.6X9.5	6.41	6.41	1.4693	1.4693			0	0	0
HSS101.6X50.8X3.2	1.66	1.02	1.4255	0.8312			0	0	0
HSS101.6X50.8X4.8	2.34	1.43	1.3897	0.8034			0	0	0
HSS101.6X50.8X6.4	2.94	1.79	1.3587	0.7798			0	0	0
HSS101.6X50.8X7.9	3.43	2.08	1.3213	0.753			0	0	0
HSS101.6X50.8X9.5	3.84	2.31	1.2842	0.7281			0	0	0
HSS101.6X63.5X3.2	1.88	1.36	1.4775	1.0243			0	0	0
HSS101.6X63.5X4.8	2.67	1.93	1.4443	0.9994			0	0	0
HSS101.6X63.5X6.4	3.38	2.43	1.4112	0.9727			0	0	0
HSS101.6X63.5X7.9	3.97	2.85	1.3785	0.9456			0	0	0
HSS101.6X63.5X9.5	4.48	3.2	1.3467	0.9214			0	0	0
HSS101.6X76.2X3.2	2.11	1.73	1.514	1.2139			0	0	0
HSS101.6X76.2X4.8	3	2.46	1.4803	1.1879			0	0	0
HSS101.6X76.2X6.4	3.81	3.12	1.4528	1.1593			0	0	0
HSS101.6X76.2X7.9	4.51	3.69	1.4241	1.133			0	0	0
HSS101.6X76.2X9.5	5.12	4.18	1.3919	1.1077			0	0	0
HSS114.3X114.3X12.7	10.19	10.19	1.6141	1.6141			0	0	0
HSS114.3X114.3X3.2	3.27	3.27	1.781	1.781			0	0	0
HSS114.3X114.3X4.8	4.71	4.71	1.7537	1.7537			0	0	0
HSS114.3X114.3X6.4	6.06	6.06	1.723	1.723			0	0	0
HSS114.3X114.3X7.9	7.26	7.26	1.6984	1.6984			0	0	0
HSS114.3X114.3X9.5	8.36	8.36	1.6701	1.6701			0	0	0
HSS127X101.6X12.7	10.92	9.34	1.7469	1.4646			0	0	0
HSS127X101.6X3.2	3.5	3.01	1.9269	1.6222			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS127X101.6X4.8	5.05	4.34	1.9018	1.597			0	0	0
HSS127X101.6X6.4	6.47	5.57	1.8675	1.5692			0	0	0
HSS127X101.6X7.9	7.81	6.65	1.8377	1.5399			0	0	0
HSS127X101.6X9.5	8.97	7.69	1.8061	1.5147			0	0	0
HSS127X127X12.7	13.12	13.12	1.8153	1.8153			0	0	0
HSS127X127X3.2	4.07	4.07	1.9848	1.9848			0	0	0
HSS127X127X4.8	5.89	5.89	1.9573	1.9573			0	0	0
HSS127X127X6.4	7.63	7.63	1.9305	1.9305			0	0	0
HSS127X127X7.9	9.15	9.15	1.9018	1.9018			0	0	0
HSS127X127X9.5	10.62	10.62	1.8729	1.8729			0	0	0
HSS127X50.8X3.2	2.37	1.24	1.7393	0.8451			0	0	0
HSS127X50.8X4.8	3.37	1.75	1.702	0.82			0	0	0
HSS127X50.8X6.4	4.27	2.2	1.6644	0.7947			0	0	0
HSS127X50.8X7.9	5.05	2.57	1.6298	0.7725			0	0	0
HSS127X50.8X9.5	5.71	2.88	1.5944	0.7464			0	0	0
HSS127X63.5X3.2	2.65	1.64	1.8017	1.0528			0	0	0
HSS127X63.5X4.8	3.79	2.33	1.7692	1.0247			0	0	0
HSS127X63.5X6.4	4.83	2.95	1.7279	0.9963			0	0	0
HSS127X76.2X12.7	8.85	6.1	1.6518	1.0929			0	0	0
HSS127X76.2X3.2	2.93	2.07	1.8474	1.245			0	0	0
HSS127X76.2X4.8	4.21	2.96	1.8206	1.2223			0	0	0
HSS127X76.2X6.4	5.38	3.77	1.7829	1.1952			0	0	0
HSS127X76.2X7.9	6.41	4.48	1.7507	1.1674			0	0	0
HSS127X76.2X9.5	7.32	5.1	1.7187	1.1439			0	0	0
HSS139.7X139.7X3.2	4.95	4.95	2.1878	2.1878			0	0	0
HSS139.7X139.7X4.8	7.2	7.2	2.1656	2.1656			0	0	0
HSS139.7X139.7X6.4	9.34	9.34	2.1317	2.1317			0	0	0
HSS139.7X139.7X7.9	11.29	11.29	2.1072	2.1072			0	0	0
HSS139.7X139.7X9.5	13.12	13.12	2.0806	2.0806			0	0	0
HSS152.4X101.6X12.7	14.58	10.98	2.0815	1.5036			0	0	0
HSS152.4X101.6X3.2	4.56	3.46	2.2612	1.66			0	0	0
HSS152.4X101.6X4.8	6.59	5	2.2346	1.6336			0	0	0
HSS152.4X101.6X6.4	8.54	6.47	2.2064	1.6079			0	0	0
HSS152.4X101.6X7.9	10.31	7.75	2.1701	1.5844			0	0	0
HSS152.4X101.6X9.5	11.9	8.97	2.141	1.5519			0	0	0
HSS152.4X127X12.7	17.21	15.19	2.1602	1.8689			0	0	0
HSS152.4X127X3.2	5.24	4.63	2.3323	2.0355			0	0	0
HSS152.4X127X4.8	7.63	6.71	2.3063	2.0068			0	0	0
HSS152.4X127X6.4	9.89	8.73	2.2767	1.9787			0	0	0
HSS152.4X127X7.9	11.9	10.5	2.2488	1.9533			0	0	0
HSS152.4X127X9.5	13.79	12.2	2.2186	1.9237			0	0	0
HSS152.4X152.4X12.7	19.77	19.77	2.2273	2.2273			0	0	0
HSS152.4X152.4X15.9	23.19	23.19	2.173	2.173			0	0	0
HSS152.4X152.4X3.2	5.92	5.92	2.397	2.397			0	0	0
HSS152.4X152.4X4.8	8.6	8.6	2.3658	2.3658			0	0	0
HSS152.4X152.4X6.4	11.23	11.23	2.336	2.336			0	0	0
HSS152.4X152.4X7.9	13.61	13.61	2.3111	2.3111			0	0	0
HSS152.4X152.4X9.5	15.81	15.81	2.28	2.28			0	0	0
HSS152.4X50.8X3.2	3.19	1.46	2.0497	0.8608			0	0	0
HSS152.4X50.8X4.8	4.58	2.07	2.02	0.8363			0	0	0
HSS152.4X50.8X6.4	5.84	2.61	1.973	0.8106			0	0	0
HSS152.4X50.8X7.9	6.96	3.07	1.9302	0.7837			0	0	0
HSS152.4X50.8X9.5	7.93	3.46	1.8929	0.7607			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS152.4X76.2X12.7	12.08	7.26	1.9685	1.1191			0	0	0
HSS152.4X76.2X3.2	3.87	2.4	2.173	1.2689			0	0	0
HSS152.4X76.2X4.8	5.59	3.45	2.1392	1.245			0	0	0
HSS152.4X76.2X6.4	7.2	4.41	2.1036	1.2171			0	0	0
HSS152.4X76.2X7.9	8.6	5.27	2.0726	1.1945			0	0	0
HSS152.4X76.2X9.5	9.89	6.03	2.0341	1.1669			0	0	0
HSS177.8X101.6X12.7	18.8	12.57	2.3996	1.5337			0	0	0
HSS177.8X101.6X3.2	5.73	3.91	2.5954	1.6901			0	0	0
HSS177.8X101.6X4.8	8.36	5.67	2.5621	1.66			0	0	0
HSS177.8X101.6X6.4	10.8	7.32	2.5281	1.6378			0	0	0
HSS177.8X101.6X7.9	13.12	8.85	2.4999	1.6132			0	0	0
HSS177.8X101.6X9.5	15.07	10.19	2.4646	1.5854			0	0	0
HSS177.8X127X12.7	21.91	17.27	2.4939	1.9112			0	0	0
HSS177.8X127X3.2	6.53	5.2	2.6745	2.0743			0	0	0
HSS177.8X127X4.8	9.52	7.57	2.645	2.0415			0	0	0
HSS177.8X127X6.4	12.39	9.82	2.614	2.0168			0	0	0
HSS177.8X127X7.9	15.01	11.9	2.5856	1.9897			0	0	0
HSS177.8X127X9.5	17.51	13.79	2.5553	1.9665			0	0	0
HSS177.8X177.8X12.7	27.89	27.89	2.6347	2.6347			0	0	0
HSS177.8X177.8X15.9	33.07	33.07	2.584	2.584			0	0	0
HSS177.8X177.8X3.2	8.12	8.12	2.7975	2.7975			0	0	0
HSS177.8X177.8X4.8	11.9	11.9	2.7793	2.7793			0	0	0
HSS177.8X177.8X6.4	15.5	15.5	2.7487	2.7487			0	0	0
HSS177.8X177.8X7.9	18.92	18.92	2.7207	2.7207			0	0	0
HSS177.8X177.8X9.5	22.09	22.09	2.6935	2.6935			0	0	0
HSS177.8X50.8X3.2	4.13	1.68	2.3561	0.8721			0	0	0
HSS177.8X50.8X4.8	5.95	2.39	2.3142	0.8466			0	0	0
HSS177.8X50.8X6.4	7.63	3.02	2.2694	0.8178			0	0	0
HSS177.8X76.2X12.7	15.81	8.48	2.2708	1.1388			0	0	0
HSS177.8X76.2X3.2	4.93	2.73	2.4857	1.2875			0	0	0
HSS177.8X76.2X4.8	7.14	3.94	2.4545	1.2625			0	0	0
HSS177.8X76.2X6.4	9.21	5.06	2.4239	1.2405			0	0	0
HSS177.8X76.2X7.9	11.11	6.05	2.3811	1.2134			0	0	0
HSS177.8X76.2X9.5	12.81	6.96	2.3487	1.1875			0	0	0
HSS203.2X101.6X12.7	23.49	14.28	2.7166	1.5568			0	0	0
HSS203.2X101.6X15.9	27.4	16.6	2.6459	1.5096			0	0	0
HSS203.2X101.6X3.2	7.02	4.36	2.9137	1.7119			0	0	0
HSS203.2X101.6X4.8	10.19	6.35	2.885	1.6836			0	0	0
HSS203.2X101.6X6.4	13.3	8.18	2.849	1.6574			0	0	0
HSS203.2X101.6X7.9	16.11	9.89	2.8139	1.6353			0	0	0
HSS203.2X101.6X9.5	18.8	11.47	2.781	1.6083			0	0	0
HSS203.2X152.4X12.7	30.51	24.9	2.9112	2.3211			0	0	0
HSS203.2X152.4X15.9	36.13	29.47	2.8554	2.273			0	0	0
HSS203.2X152.4X4.8	13	10.68	3.0614	2.4546			0	0	0
HSS203.2X152.4X6.4	16.9	13.91	3.0317	2.433			0	0	0
HSS203.2X152.4X7.9	20.63	16.9	2.9973	2.3994			0	0	0
HSS203.2X152.4X9.5	24.1	19.77	2.9677	2.3767			0	0	0
HSS203.2X203.2X12.7	37.53	37.53	3.042	3.042			0	0	0
HSS203.2X203.2X15.9	44.73	44.73	2.9817	2.9817			0	0	0
HSS203.2X203.2X3.2	10.68	10.68	3.2146	3.2146			0	0	0
HSS203.2X203.2X4.8	15.68	15.68	3.1819	3.1819			0	0	0
HSS203.2X203.2X6.4	20.5	20.5	3.1543	3.1543			0	0	0
HSS203.2X203.2X7.9	25.08	25.08	3.1251	3.1251			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS203.2X203.2X9.5	29.41	29.41	3.0999	3.0999			0	0	0
HSS203.2X50.8X3.2	5.19	1.9	2.6512	0.8779			0	0	0
HSS203.2X50.8X4.8	7.51	2.7	2.6104	0.8529			0	0	0
HSS203.2X50.8X6.4	9.7	3.43	2.5805	0.8262			0	0	0
HSS203.2X50.8X7.9	11.59	4.06	2.5301	0.8029			0	0	0
HSS203.2X50.8X9.5	13.43	4.61	2.4853	0.776			0	0	0
HSS203.2X76.2X12.7	20.02	9.64	2.5804	1.1528			0	0	0
HSS203.2X76.2X3.2	6.1	3.07	2.7979	1.3061			0	0	0
HSS203.2X76.2X4.8	8.85	4.43	2.772	1.2791			0	0	0
HSS203.2X76.2X6.4	11.47	5.7	2.7291	1.253			0	0	0
HSS203.2X76.2X7.9	13.91	6.83	2.69	1.2284			0	0	0
HSS203.2X76.2X9.5	16.11	7.87	2.6555	1.2051			0	0	0
HSS228.6X127X12.7	32.53	21.48	3.1505	1.9738			0	0	0
HSS228.6X127X15.9	38.51	25.32	3.0837	1.9255			0	0	0
HSS228.6X127X4.8	13.79	9.28	3.3119	2.1069			0	0	0
HSS228.6X127X6.4	18.12	12.02	3.2726	2.0792			0	0	0
HSS228.6X127X7.9	22.03	14.58	3.2407	2.0511			0	0	0
HSS228.6X127X9.5	25.69	17.09	3.2104	2.0238			0	0	0
HSS228.6X177.8X12.7	40.52	33.99	3.3216	2.7208			0	0	0
HSS228.6X177.8X15.9	48.27	40.52	3.2537	2.6686			0	0	0
HSS228.6X177.8X4.8	16.9	14.28	3.4714	2.871			0	0	0
HSS228.6X177.8X6.4	22.21	18.67	3.4417	2.8381			0	0	0
HSS228.6X177.8X7.9	27.09	22.82	3.4146	2.8109			0	0	0
HSS228.6X177.8X9.5	31.79	26.73	3.3815	2.7818			0	0	0
HSS228.6X228.6X12.7	48.39	48.39	3.4593	3.4593			0	0	0
HSS228.6X228.6X15.9	58.09	58.09	3.3935	3.3935			0	0	0
HSS228.6X228.6X3.2	13.61	13.61	3.6184	3.6184			0	0	0
HSS228.6X228.6X6.4	26.18	26.18	3.5661	3.5661			0	0	0
HSS228.6X228.6X7.9	32.1	32.1	3.5351	3.5351			0	0	0
HSS228.6X228.6X74.8	20.02	20.02	3.5894	3.5894			0	0	0
HSS228.6X228.6X9.5	37.77	37.77	3.5075	3.5075			0	0	0
HSS228.6X76.2X12.7	24.59	10.8	2.8798	1.1641			0	0	0
HSS228.6X76.2X4.8	10.8	4.92	3.0772	1.2902			0	0	0
HSS228.6X76.2X6.4	13.97	6.35	3.036	1.2651			0	0	0
HSS228.6X76.2X7.9	16.9	7.63	2.994	1.239			0	0	0
HSS228.6X76.2X9.5	19.71	8.79	2.9578	1.2154			0	0	0
HSS254X101.6X12.7	34.11	17.57	3.3358	1.5965			0	0	0
HSS254X101.6X15.9	40.28	20.63	3.2623	1.5446			0	0	0
HSS254X101.6X3.2	10.01	5.26	3.5514	1.7477			0	0	0
HSS254X101.6X4.8	14.58	7.69	3.5228	1.7267			0	0	0
HSS254X101.6X6.4	18.98	10.01	3.4802	1.6942			0	0	0
HSS254X101.6X7.9	23.13	12.08	3.4442	1.6703			0	0	0
HSS254X101.6X9.5	26.97	13.97	3.4046	1.6443			0	0	0
HSS254X127X4.8	16.29	10.13	3.6337	2.1262			0	0	0
HSS254X127X6.4	21.3	13.18	3.5957	2.102			0	0	0
HSS254X127X7.9	26	15.99	3.5687	2.0793			0	0	0
HSS254X127X9.5	30.39	18.67	3.5207	2.0489			0	0	0
HSS254X152.4X12.7	43.02	30.08	3.5596	2.3863			0	0	0
HSS254X152.4X15.9	51.32	35.82	3.4985	2.3323			0	0	0
HSS254X152.4X4.8	18	12.69	3.7326	2.5222			0	0	0
HSS254X152.4X6.4	23.62	16.6	3.6931	2.4954			0	0	0
HSS254X152.4X7.9	28.8	20.2	3.6701	2.4678			0	0	0
HSS254X152.4X9.5	33.81	23.68	3.6286	2.4365			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS254X203.2X12.7	51.87	44.49	3.7406	3.1429			0	0	0
HSS254X203.2X15.9	62.24	53.27	3.6675	3.0809			0	0	0
HSS254X203.2X4.8	21.42	18.43	3.8812	3.2776			0	0	0
HSS254X203.2X6.4	28.07	24.23	3.8486	3.25			0	0	0
HSS254X203.2X7.9	34.42	29.6	3.8247	3.2233			0	0	0
HSS254X203.2X9.5	40.52	34.78	3.784	3.188			0	0	0
HSS254X254X12.7	60.72	60.72	3.8654	3.8654			0	0	0
HSS254X254X15.9	73.23	73.23	3.8186	3.8186			0	0	0
HSS254X254X4.8	24.78	24.78	3.9997	3.9997			0	0	0
HSS254X254X6.4	32.71	32.71	3.9675	3.9675			0	0	0
HSS254X254X7.9	40.09	40.09	3.937	3.937			0	0	0
HSS254X254X9.5	47.17	47.17	3.9115	3.9115			0	0	0
HSS254X50.8X3.2	7.63	2.33	3.2559	0.8909			0	0	0
HSS254X50.8X4.8	11.11	3.34	3.2114	0.8648			0	0	0
HSS254X50.8X6.4	14.4	4.26	3.1691	0.8376			0	0	0
HSS254X50.8X7.9	17.51	5.06	3.1222	0.8108			0	0	0
HSS254X50.8X9.5	20.32	5.76	3.0734	0.7882			0	0	0
HSS254X76.2X3.2	8.79	3.74	3.4126	1.3279			0	0	0
HSS254X76.2X4.8	12.81	5.41	3.383	1.3017			0	0	0
HSS254X76.2X6.4	16.72	7.02	3.3409	1.2751			0	0	0
HSS254X76.2X7.9	20.32	8.42	3.3023	1.2532			0	0	0
HSS254X76.2X9.5	23.68	9.7	3.2594	1.2238			0	0	0
HSS254X88.9X12.7	31.92	14.71	3.2602	1.3888			0	0	0
HSS254X88.9X3.2	9.4	4.48	3.4898	1.5428			0	0	0
HSS254X88.9X4.8	13.73	6.53	3.4524	1.5142			0	0	0
HSS254X88.9X6.4	17.88	8.42	3.414	1.4851			0	0	0
HSS254X88.9X7.9	21.72	10.19	3.3744	1.4614			0	0	0
HSS254X88.9X9.5	25.32	11.78	3.3393	1.4373			0	0	0
HSS304.8X101.6X12.7	46.68	20.87	3.9438	1.6174			0	0	0
HSS304.8X101.6X15.9	55.47	24.47	3.862	1.5674			0	0	0
HSS304.8X101.6X4.8	19.59	8.97	4.1368	1.7492			0	0	0
HSS304.8X101.6X6.4	25.63	11.72	4.0929	1.7198			0	0	0
HSS304.8X101.6X7.9	31.31	14.22	4.0537	1.6972			0	0	0
HSS304.8X101.6X9.5	36.68	16.6	4.0183	1.6649			0	0	0
HSS304.8X152.4X12.7	57.42	35.21	4.2126	2.4396			0	0	0
HSS304.8X152.4X15.9	68.96	42.11	4.1431	2.3876			0	0	0
HSS304.8X152.4X4.8	23.68	14.71	4.3757	2.5653			0	0	0
HSS304.8X152.4X6.4	31.12	19.28	4.3384	2.5423			0	0	0
HSS304.8X152.4X7.9	38.08	23.62	4.3072	2.5142			0	0	0
HSS304.8X152.4X9.5	44.79	27.7	4.2696	2.4843			0	0	0
HSS304.8X203.2X12.7	68.35	51.5	4.4057	3.2167			0	0	0
HSS304.8X203.2X15.9	82.38	61.63	4.3525	3.1678			0	0	0
HSS304.8X203.2X4.8	27.83	21.11	4.5526	3.3464			0	0	0
HSS304.8X203.2X6.4	36.61	27.83	4.5323	3.3199			0	0	0
HSS304.8X203.2X7.9	44.91	34.11	4.4918	3.2867			0	0	0
HSS304.8X203.2X9.5	53.03	40.09	4.4531	3.2567			0	0	0
HSS304.8X254X12.7	78.72	69.57	4.5461	3.953			0	0	0
HSS304.8X254X6.4	42.11	37.22	4.6698	4.0703			0	0	0
HSS304.8X254X7.9	51.69	45.71	4.6545	4.048			0	0	0
HSS304.8X254X9.5	61.02	54.01	4.6072	4.0033			0	0	0
HSS304.8X304.8X12.7	89.7	89.7	4.6706	4.6706			0	0	0
HSS304.8X304.8X15.9	109.23	109.23	4.614	4.614			0	0	0
HSS304.8X304.8X4.8	36	36	4.8157	4.8157			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS304.8X304.8X6.4	47.6	47.6	4.786	4.786			0	0	0
HSS304.8X304.8X7.9	58.58	58.58	4.7705	4.7705			0	0	0
HSS304.8X304.8X9.5	68.96	68.96	4.7352	4.7352			0	0	0
HSS304.8X50.8X4.8	15.5	3.97	3.804	0.873			0	0	0
HSS304.8X50.8X6.4	20.08	5.08	3.7547	0.8465			0	0	0
HSS304.8X50.8X7.9	24.47	6.05	3.7009	0.8189			0	0	0
HSS304.8X76.2X4.8	17.51	6.41	3.9793	1.3178			0	0	0
HSS304.8X76.2X6.4	22.88	8.3	3.9416	1.2935			0	0	0
HSS304.8X76.2X7.9	27.89	10.01	3.8957	1.2661			0	0	0
HSS304.8X88.9X7.9	29.6	12.08	3.98	1.4823			0	0	0
HSS304.8X88.9X9.5	34.72	13.97	3.9492	1.46			0	0	0
HSS355.6X101.6X12.7	61.02	24.1	4.553	1.6387			0	0	0
HSS355.6X101.6X15.9	73.23	28.5	4.4559	1.5845			0	0	0
HSS355.6X101.6X4.8	25.32	10.31	4.7535	1.7708			0	0	0
HSS355.6X101.6X6.4	33.2	13.49	4.7088	1.747			0	0	0
HSS355.6X101.6X7.9	40.58	16.42	4.6661	1.7189			0	0	0
HSS355.6X101.6X9.5	47.78	19.1	4.6245	1.6886			0	0	0
HSS355.6X152.4X12.7	73.84	40.4	4.8291	2.4703			0	0	0
HSS355.6X152.4X15.9	88.48	48.39	4.78	2.434			0	0	0
HSS355.6X152.4X4.8	30.08	16.72	5.0169	2.6058			0	0	0
HSS355.6X152.4X6.4	39.6	22.03	4.9778	2.5789			0	0	0
HSS355.6X152.4X7.9	48.57	26.91	4.9459	2.5527			0	0	0
HSS355.6X152.4X9.5	57.3	31.61	4.9004	2.5234			0	0	0
HSS355.6X254X12.7	98.86	78.72	5.2384	4.0378			0	0	0
HSS355.6X254X15.9	120.22	95.2	5.1677	3.9724			0	0	0
HSS355.6X254X6.4	52.42	41.8	5.3561	4.1488			0	0	0
HSS355.6X254X7.9	64.69	51.38	5.3209	4.115			0	0	0
HSS355.6X254X9.5	76.28	60.72	5.2906	4.087			0	0	0
HSS355.6X355.6X12.7	123.88	123.88	5.4884	5.4884			0	0	0
HSS355.6X355.6X15.9	150.73	150.73	5.4451	5.4451			0	0	0
HSS355.6X355.6X7.9	80.55	80.55	5.5953	5.5953			0	0	0
HSS355.6X355.6X9.5	95.2	95.2	5.5447	5.5447			0	0	0
HSS406.4X101.6X12.7	77.5	27.4	5.1373	1.6544			0	0	0
HSS406.4X101.6X15.9	92.76	32.53	5.0713	1.6073			0	0	0
HSS406.4X101.6X4.8	31.67	11.72	5.3429	1.7838			0	0	0
HSS406.4X101.6X6.4	41.68	15.19	5.3064	1.7561			0	0	0
HSS406.4X101.6X7.9	51.08	18.49	5.264	1.7284			0	0	0
HSS406.4X101.6X9.5	60.23	21.72	5.2239	1.7008			0	0	0
HSS406.4X203.2X12.7	106.18	65.3	5.7002	3.3148			0	0	0
HSS406.4X203.2X15.9	128.76	79.33	5.6262	3.2626			0	0	0
HSS406.4X203.2X6.4	56.39	35.03	5.833	3.4299			0	0	0
HSS406.4X203.2X7.9	69.57	43.02	5.8041	3.3997			0	0	0
HSS406.4X203.2X9.5	82.38	50.77	5.7669	3.3662			0	0	0
HSS406.4X304.8X12.7	134.86	111.06	6.0543	4.8571			0	0	0
HSS406.4X304.8X15.9	164.76	134.86	6.0073	4.8094			0	0	0
HSS406.4X304.8X7.9	87.87	72.01	6.1692	4.9553			0	0	0
HSS406.4X304.8X9.5	103.74	85.43	6.116	4.9074			0	0	0
HSS406.4X406.4X12.7	164.15	164.15	6.3094	6.3094			0	0	0
HSS406.4X406.4X15.9	200.16	200.16	6.2524	6.2524			0	0	0
HSS406.4X406.4X7.9	106.18	106.18	6.3878	6.3878			0	0	0
HSS406.4X406.4X9.5	125.71	125.71	6.3623	6.3623			0	0	0
HSS457.2X152.4X12.7	112.28	50.71	6.0614	2.5311			0	0	0
HSS457.2X152.4X15.9	134.86	61.02	5.9879	2.4787			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS457.2X152.4X6.4	59.38	27.28	6.2205	2.6383			0	0	0
HSS457.2X152.4X7.9	73.23	33.5	6.1925	2.6095			0	0	0
HSS457.2X152.4X9.5	86.65	39.48	6.1459	2.5761			0	0	0
HSS50.8X25.4X3.2	0.37	0.22	0.6802	0.3897			0	0	0
HSS50.8X25.4X4.8	0.48	0.29	0.6444	0.364			0	0	0
HSS50.8X38.1X3.2	0.47	0.39	0.7265	0.5818			0	0	0
HSS50.8X38.1X4.8	0.64	0.52	0.6966	0.5534			0	0	0
HSS50.8X50.8X3.2	0.58	0.58	0.7601	0.7601			0	0	0
HSS50.8X50.8X4.8	0.8	0.8	0.7341	0.7341			0	0	0
HSS50.8X50.8X6.4	0.96	0.96	0.7035	0.7035			0	0	0
HSS508X101.6X12.7	114.72	33.99	6.3301	1.6738			0	0	0
HSS508X101.6X6.4	61.63	18.67	6.5173	1.7833			0	0	0
HSS508X101.6X7.9	75.67	22.88	6.4615	1.7505			0	0	0
HSS508X101.6X9.5	89.09	26.79	6.4096	1.7262			0	0	0
HSS508X203.2X12.7	151.95	79.33	6.9466	3.3916			0	0	0
HSS508X203.2X15.9	184.9	96.42	6.9002	3.3478			0	0	0
HSS508X203.2X7.9	98.86	51.99	7.084	3.4753			0	0	0
HSS508X203.2X9.5	117.17	61.63	7.0227	3.4404			0	0	0
HSS508X304.8X12.7	187.95	131.81	7.3913	4.9817			0	0	0
HSS508X304.8X15.9	230.06	161.71	7.3281	4.9274			0	0	0
HSS508X304.8X7.9	122.05	86.04	7.4593	5.0565			0	0	0
HSS508X304.8X9.5	144.02	101.91	7.4595	5.0423			0	0	0
HSS57.2X50.8X3.2	0.7	0.64	0.8486	0.7744			0	0	0
HSS57.2X50.8X4.8	0.95	0.88	0.8196	0.7465			0	0	0
HSS57.2X57.2X3.2	0.76	0.76	0.8623	0.8623			0	0	0
HSS57.2X57.2X4.8	1.04	1.04	0.8343	0.8343			0	0	0
HSS57.2X57.2X6.4	1.28	1.28	0.8065	0.8065			0	0	0
HSS63.5X25.4X3.2	0.53	0.27	0.8329	0.3987			0	0	0
HSS63.5X25.4X4.8	0.71	0.36	0.796	0.3744			0	0	0
HSS63.5X38.1X3.2	0.67	0.47	0.8916	0.5979			0	0	0
HSS63.5X38.1X4.8	0.92	0.63	0.8606	0.5718			0	0	0
HSS63.5X38.1X6.4	1.11	0.76	0.8263	0.5455			0	0	0
HSS63.5X50.8X3.2	0.81	0.7	0.9337	0.7845			0	0	0
HSS63.5X50.8X4.8	1.12	0.96	0.9039	0.7572			0	0	0
HSS63.5X50.8X6.4	1.37	1.17	0.8756	0.7318			0	0	0
HSS63.5X63.5X3.2	0.95	0.95	0.9655	0.9655			0	0	0
HSS63.5X63.5X4.8	1.32	1.32	0.9361	0.9361			0	0	0
HSS63.5X63.5X6.4	1.63	1.63	0.9097	0.9097			0	0	0
HSS63.5X63.5X7.9	1.88	1.88	0.8792	0.8792			0	0	0
HSS76.2X25.4X3.2	0.73	0.33	0.9861	0.4052			0	0	0
HSS76.2X25.4X4.8	0.99	0.43	0.9477	0.3812			0	0	0
HSS76.2X38.1X3.2	0.9	0.55	1.0526	0.6098			0	0	0
HSS76.2X38.1X4.8	1.24	0.75	1.018	0.5832			0	0	0
HSS76.2X38.1X6.4	1.51	0.91	0.9835	0.5593			0	0	0
HSS76.2X50.8X3.2	1.06	0.81	1.1024	0.8043			0	0	0
HSS76.2X50.8X4.8	1.48	1.12	1.072	0.7778			0	0	0
HSS76.2X50.8X6.4	1.83	1.38	1.0405	0.7509			0	0	0
HSS76.2X50.8X7.9	2.11	1.58	1.0053	0.7254			0	0	0
HSS76.2X63.5X3.2	1.23	1.09	1.1374	0.9873			0	0	0
HSS76.2X63.5X4.8	1.73	1.52	1.1123	0.9658			0	0	0
HSS76.2X63.5X6.4	2.16	1.9	1.0769	0.9329			0	0	0
HSS76.2X63.5X7.9	2.51	2.2	1.0547	0.9094			0	0	0
HSS76.2X76.2X3.2	1.4	1.4	1.17	1.17			0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
HSS76.2X76.2X4.8	1.97	1.97	1.1384	1.1384			0	0	0
HSS76.2X76.2X6.4	2.48	2.48	1.1153	1.1153			0	0	0
HSS76.2X76.2X7.9	2.9	2.9	1.0839	1.0839			0	0	0
HSS76.2X76.2X9.5	3.25	3.25	1.0541	1.0541			0	0	0
HSS88.9X38.1X3.2	1.15	0.63	1.2112	0.6198			0	0	0
HSS88.9X38.1X4.8	1.6	0.87	1.1728	0.5936			0	0	0
HSS88.9X38.1X6.4	1.98	1.06	1.1374	0.5698			0	0	0
HSS88.9X50.8X3.2	1.34	0.91	1.2635	0.8173			0	0	0
HSS88.9X50.8X4.8	1.89	1.27	1.2393	0.7963			0	0	0
HSS88.9X50.8X6.4	2.36	1.58	1.1961	0.7658			0	0	0
HSS88.9X63.5X3.2	1.54	1.23	1.3103	1.0108			0	0	0
HSS88.9X63.5X4.8	2.18	1.73	1.2778	0.9826			0	0	0
HSS88.9X63.5X6.4	2.74	2.16	1.2487	0.9572			0	0	0
HSS88.9X63.5X7.9	3.2	2.53	1.2142	0.9299			0	0	0
HSS88.9X63.5X9.5	3.59	2.83	1.1834	0.9022			0	0	0
HSS88.9X88.9X3.2	1.93	1.93	1.3736	1.3736			0	0	0
HSS88.9X88.9X4.8	2.76	2.76	1.3441	1.3441			0	0	0
HSS88.9X88.9X6.4	3.5	3.5	1.3158	1.3158			0	0	0
HSS88.9X88.9X7.9	4.14	4.14	1.2881	1.2881			0	0	0
HSS88.9X88.9X9.5	4.69	4.69	1.2591	1.2591			0	0	0
L102X102X11.1	3.1	3.1	1.2214	1.2214	0.25	0.378	0.8583	-0.8583	1.5902
L102X102X12.7	3.5	3.5	1.2137	1.2137	0.37	0.374	0.8268	-0.8268	1.5346
L102X102X15.9	4.28	4.28	1.2002	1.2002	0.69	0.374	0.7874	-0.7874	1.4263
L102X102X19	5.02	5.02	1.1832	1.1832	1.14	0.378	0.7362	-0.7362	1.3218
L102X102X6.4	1.82	1.82	1.245	1.245	0.05	0.376	0.9291	-0.9291	1.7613
L102X102X7.9	2.26	2.26	1.2369	1.2369	0.1	0.3764	0.8976	-0.8976	1.703
L102X102X9.5	2.69	2.69	1.2281	1.2281	0.16	0.3768	0.878	-0.878	1.6458
L102X76X12.7	3.36	1.99	1.242	0.8587	0.26	0.374	0.6772	-0.689	1.09
L102X76X15.9	4.08	2.45	1.2279	0.8472	0.48	0.374	0.6339	-0.6378	0.9976
L102X76X6.4	1.77	1.03	1.2732	0.8876	0.04	0.376	0.7756	-0.7874	1.285
L102X76X7.9	2.19	1.28	1.2678	0.8797	0.07	0.3764	0.748	-0.7559	1.2347
L102X76X9.5	2.6	1.52	1.2565	0.8704	0.11	0.3768	0.7244	-0.7362	1.1854
L102X89X12.7	3.46	2.69	1.2311	1.0377	0.3	0.374	0.7579	-0.7677	1.3089
L102X89X6.4	1.81	1.4	1.2609	1.0686	0.04	0.376	0.8524	-0.8661	1.5194
L102X89X7.9	2.24	1.74	1.2535	1.0594	0.08	0.3764	0.8287	-0.8386	1.4652
L102X89X9.5	2.66	2.06	1.245	1.0498	0.14	0.3768	0.8012	-0.8071	1.412
L127X127X11.1	5	5	1.5397	1.5397	0.51	0.5	1.0984	-1.0984	2.0758
L127X127X12.7	5.66	5.66	1.5355	1.5355	0.74	0.5	1.0787	-1.0787	2.0187
L127X127X15.9	6.96	6.96	1.5174	1.5174	1.4	0.5039	1.0315	-1.0315	1.9068
L127X127X19	8.12	8.12	1.4997	1.4997	2.34	0.5	0.9803	-0.9803	1.7983
L127X127X22.2	9.34	9.34	1.4919	1.4919	3.52	0.5079	0.9409	-0.9409	1.6959
L127X127X7.9	3.65	3.65	1.5578	1.5578	0.19	0.5024	1.1496	-1.1496	2.1909
L127X127X9.5	4.33	4.33	1.5516	1.5516	0.33	0.4988	1.1299	-1.1299	2.1327
L127X76X11.1	4.53	1.82	1.5922	0.8308	0.3	0.437	0.7795	-0.7795	1.1647
L127X76X12.7	5.12	2.08	1.5866	0.824	0.44	0.437	0.7559	-0.7598	1.1202
L127X76X6.4	2.68	1.05	1.6214	0.8532	0.06	0.439	0.8504	-0.8583	1.3014
L127X76X7.9	3.32	1.31	1.6125	0.8462	0.12	0.4394	0.8268	-0.8307	1.2549
L127X76X9.5	3.93	1.57	1.6012	0.8374	0.2	0.4398	0.8031	-0.811	1.2091
L127X89X12.7	5.33	2.79	1.5809	1.0016	0.49	0.437	0.8484	-0.8504	1.3452
L127X89X15.9	6.53	3.43	1.5596	0.9873	0.92	0.4331	0.8012	-0.811	1.2524
L127X89X19	7.63	4.07	1.5429	0.9724	1.53	0.437	0.7579	-0.7598	1.1626
L127X89X6.4	2.78	1.42	1.6061	1.0293	0.07	0.439	0.9469	-0.9488	1.5389
L127X89X7.9	3.45	1.77	1.6044	1.0257	0.13	0.4394	0.9193	-0.9291	1.4892

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
L127X89X9.5	4.09	2.12	1.5942	1.0152	0.22	0.4398	0.8957	-0.9016	1.4403
L152X102X11.1	6.83	3.24	1.9183	1.1516	0.57	0.5	1.0512	-1.0433	1.6535
L152X102X12.7	7.69	3.69	1.9097	1.1454	0.84	0.5	1.0276	-1.0118	1.6058
L152X102X14.3	8.6	4.13	1.9002	1.1368	1.18	0.4961	1.0079	-0.9921	1.5588
L152X102X15.9	9.46	4.56	1.8931	1.1293	1.59	0.5039	0.9764	-0.9606	1.5123
L152X102X19	11.11	5.42	1.8786	1.1145	2.65	0.5	0.937	-0.9213	1.4216
L152X102X22.2	12.69	6.29	1.8586	1.1016	4.02	0.5079	0.8898	-0.874	1.336
L152X102X7.9	4.96	2.33	1.9431	1.1693	0.22	0.5024	1.0984	-1.0906	1.7496
L152X102X9.5	5.89	2.79	1.9267	1.1592	0.37	0.4988	1.0748	-1.063	1.701
L152X152X11.1	7.26	7.26	1.8612	1.8612	0.89	0.5	1.3425	-1.3425	2.5659
L152X152X12.7	8.24	8.24	1.8574	1.8574	1.31	0.5	1.3228	-1.3228	2.5078
L152X152X14.3	9.15	9.15	1.8474	1.8474	1.84	0.4961	1.2913	-1.2913	2.4505
L152X152X15.9	10.13	10.13	1.8356	1.8356	2.49	0.5039	1.2717	-1.2717	2.3936
L152X152X19	11.9	11.9	1.8225	1.8225	4.17	0.5	1.2205	-1.2205	2.2822
L152X152X22.2	13.73	13.73	1.8104	1.8104	6.33	0.5079	1.1811	-1.1811	2.1765
L152X152X25.4	15.38	15.38	1.7914	1.7914	9.16	0.5	1.1339	-1.1339	2.0705
L152X152X7.9	5.26	5.26	1.881	1.881	0.34	0.5024	1.3937	-1.3937	2.6824
L152X152X9.5	6.29	6.29	1.8737	1.8737	0.57	0.4988	1.374	-1.374	2.6235
L152X89X12.7	7.51	2.88	1.9218	0.9699	0.77	0.5	0.9193	-0.9213	1.3694
L152X89X7.9	4.84	1.82	1.9451	0.9916	0.2	0.5024	0.9941	-0.9921	1.5052
L152X89X9.5	5.74	2.18	1.9363	0.9851	0.34	0.4988	0.9705	-0.9724	1.4592
L178X102X11.1	9.03	3.31	2.2562	1.1177	0.85	0.5	1.122	-1.122	1.6733
L178X102X12.7	10.19	3.77	2.2528	1.1111	1.26	0.5	1.0984	-1.1024	1.628
L178X102X15.9	12.51	4.69	2.2347	1.0948	2.39	0.5039	1.0512	-1.0551	1.539
L178X102X19	14.83	5.6	2.2083	1.0793	4.01	0.5	1.0079	-1.0039	1.4523
L178X102X9.5	7.81	2.84	2.2638	1.1259	0.55	0.4988	1.1457	-1.1535	1.7184
L203X102X11.1	11.59	3.42	2.5826	1.0858	1.22	0.5	1.1772	-1.185	1.6876
L203X102X12.7	13.12	3.91	2.5831	1.0792	1.8	0.5	1.1535	-1.1575	1.644
L203X102X14.3	14.58	4.39	2.5733	1.0709	2.53	0.4961	1.1299	-1.1378	1.6009
L203X102X15.9	16.11	4.86	2.5643	1.0649	3.43	0.5039	1.1063	-1.1063	1.5582
L203X102X19	18.92	5.82	2.545	1.0503	5.79	0.5	1.0591	-1.0551	1.4744
L203X102X22.2	21.72	6.77	2.53	1.0353	8.85	0.5079	1.0118	-1.0079	1.395
L203X102X25.4	24.29	7.75	2.5056	1.0225	12.89	0.5	0.9685	-0.9646	1.3153
L203X152X11.1	12.88	7.51	2.5662	1.7957	1.53	0.5	1.5512	-1.5669	2.6181
L203X152X12.7	14.58	8.54	2.5558	1.7856	2.27	0.5	1.5315	-1.5354	2.5667
L203X152X14.3	16.17	9.52	2.5501	1.7767	3.19	0.4961	1.5039	-1.5157	2.5158
L203X152X15.9	17.88	10.5	2.5399	1.772	4.33	0.5039	1.4803	-1.4961	2.4653
L203X152X19	21.11	12.51	2.5188	1.7538	7.3	0.5	1.4331	-1.4449	2.3661
L203X152X22.2	24.29	14.4	2.5075	1.7404	11.16	0.5079	1.3937	-1.3976	2.2716
L203X152X25.4	27.28	16.17	2.4863	1.7185	16.26	0.5	1.3425	-1.3465	2.1764
L203X203X12.7	15.07	15.07	2.4937	2.4937	3.22	0.6299	1.8268	-1.8268	3.5055
L203X203X14.3	16.78	16.78	2.4878	2.4878	4.54	0.626	1.8071	-1.8071	3.4467
L203X203X15.9	18.61	18.61	2.48	2.48	6.17	0.626	1.7874	-1.7874	3.3885
L203X203X19	22.03	22.03	2.4655	2.4655	10.44	0.6299	1.7362	-1.7362	3.2733
L203X203X22.2	25.32	25.32	2.449	2.449	16	0.626	1.685	-1.685	3.1637
L203X203X25.4	28.5	28.5	2.4298	2.4298	23.36	0.6299	1.6378	-1.6378	3.0524
L203X203X28.6	31.61	31.61	2.4198	2.4198	32.52	0.626	1.5945	-1.5945	2.9433
L51X51X3.2	0.23	0.23	0.6203	0.6203	0.0007928	0.25	0.4646	-0.4646	0.8748
L51X51X4.8	0.34	0.34	0.6131	0.6131	0.002533	0.2496	0.4409	-0.4409	0.8175
L51X51X6.4	0.44	0.44	0.6054	0.6054	0.01	0.25	0.4134	-0.4134	0.7617
L51X51X7.9	0.54	0.54	0.597	0.597	0.01	0.2504	0.3898	-0.3898	0.7079
L51X51X9.5	0.63	0.63	0.5892	0.5892	0.02	0.2508	0.3661	-0.3661	0.6561
L64X38X4.8	0.5	0.2	0.8004	0.417	0.003056	0.2496	0.4031	-0.4114	0.604

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
L64X38X6.4	0.65	0.26	0.7916	0.411	0.01	0.25	0.378	-0.3839	0.5594
L64X51X4.8	0.53	0.35	0.7908	0.5985	0.003814	0.2496	0.4921	-0.4941	0.8352
L64X51X6.4	0.69	0.45	0.7831	0.5901	0.01	0.25	0.4685	-0.4705	0.7844
L64X51X7.9	0.84	0.56	0.7737	0.5817	0.02	0.2504	0.4449	-0.4469	0.7351
L64X51X9.5	0.98	0.66	0.7675	0.5759	0.03	0.2508	0.4213	-0.4232	0.6873
L64X64X12.7	1.29	1.29	0.7344	0.7344	0.08	0.252	0.4469	-0.4469	0.7968
L64X64X4.8	0.53	0.53	0.7713	0.7713	0.01	0.2496	0.565	-0.565	1.0665
L64X64X6.4	0.7	0.7	0.7624	0.7624	0.01	0.25	0.5374	-0.5374	1.0094
L64X64X7.9	0.85	0.85	0.7567	0.7567	0.02	0.2504	0.5138	-0.5138	0.9538
L64X64X9.5	1.01	1.01	0.7487	0.7487	0.04	0.2508	0.4902	-0.4902	0.8998
L76X51X12.7	1.78	0.88	0.921	0.5433	0.09	0.315	0.4213	-0.4213	0.6624
L76X51X4.8	0.74	0.35	0.9614	0.5766	0.01	0.3126	0.5394	-0.5472	0.8456
L76X51X6.4	0.97	0.46	0.9535	0.5696	0.01	0.313	0.5118	-0.5197	0.798
L76X51X7.9	1.19	0.57	0.944	0.5611	0.02	0.3134	0.4882	-0.4882	0.7517
L76X51X9.5	1.39	0.68	0.9377	0.5543	0.04	0.3138	0.4646	-0.4685	0.7067
L76X64X11.1	1.66	1.19	0.9183	0.7265	0.08	0.378	0.5256	-0.5276	0.8753
L76X64X12.7	1.86	1.34	0.911	0.719	0.11	0.374	0.5059	-0.5039	0.8768
L76X64X4.8	0.76	0.54	0.948	0.7531	0.01	0.3756	0.624	-0.626	1.0782
L76X64X6.4	1	0.71	0.9374	0.7461	0.02	0.376	0.5965	-0.5984	1.0254
L76X64X7.9	1.23	0.87	0.9309	0.739	0.03	0.3764	0.5728	-0.5748	0.9738
L76X64X9.5	1.45	1.03	0.923	0.7294	0.05	0.3768	0.5492	-0.5512	0.9643
L76X76X11.1	1.7	1.7	0.9019	0.9019	0.1	0.378	0.5945	-0.5945	1.0889
L76X76X12.7	1.91	1.91	0.8931	0.8931	0.14	0.374	0.5709	-0.5709	1.0942
L76X76X4.8	0.78	0.78	0.9332	0.9332	0.01	0.3756	0.689	-0.689	1.3125
L76X76X6.4	1.02	1.02	0.9243	0.9243	0.02	0.376	0.6654	-0.6654	1.2545
L76X76X7.9	1.26	1.26	0.9171	0.9171	0.04	0.3764	0.6417	-0.6417	1.1978
L76X76X9.5	1.48	1.48	0.9109	0.9109	0.07	0.3768	0.6142	-0.6142	1.1902
L89X64X12.7	2.52	1.39	1.0812	0.7001	0.16	0.374	0.5492	-0.5492	0.8985
L89X64X6.4	1.36	0.73	1.1173	0.7317	0.02	0.376	0.6437	-0.6516	1.0396
L89X64X7.9	1.67	0.9	1.1111	0.725	0.04	0.3764	0.6161	-0.6201	0.991
L89X64X9.5	1.96	1.07	1.1003	0.7167	0.07	0.3768	0.5965	-0.6004	0.9812
L89X76X11.1	2.32	1.75	1.0782	0.8854	0.13	0.378	0.6535	-0.6594	1.1164
L89X76X12.7	2.61	1.97	1.0699	0.8763	0.19	0.374	0.6299	-0.6319	1.1205
L89X76X6.4	1.39	1.04	1.1019	0.9067	0.03	0.376	0.7283	-0.7303	1.2717
L89X76X7.9	1.72	1.28	1.0924	0.8997	0.05	0.3764	0.7008	-0.6988	1.2186
L89X76X9.5	2.03	1.52	1.0854	0.8897	0.09	0.3768	0.6772	-0.6791	1.2105
L89X89X11.1	2.36	2.36	1.0607	1.0607	0.16	0.378	0.7185	-0.7185	1.3352
L89X89X12.7	2.66	2.66	1.0557	1.0557	0.24	0.374	0.6988	-0.6988	1.3429
L89X89X6.4	1.41	1.41	1.0828	1.0828	0.03	0.376	0.7972	-0.7972	1.5039
L89X89X7.9	1.74	1.74	1.0822	1.0822	0.06	0.3764	0.7697	-0.7697	1.4463
L89X89X9.5	2.06	2.06	1.0704	1.0704	0.11	0.3768	0.75	-0.75	1.4401
Rufter									
Truss									
W1000X222	598.03	62.24	14.9487	2.2846	79592.76	1.1811	0	0	0
W1000X249	695.67	76.28	15.3436	2.3982	99854.16	1.1772	0	0	0
W1000X272	775	88.48	15.728	2.4936	117881.27	1.1811	0	0	0
W1000X296	866.54	137.3	15.9236	3.4379	248612.26	1.1811	0	0	0
W1000X314	903.15	104.96	15.7897	2.5024	140514.58	1.1772	0	0	0
W1000X321	964.18	156.22	16.2094	3.5534	285840.48	1.1811	0	0	0
W1000X350	1012.99	117.78	15.8624	2.5356	161215.05	1.1811	0	0	0
W1000X371	1122.84	181.85	16.3351	3.5482	335575.88	1.1772	0	0	0
W1000X393	1128.94	131.81	15.8326	2.5234	178755.92	1.1811	0	0	0
W1000X412	1250.99	203.82	16.3935	3.572	377413.88	1.1811	0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
W1000X415	1189.96	139.74	15.7795	2.5168	192115.2	1.1772	0	0	0
W1000X438	1269.29	150.12	15.9451	2.5541	207596.52	1.1811	0	0	0
W1000X443	1330.32	214.8	16.308	3.5354	392892.77	1.1811	0	0	0
W1000X483	1458.47	239.21	16.4218	3.5782	446484.41	1.1772	0	0	0
W1000X486	1409.65	170.26	15.9816	2.5808	236082.93	1.1811	0	0	0
W1000X494	1427.96	172.09	15.919	2.5678	243071.78	1.1811	0	0	0
W1000X539	1641.54	269.72	16.4903	3.6066	508466.23	1.1772	0	0	0
W1000X554	1678.15	277.05	16.3866	3.592	528911.49	1.1772	0	0	0
W1000X584	1708.66	211.75	16.0297	2.6308	307905.02	1.1811	0	0	0
W1000X591	1800.2	300.24	16.5241	3.6276	574707.22	1.1811	0	0	0
W1000X642	1958.86	327.7	16.5656	3.6476	632552.92	1.1811	0	0	0
W1000X748	2318.9	394.21	16.7567	3.7121	788543.72	1.1772	0	0	0
W1000X883	2758.27	480.87	17.0477	3.812	985414.49	1.1811	0	0	0
W100X19.3	6.29	2.92	1.7174	1.0051	14.04	0.2496	0	0	0
W1100X343	1098.43	156.83	17.5261	3.4264	347834.93	0.7913	0	0	0
W1100X390	1269.29	181.85	17.6422	3.4571	408816.48	0.7795	0	0	0
W1100X433	1409.65	205.04	17.75	3.4901	459251.23	0.7795	0	0	0
W1100X499	1617.13	236.16	17.7449	3.49	532059.25	0.7874	0	0	0
W130X23.8	9.64	4.58	2.1314	1.2633	40.35	0.3016	0	0	0
W130X28.1	11.59	5.53	2.1694	1.2809	51.16	0.2992	0	0	0
W150X13	5.73	1.56	2.4281	0.8873	15.72	0.25	0	0	0
W150X13.5	6.22	1.72	2.4737	0.9059	17.7	0.2496	0	0	0
W150X18	8.3	2.32	2.4954	0.9161	24.92	0.2516	0	0	0
W150X22.5	10.8	4.75	2.5608	1.4501	76.03	0.252	0	0	0
W150X24	11.72	3.39	2.6053	0.9654	38.01	0.248	0	0	0
W150X29.8	15.01	6.71	2.6522	1.5052	112.43	0.2492	0	0	0
W150X37.1	18.92	8.54	2.6943	1.5259	148.7	0.248	0	0	0
W200X100	70.18	32.71	3.7137	2.1222	1435.39	0.3976	0	0	0
W200X15	8.85	1.66	3.223	0.8403	30.67	0.2988	0	0	0
W200X19.3	11.41	2.15	3.2113	0.8441	41.21	0.3	0	0	0
W200X22.5	13.61	2.67	3.2923	0.8773	51.64	0.2992	0	0	0
W200X26.6	17.03	4.66	3.4346	1.2321	120.68	0.3	0	0	0
W200X31.3	20.38	5.69	3.4958	1.2606	152.02	0.2992	0	0	0
W200X35.9	23.13	8.54	3.4158	1.6076	258.82	0.3937	0	0	0
W200X41.7	27.22	10.13	3.4478	1.622	312.61	0.3937	0	0	0
W200X46.1	30.39	14.1	3.4717	2.0131	526.34	0.3976	0	0	0
W200X52	34.72	16.11	3.5114	2.0312	620.81	0.3937	0	0	0
W200X59	39.79	18.49	3.533	2.0465	727.72	0.3937	0	0	0
W200X71	49	22.88	3.6121	2.0759	930.88	0.3937	0	0	0
W200X86	59.8	27.89	3.6568	2.1001	1183.62	0.3898	0	0	0
W250X101	85.43	40.09	4.4564	2.5994	3083.5	0.5	0	0	0
W250X115	97.64	45.89	4.4794	2.6087	3631.79	0.5	0	0	0
W250X131	112.89	53.09	4.5257	2.6217	4339.24	0.5	0	0	0
W250X149	129.98	61.02	4.6088	2.6588	5096.91	0.5	0	0	0
W250X167	147.07	68.96	4.6677	2.6795	6052.54	0.5	0	0	0
W250X17.9	12.57	1.74	3.9023	0.7852	51.43	0.302	0	0	0
W250X22.3-1	15.99	2.3	3.9508	0.8079	68.99	0.3008	0	0	0
W250X25.3-1	18.67	2.8	4.0515	0.8441	85.29	0.3	0	0	0
W250X28.4-1	21.6	3.35	4.1379	0.8743	102.09	0.3031	0	0	0
W250X32.7	26	6.1	4.2619	1.3256	275.54	0.3016	0	0	0
W250X38.5	31.31	7.51	4.3485	1.3613	347.23	0.2992	0	0	0
W250X44.8	36.61	8.85	4.3878	1.3747	421.87	0.2992	0	0	0
W250X49.1-1	38.81	13.97	4.1987	1.94	783.53	0.5	0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
W250X58-1	46.81	17.21	4.2631	1.9764	996.75	0.5	0	0	0
W250X67-1	54.92	20.32	4.3136	2.0026	1204.18	0.5	0	0	0
W250X73	60.41	28.32	4.3421	2.5476	2076.22	0.5	0	0	0
W250X80	66.52	31.31	4.3757	2.5533	2311.46	0.5039	0	0	0
W250X89-1	74.45	35.03	4.394	2.5626	2661.86	0.5	0	0	0
W310X107-1	108.01	49.19	5.3165	3.0421	6512.97	0.6024	0	0	0
W310X117-1	119	54.31	5.3404	3.0479	7370.66	0.5945	0	0	0
W310X129-1	131.81	60.41	5.379	3.065	8179.95	0.5984	0	0	0
W310X143-1	147.07	67.74	5.4362	3.0884	9533.21	0.5984	0	0	0
W310X158-1	164.15	75.06	5.47	3.1047	10644.95	0.6024	0	0	0
W310X179-1	186.12	85.43	5.5123	3.1357	12346.09	0.5906	0	0	0
W310X202-1	214.19	98.25	5.5786	3.1641	14649.09	0.5984	0	0	0
W310X21-1	17.39	1.9	4.6197	0.7536	80.27	0.2984	0	0	0
W310X226-1	242.87	111.06	5.6589	3.1893	17335.64	0.5984	0	0	0
W310X23.8-1	20.08	2.26	4.6769	0.7724	95.72	0.302	0	0	0
W310X253-1	275.22	125.71	5.7417	3.2121	20156.01	0.6024	0	0	0
W310X28.3-1	24.71	2.98	4.833	0.8233	132.72	0.2996	0	0	0
W310X283-1	311.22	142.8	5.813	3.2434	23932.52	0.5906	0	0	0
W310X313-1	347.84	159.27	5.8833	3.2744	27123.66	0.5984	0	0	0
W310X32.7-1	29.29	3.66	4.9057	0.8482	161.33	0.2992	0	0	0
W310X342	386.28	176.97	5.9853	3.3106	31629.97	0.5984	0	0	0
W310X375	427.78	195.89	6.0533	3.3447	35538.33	0.5984	0	0	0
W310X38.7-1	37.22	8.18	5.1613	1.503	606.77	0.3012	0	0	0
W310X415	480.87	220.3	6.1538	3.3836	41846.51	0.6024	0	0	0
W310X44.5-1	43.08	9.58	5.2049	1.5199	719.26	0.2992	0	0	0
W310X454	537.01	244.09	6.3054	3.4263	47824.55	0.5906	0	0	0
W310X500	602.91	274	6.4077	3.4678	56758.74	0.5906	0	0	0
W310X52	51.2	11.47	5.2666	1.5419	886.22	0.2992	0	0	0
W310X60	57	16.78	5.1262	1.9436	1419.19	0.5039	0	0	0
W310X67	64.07	18.98	5.1573	1.9533	1644.31	0.5039	0	0	0
W310X74	72.01	21.3	5.1789	1.9622	1879.51	0.5	0	0	0
W310X79-1	78.11	29.11	5.2119	2.4745	3173.91	0.6063	0	0	0
W310X86-1	86.65	32.53	5.2821	2.5041	3575.06	0.5984	0	0	0
W310X97	97.03	44.12	5.2892	3.0205	5764.77	0.5945	0	0	0
W360X101-1	114.72	36.92	6.0139	2.4609	5306.41	0.5906	0	0	0
W360X1086	1659.85	817.72	8.1455	4.6751	360160.93	0.5906	0	0	0
W360X110-1	125.71	40.52	6.0321	2.4767	6098.21	0.5984	0	0	0
W360X122-1	139.13	44.79	6.0581	2.4819	6657.61	0.5945	0	0	0
W360X134	156.83	75.67	6.1407	3.6996	15901.42	0.6024	0	0	0
W360X147	172.7	83.6	6.1718	3.7106	18263.81	0.6024	0	0	0
W360X162-1	192.22	92.76	6.231	3.741	20108.64	0.6024	0	0	0
W360X179-1	211.75	101.91	6.2468	3.7422	22786.64	0.5984	0	0	0
W360X196-1	233.72	112.89	6.2844	3.7598	25373.04	0.5984	0	0	0
W360X216-1	259.96	133.03	6.3349	3.9868	31890.54	0.5984	0	0	0
W360X237-1	286.81	145.85	6.3822	4.0019	35809.49	0.6024	0	0	0
W360X262-1	319.76	162.93	6.4303	4.0244	40828.21	0.5984	0	0	0
W360X287-1	355.16	180.02	6.5044	4.0536	46078.18	0.5984	0	0	0
W360X314-1	389.94	197.72	6.5584	4.0772	51176.54	0.6024	0	0	0
W360X32.9-1	33.2	4.39	5.5344	1.0375	311.73	0.4012	0	0	0
W360X347-1	435.71	220.91	6.6208	4.0985	58687.91	0.5984	0	0	0
W360X382-1	486.97	245.93	6.7158	4.1299	67866.98	0.5984	0	0	0
W360X39-1	40.21	5.54	5.6458	1.0767	407.96	0.3976	0	0	0
W360X421-1	541.89	274	6.7958	4.1581	77023.97	0.5984	0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
W360X44-1	47.29	8.97	5.7311	1.4883	883.45	0.3984	0	0	0
W360X463-1	602.91	303.9	6.8766	4.1954	87696.94	0.5984	0	0	0
W360X509-1	671.26	338.07	6.964	4.231	103102.87	0.6024	0	0	0
W360X51-1	54.62	10.62	5.8416	1.5268	1067.48	0.3976	0	0	0
W360X551-1	738.39	369.8	7.059	4.2727	115796.24	0.5984	0	0	0
W360X57.8-1	61.63	12.08	5.8568	1.5426	1230.35	0.3976	0	0	0
W360X592-1	799.41	402.15	7.1641	4.3056	130125.67	0.5906	0	0	0
W360X634-1	866.54	433.88	7.2722	4.3456	144487.7	0.5906	0	0	0
W360X64	69.57	17.27	5.8255	1.8932	1960.65	0.5866	0	0	0
W360X677-1	933.66	468.05	7.3197	4.3787	158707.99	0.6024	0	0	0
W360X72	78.11	19.59	5.8512	1.9092	2244.26	0.5945	0	0	0
W360X744-1	1049.61	521.75	7.4778	4.4295	186122.3	0.5945	0	0	0
W360X79	87.26	22.03	5.8762	1.9192	2538.36	0.5906	0	0	0
W360X818	1177.76	582.78	7.6167	4.4641	217364.77	0.5906	0	0	0
W360X900	1318.11	652.95	7.788	4.5411	256529.31	0.5906	0	0	0
W360X91-1	101.91	32.77	5.9877	2.449	4724.54	0.5945	0	0	0
W360X990	1482.88	732.28	7.9672	4.6265	306267.77	0.5906	0	0	0
W410X100	129.98	35.52	6.9884	2.4677	7184.1	0.4055	0	0	0
W410X114	150.12	41.13	7.0034	2.4686	8604.31	0.4016	0	0	0
W410X132	175.14	48.09	7.044	2.4937	10385.62	0.4055	0	0	0
W410X149	197.72	54.92	7.1119	2.5128	11822.98	0.4055	0	0	0
W410X38.8-1	44.18	5.48	6.2563	1.1178	567.99	0.4031	0	0	0
W410X46.1-1	54.01	7.02	6.4072	1.1653	735.75	0.4016	0	0	0
W410X53-1	64.07	10.8	6.4922	1.5203	1473.94	0.4016	0	0	0
W410X60	73.23	12.69	6.6329	1.5634	1731.74	0.4016	0	0	0
W410X67	82.38	14.52	6.6392	1.5732	1995.36	0.4016	0	0	0
W410X75	92.15	16.29	6.6933	1.5919	2293.45	0.4016	0	0	0
W410X85	104.96	18.92	6.7344	1.6028	2663.2	0.4016	0	0	0
W460X106	145.85	24.71	7.4776	1.6976	4713.3	0.3976	0	0	0
W460X113	162.93	42.23	7.7222	2.6103	11528.68	0.3976	0	0	0
W460X128	186.12	48.39	7.7829	2.6311	13651.55	0.3976	0	0	0
W460X144	211.14	55.29	7.8311	2.6553	15565.32	0.4016	0	0	0
W460X158	230.06	60.47	7.8298	2.6578	17285.62	0.3976	0	0	0
W460X177	261.79	68.96	7.9088	2.6835	20526.01	0.4016	0	0	0
W460X193	289.86	76.89	8.0005	2.698	22888.24	0.3976	0	0	0
W460X213	322.21	85.43	8.0748	2.7163	25358.86	0.4016	0	0	0
W460X235	355.77	94.59	8.114	2.7322	28828.72	0.3976	0	0	0
W460X260	397.87	106.18	8.1993	2.7586	33428.03	0.3976	0	0	0
W460X286	441.81	119	8.2914	2.7954	38542.07	0.3976	0	0	0
W460X315	490.02	131.81	8.3309	2.8114	44049.5	0.4016	0	0	0
W460X349	549.21	148.9	8.4485	2.8491	50703.18	0.4016	0	0	0
W460X384	610.24	165.98	8.5111	2.8734	58168.85	0.4016	0	0	0
W460X421	677.36	184.9	8.6128	2.9081	65825.4	0.5	0	0	0
W460X464	756.69	206.87	8.7211	2.9464	75502.05	0.5	0	0	0
W460X52-1	66.52	8.06	7.0295	1.2185	1135.18	0.4016	0	0	0
W460X60-1	78.11	10.01	7.2068	1.2725	1442.01	0.4016	0	0	0
W460X68-1	90.93	11.72	7.2578	1.2913	1725.08	0.4055	0	0	0
W460X74-1	101.3	16.6	7.3788	1.6524	3069.6	0.4016	0	0	0
W460X82-1	112.28	18.49	7.3905	1.6615	3410.14	0.4016	0	0	0
W460X89-1	123.27	20.63	7.4663	1.6857	3837.21	0.4016	0	0	0
W460X97	133.03	22.52	7.4885	1.695	4274.19	0.3976	0	0	0
W530X101-1	159.88	24.41	8.6032	1.7978	6724.48	0.5039	0	0	0
W530X109	172.09	26.61	8.6178	1.8106	7386.88	0.5	0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
W530X123	195.89	30.51	8.6735	1.8294	8566.45	0.5039	0	0	0
W530X138	220.91	34.72	8.7129	1.8461	9906.49	0.5	0	0	0
W530X150	253.25	61.63	9.0298	2.8836	26236.83	0.5	0	0	0
W530X165	278.88	68.35	9.0514	2.9007	28703.46	0.5079	0	0	0
W530X182	306.95	75.67	9.0651	2.9129	32814.79	0.5	0	0	0
W530X196	333.19	82.38	9.1148	2.9356	35639.5	0.5	0	0	0
W530X219	372.86	92.76	9.1591	2.9533	41204.75	0.5	0	0	0
W530X248	432.05	108.01	9.3588	2.9844	48338.61	0.5	0	0	0
W530X272	475.99	119	9.3942	3.0007	54585.53	0.5	0	0	0
W530X300	530.3	133.03	9.4572	3.0243	61972.11	0.5	0	0	0
W530X66-1	95.2	10.19	8.0526	1.2619	2104.18	0.5	0	0	0
W530X72	106.79	14.89	8.2439	1.656	3933.99	0.5	0	0	0
W530X74-1	109.84	12.2	8.1876	1.304	2554.18	0.5039	0	0	0
W530X82-1	125.71	18.43	8.3737	1.7225	4990.97	0.4961	0	0	0
W530X85-1	128.76	14.83	8.3602	1.3501	3218.04	0.5	0	0	0
W530X92-1	144.02	21.72	8.5306	1.7718	5915.65	0.5039	0	0	0
W610X101-1	176.97	24.47	9.5317	1.8691	9444.99	0.5039	0	0	0
W610X113-1	200.16	28.62	9.6658	1.9148	11063.67	0.5	0	0	0
W610X125-1	223.96	32.59	9.8041	1.9573	12816.78	0.5	0	0	0
W610X140-1	253.86	37.53	9.848	1.9827	14827.4	0.5079	0	0	0
W610X153	280.1	41.5	9.9679	1.9836	16541.94	0.5	0	0	0
W610X155	289.25	62.24	10.0491	2.9077	35762.91	0.5	0	0	0
W610X174	327.09	71.4	10.1309	2.9424	40785.7	0.5	0	0	0
W610X195	369.8	81.77	10.1959	2.9731	47711.05	0.5	0	0	0
W610X217	418.01	93.37	10.3382	3.0201	54472.38	0.5	0	0	0
W610X241	468.05	104.96	10.4018	3.043	63061.44	0.5	0	0	0
W610X262	510.77	114.72	10.4652	3.0389	68364.74	0.5	0	0	0
W610X285	558.98	125.71	10.5279	3.0635	77204.47	0.5	0	0	0
W610X307	605.97	137.3	10.597	3.087	83630.33	0.5	0	0	0
W610X341	677.36	153.78	10.657	3.111	95474.22	0.5	0	0	0
W610X372	744.49	170.87	10.744	3.1373	107633.01	0.5	0	0	0
W610X415	836.03	192.84	10.8363	3.1732	124237.34	0.5	0	0	0
W610X455	921.46	214.19	10.9146	3.202	140191.19	0.5	0	0	0
W610X498	1019.1	237.99	11.0008	3.2385	159559.84	0.5	0	0	0
W610X551	1128.94	267.28	11.0919	3.2633	186181.7	0.5	0	0	0
W610X82-1	134.25	13.3	9.1084	1.3365	3849	0.5079	0	0	0
W610X92-1	153.17	15.68	9.2439	1.3812	4599.53	0.5	0	0	0
W690X125-1	244.09	33.2	10.7706	2.0734	18146.69	0.5984	0	0	0
W690X140-1	278.27	38.81	10.8824	2.1197	21194.17	0.5945	0	0	0
W690X152-1	305.12	43.39	10.9838	2.1508	23861.16	0.5984	0	0	0
W690X170-1	342.95	49.31	11.0195	2.1745	27852.82	0.6024	0	0	0
W690X192-1	394.82	57.61	11.2151	2.2059	32140.38	0.6024	0	0	0
W690X217	463.78	97.64	11.4504	3.1972	78215.36	0.7835	0	0	0
W690X240	515.04	109.23	11.5233	3.2328	87034.45	0.7913	0	0	0
W690X265	569.96	122.05	11.5547	3.2499	98192.62	0.7913	0	0	0
W690X289	628.54	136.08	11.7359	3.2965	110060.93	0.7913	0	0	0
W690X323	713.98	153.78	11.8142	3.3201	126495.4	0.7913	0	0	0
W690X350	775	167.82	11.8227	3.3274	141362.59	0.7913	0	0	0
W690X384	854.33	186.73	11.9188	3.3618	159930.4	0.7874	0	0	0
W690X419	933.66	206.26	11.9643	3.3883	180050.94	0.7913	0	0	0
W690X457	1031.3	227.01	12.0477	3.4115	197312.37	0.7913	0	0	0
W690X500	1128.94	252.03	12.1347	3.4484	227434.03	0.7913	0	0	0
W690X548	1238.78	278.88	12.1904	3.4665	255003.4	0.7913	0	0	0

Name	Z33 in3	Z22 in3	R33 in	R22 in	Cw in6	Fillet Radius in	CG Offset 3 in	CG Offset 2 in	PNA Offset 3 in
W690X802	1891.74	436.93	12.6894	3.6349	444906.34	0.7913	0	0	0
W760X134-1	283.15	34.72	11.6947	2.0898	23808.4	0.6496	0	0	0
W760X147-1	311.83	38.63	11.73	2.1019	27271.19	0.6496	0	0	0
W760X161-1	346	43.88	11.8589	2.1441	31019.67	0.6496	0	0	0
W760X173-1	377.74	49.19	11.9908	2.1887	34970.9	0.6496	0	0	0
W760X185-1	408.25	54.01	12.1279	2.2286	38519.21	0.6496	0	0	0
W760X196-1	436.93	58.4	12.1984	2.2493	41591.06	0.6496	0	0	0
W760X220	499.78	67.74	12.3833	2.2831	49838.39	0.6496	0	0	0
W760X257	607.19	123.27	12.7314	3.4303	129482.07	0.7795	0	0	0
W760X284	677.36	137.91	12.8059	3.4625	145699.88	0.7795	0	0	0
W760X314	750.59	155	12.8612	3.485	166219.86	0.7795	0	0	0
W760X350	848.23	175.14	12.995	3.5135	191770.64	0.7913	0	0	0
W760X389	945.87	195.89	13.0373	3.5276	216593.71	0.7913	0	0	0
W760X434	1061.81	222.74	13.1588	3.5765	251878.25	0.7913	0	0	0
W760X484	1189.96	252.03	13.23	3.5946	287177.59	0.7874	0	0	0
W760X531	1318.11	278.88	13.3463	3.6409	325254.08	0.7913	0	0	0
W760X582	1452.37	310	13.4189	3.6707	364389.43	0.7874	0	0	0
W840X176-1	414.96	51.32	13.047	2.3202	48500.88	0.7008	0	0	0
W840X193-1	466.83	59.5	13.2318	2.3857	56270.48	0.7047	0	0	0
W840X210-1	513.82	67.13	13.39	2.4288	63627.62	0.7008	0	0	0
W840X226-1	558.98	73.84	13.4805	2.4684	72772.05	0.7008	0	0	0
W840X251	628.54	84.21	13.7128	2.5036	82102.69	0.7008	0	0	0
W840X299	775	147.07	14.0177	3.5627	196738.31	0.7913	0	0	0
W840X329	854.33	164.15	14.0609	3.5897	223196.8	0.7795	0	0	0
W840X359	939.77	181.85	14.1271	3.6197	252980.79	0.7874	0	0	0
W840X392	1037.4	201.99	14.3399	3.6674	279067.28	0.7874	0	0	0
W840X433	1159.45	225.79	14.3857	3.6827	316998.38	0.7913	0	0	0
W840X473	1269.29	250.2	14.4234	3.7092	356735.66	0.7913	0	0	0
W840X527	1421.85	281.93	14.5463	3.7476	408138.86	0.7913	0	0	0
W840X576	1562.21	311.83	14.5943	3.7701	457129.75	0.7913	0	0	0
W920X201	508.94	59.68	14.0004	2.3772	69133.46	0.748	0	0	0
W920X223-1	580.95	70.79	14.275	2.4637	82986.53	0.748	0	0	0
W920X238-1	622.44	77.5	14.4115	2.5084	89931.01	0.752	0	0	0
W920X253-1	665.16	83.6	14.4812	2.5263	97531.09	0.752	0	0	0
W920X271-1	720.08	90.93	14.5103	2.5399	107164.9	0.748	0	0	0
W920X289	768.9	97.64	14.5699	2.5633	115079.41	0.752	0	0	0
W920X313	836.03	106.79	14.6038	2.5774	128457.52	0.752	0	0	0
W920X344	964.18	175.75	15.1204	3.7113	292566.49	0.9488	0	0	0
W920X345	933.66	122.05	14.8432	2.6239	146098.64	0.748	0	0	0
W920X368	1031.3	189.78	15.1718	3.7296	315489.64	0.9488	0	0	0
W920X381	1037.4	137.3	14.9309	2.6489	166973.06	0.752	0	0	0
W920X390	1098.43	203.82	15.2275	3.7591	345999.82	0.9488	0	0	0
W920X420	1189.96	222.74	15.3757	3.8022	378622.2	0.9488	0	0	0
W920X449	1281.5	241.04	15.3978	3.8222	412987.56	0.9488	0	0	0
W920X491	1409.65	264.84	15.51	3.8284	454839.76	0.9488	0	0	0
W920X537	1550	292.91	15.5715	3.8468	504780.17	0.9488	0	0	0
W920X588	1708.66	325.26	15.7032	3.884	570436.65	0.9488	0	0	0
W920X656	1910.04	367.97	15.7339	3.9111	665045.5	0.9488	0	0	0
W920X725	2129.73	411.91	15.8712	3.9668	746922.01	0.9488	0	0	0
W920X787	2331.11	454.02	15.9128	3.9951	838532.11	0.9488	0	0	0
W920X970	2910.83	580.95	16.2404	4.0927	1124624.02	0.9488	0	0	0

Table 2.2 - Frame Section Property Definitions - Summary (Part 4 of 4)

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
Column								
HSS101.6X101.6X12.7	0			1	1	1	1	1
HSS101.6X101.6X3.2	0			1	1	1	1	1
HSS101.6X101.6X4.8	0			1	1	1	1	1
HSS101.6X101.6X6.4	0			1	1	1	1	1
HSS101.6X101.6X7.9	0			1	1	1	1	1
HSS101.6X101.6X9.5	0			1	1	1	1	1
HSS101.6X50.8X3.2	0			1	1	1	1	1
HSS101.6X50.8X4.8	0			1	1	1	1	1
HSS101.6X50.8X6.4	0			1	1	1	1	1
HSS101.6X50.8X7.9	0			1	1	1	1	1
HSS101.6X50.8X9.5	0			1	1	1	1	1
HSS101.6X63.5X3.2	0			1	1	1	1	1
HSS101.6X63.5X4.8	0			1	1	1	1	1
HSS101.6X63.5X6.4	0			1	1	1	1	1
HSS101.6X63.5X7.9	0			1	1	1	1	1
HSS101.6X63.5X9.5	0			1	1	1	1	1
HSS101.6X76.2X3.2	0			1	1	1	1	1
HSS101.6X76.2X4.8	0			1	1	1	1	1
HSS101.6X76.2X6.4	0			1	1	1	1	1
HSS101.6X76.2X7.9	0			1	1	1	1	1
HSS101.6X76.2X9.5	0			1	1	1	1	1
HSS114.3X114.3X12.7	0			1	1	1	1	1
HSS114.3X114.3X3.2	0			1	1	1	1	1
HSS114.3X114.3X4.8	0			1	1	1	1	1
HSS114.3X114.3X6.4	0			1	1	1	1	1
HSS114.3X114.3X7.9	0			1	1	1	1	1
HSS114.3X114.3X9.5	0			1	1	1	1	1
HSS127X101.6X12.7	0			1	1	1	1	1
HSS127X101.6X3.2	0			1	1	1	1	1
HSS127X101.6X4.8	0			1	1	1	1	1
HSS127X101.6X6.4	0			1	1	1	1	1
HSS127X101.6X7.9	0			1	1	1	1	1
HSS127X101.6X9.5	0			1	1	1	1	1
HSS127X127X12.7	0			1	1	1	1	1
HSS127X127X3.2	0			1	1	1	1	1
HSS127X127X4.8	0			1	1	1	1	1
HSS127X127X6.4	0			1	1	1	1	1
HSS127X127X7.9	0			1	1	1	1	1
HSS127X127X9.5	0			1	1	1	1	1
HSS127X50.8X3.2	0			1	1	1	1	1
HSS127X50.8X4.8	0			1	1	1	1	1
HSS127X50.8X6.4	0			1	1	1	1	1
HSS127X50.8X7.9	0			1	1	1	1	1
HSS127X50.8X9.5	0			1	1	1	1	1
HSS127X63.5X3.2	0			1	1	1	1	1
HSS127X63.5X4.8	0			1	1	1	1	1
HSS127X63.5X6.4	0			1	1	1	1	1
HSS127X76.2X12.7	0			1	1	1	1	1
HSS127X76.2X3.2	0			1	1	1	1	1
HSS127X76.2X4.8	0			1	1	1	1	1
HSS127X76.2X6.4	0			1	1	1	1	1
HSS127X76.2X7.9	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
HSS127X76.2X9.5	0			1	1	1	1	1
HSS139.7X139.7X3.2	0			1	1	1	1	1
HSS139.7X139.7X4.8	0			1	1	1	1	1
HSS139.7X139.7X6.4	0			1	1	1	1	1
HSS139.7X139.7X7.9	0			1	1	1	1	1
HSS139.7X139.7X9.5	0			1	1	1	1	1
HSS152.4X101.6X12.7	0			1	1	1	1	1
HSS152.4X101.6X3.2	0			1	1	1	1	1
HSS152.4X101.6X4.8	0			1	1	1	1	1
HSS152.4X101.6X6.4	0			1	1	1	1	1
HSS152.4X101.6X7.9	0			1	1	1	1	1
HSS152.4X101.6X9.5	0			1	1	1	1	1
HSS152.4X127X12.7	0			1	1	1	1	1
HSS152.4X127X3.2	0			1	1	1	1	1
HSS152.4X127X4.8	0			1	1	1	1	1
HSS152.4X127X6.4	0			1	1	1	1	1
HSS152.4X127X7.9	0			1	1	1	1	1
HSS152.4X127X9.5	0			1	1	1	1	1
HSS152.4X152.4X12.7	0			1	1	1	1	1
HSS152.4X152.4X15.9	0			1	1	1	1	1
HSS152.4X152.4X3.2	0			1	1	1	1	1
HSS152.4X152.4X4.8	0			1	1	1	1	1
HSS152.4X152.4X6.4	0			1	1	1	1	1
HSS152.4X152.4X7.9	0			1	1	1	1	1
HSS152.4X152.4X9.5	0.0002			1	1	1	1	1
HSS152.4X50.8X3.2	0			1	1	1	1	1
HSS152.4X50.8X4.8	0			1	1	1	1	1
HSS152.4X50.8X6.4	0			1	1	1	1	1
HSS152.4X50.8X7.9	0			1	1	1	1	1
HSS152.4X50.8X9.5	0			1	1	1	1	1
HSS152.4X76.2X12.7	0			1	1	1	1	1
HSS152.4X76.2X3.2	0			1	1	1	1	1
HSS152.4X76.2X4.8	0			1	1	1	1	1
HSS152.4X76.2X6.4	0			1	1	1	1	1
HSS152.4X76.2X7.9	0			1	1	1	1	1
HSS152.4X76.2X9.5	2.84E-06			1	1	1	1	1
HSS177.8X101.6X12.7	2.325E-06			1	1	1	1	1
HSS177.8X101.6X3.2	0			1	1	1	1	1
HSS177.8X101.6X4.8	0			1	1	1	1	1
HSS177.8X101.6X6.4	1.551E-06			1	1	1	1	1
HSS177.8X101.6X7.9	0			1	1	1	1	1
HSS177.8X101.6X9.5	0			1	1	1	1	1
HSS177.8X127X12.7	0			1	1	1	1	1
HSS177.8X127X3.2	0			1	1	1	1	1
HSS177.8X127X4.8	0			1	1	1	1	1
HSS177.8X127X6.4	0			1	1	1	1	1
HSS177.8X127X7.9	0			1	1	1	1	1
HSS177.8X127X9.5	0			1	1	1	1	1
HSS177.8X177.8X12.7	0			1	1	1	1	1
HSS177.8X177.8X15.9	0			1	1	1	1	1
HSS177.8X177.8X3.2	0			1	1	1	1	1
HSS177.8X177.8X4.8	0			1	1	1	1	1
HSS177.8X177.8X6.4	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
HSS177.8X177.8X7.9	0			1	1	1	1	1
HSS177.8X177.8X9.5	0			1	1	1	1	1
HSS177.8X50.8X3.2	0			1	1	1	1	1
HSS177.8X50.8X4.8	0			1	1	1	1	1
HSS177.8X50.8X6.4	0			1	1	1	1	1
HSS177.8X76.2X12.7	0			1	1	1	1	1
HSS177.8X76.2X3.2	0			1	1	1	1	1
HSS177.8X76.2X4.8	0			1	1	1	1	1
HSS177.8X76.2X6.4	0			1	1	1	1	1
HSS177.8X76.2X7.9	0			1	1	1	1	1
HSS177.8X76.2X9.5	0			1	1	1	1	1
HSS203.2X101.6X12.7	3.756E-06			1	1	1	1	1
HSS203.2X101.6X15.9	-0.0001			1	1	1	1	1
HSS203.2X101.6X3.2	0			1	1	1	1	1
HSS203.2X101.6X4.8	0			1	1	1	1	1
HSS203.2X101.6X6.4	-0.0002			1	1	1	1	1
HSS203.2X101.6X7.9	2.838E-06			1	1	1	1	1
HSS203.2X101.6X9.5	0			1	1	1	1	1
HSS203.2X152.4X12.7	0			1	1	1	1	1
HSS203.2X152.4X15.9	0			1	1	1	1	1
HSS203.2X152.4X4.8	0			1	1	1	1	1
HSS203.2X152.4X6.4	0			1	1	1	1	1
HSS203.2X152.4X7.9	0			1	1	1	1	1
HSS203.2X152.4X9.5	0			1	1	1	1	1
HSS203.2X203.2X12.7	0			1	1	1	1	1
HSS203.2X203.2X15.9	0			1	1	1	1	1
HSS203.2X203.2X3.2	0			1	1	1	1	1
HSS203.2X203.2X4.8	0			1	1	1	1	1
HSS203.2X203.2X6.4	0			1	1	1	1	1
HSS203.2X203.2X7.9	0			1	1	1	1	1
HSS203.2X203.2X9.5	0			1	1	1	1	1
HSS203.2X50.8X3.2	0			1	1	1	1	1
HSS203.2X50.8X4.8	0			1	1	1	1	1
HSS203.2X50.8X6.4	1.508E-05			1	1	1	1	1
HSS203.2X50.8X7.9	0			1	1	1	1	1
HSS203.2X50.8X9.5	-2.918E-05			1	1	1	1	1
HSS203.2X76.2X12.7	0			1	1	1	1	1
HSS203.2X76.2X3.2	0			1	1	1	1	1
HSS203.2X76.2X4.8	-0.0001			1	1	1	1	1
HSS203.2X76.2X6.4	0			1	1	1	1	1
HSS203.2X76.2X7.9	0			1	1	1	1	1
HSS203.2X76.2X9.5	7.35E-06			1	1	1	1	1
HSS228.6X127X12.7	0			1	1	1	1	1
HSS228.6X127X15.9	0			1	1	1	1	1
HSS228.6X127X4.8	1.886E-06			1	1	1	1	1
HSS228.6X127X6.4	0			1	1	1	1	1
HSS228.6X127X7.9	0			1	1	1	1	1
HSS228.6X127X9.5	0			1	1	1	1	1
HSS228.6X177.8X12.7	0			1	1	1	1	1
HSS228.6X177.8X15.9	0			1	1	1	1	1
HSS228.6X177.8X4.8	0			1	1	1	1	1
HSS228.6X177.8X6.4	0			1	1	1	1	1
HSS228.6X177.8X7.9	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
HSS228.6X177.8X9.5	0			1	1	1	1	1
HSS228.6X228.6X12.7	0			1	1	1	1	1
HSS228.6X228.6X15.9	0			1	1	1	1	1
HSS228.6X228.6X3.2	0			1	1	1	1	1
HSS228.6X228.6X6.4	0			1	1	1	1	1
HSS228.6X228.6X7.9	0			1	1	1	1	1
HSS228.6X228.6X74.8	0			1	1	1	1	1
HSS228.6X228.6X9.5	0			1	1	1	1	1
HSS228.6X76.2X12.7	0			1	1	1	1	1
HSS228.6X76.2X4.8	0			1	1	1	1	1
HSS228.6X76.2X6.4	0			1	1	1	1	1
HSS228.6X76.2X7.9	0			1	1	1	1	1
HSS228.6X76.2X9.5	0			1	1	1	1	1
HSS254X101.6X12.7	0			1	1	1	1	1
HSS254X101.6X15.9	0			1	1	1	1	1
HSS254X101.6X3.2	0			1	1	1	1	1
HSS254X101.6X4.8	0			1	1	1	1	1
HSS254X101.6X6.4	0			1	1	1	1	1
HSS254X101.6X7.9	0			1	1	1	1	1
HSS254X101.6X9.5	0			1	1	1	1	1
HSS254X127X4.8	0			1	1	1	1	1
HSS254X127X6.4	0			1	1	1	1	1
HSS254X127X7.9	0			1	1	1	1	1
HSS254X127X9.5	0			1	1	1	1	1
HSS254X152.4X12.7	0			1	1	1	1	1
HSS254X152.4X15.9	2.568E-06			1	1	1	1	1
HSS254X152.4X4.8	0			1	1	1	1	1
HSS254X152.4X6.4	0			1	1	1	1	1
HSS254X152.4X7.9	0			1	1	1	1	1
HSS254X152.4X9.5	0			1	1	1	1	1
HSS254X203.2X12.7	0			1	1	1	1	1
HSS254X203.2X15.9	0			1	1	1	1	1
HSS254X203.2X4.8	0			1	1	1	1	1
HSS254X203.2X6.4	0			1	1	1	1	1
HSS254X203.2X7.9	0			1	1	1	1	1
HSS254X203.2X9.5	0			1	1	1	1	1
HSS254X254X12.7	0			1	1	1	1	1
HSS254X254X15.9	0			1	1	1	1	1
HSS254X254X4.8	0			1	1	1	1	1
HSS254X254X6.4	0			1	1	1	1	1
HSS254X254X7.9	0			1	1	1	1	1
HSS254X254X9.5	0			1	1	1	1	1
HSS254X50.8X3.2	0			1	1	1	1	1
HSS254X50.8X4.8	0			1	1	1	1	1
HSS254X50.8X6.4	0			1	1	1	1	1
HSS254X50.8X7.9	0			1	1	1	1	1
HSS254X50.8X9.5	0			1	1	1	1	1
HSS254X76.2X3.2	0			1	1	1	1	1
HSS254X76.2X4.8	0			1	1	1	1	1
HSS254X76.2X6.4	0			1	1	1	1	1
HSS254X76.2X7.9	0			1	1	1	1	1
HSS254X76.2X9.5	0			1	1	1	1	1
HSS254X88.9X12.7	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
HSS254X88.9X3.2	0			1	1	1	1	1
HSS254X88.9X4.8	0			1	1	1	1	1
HSS254X88.9X6.4	0			1	1	1	1	1
HSS254X88.9X7.9	0			1	1	1	1	1
HSS254X88.9X9.5	0			1	1	1	1	1
HSS304.8X101.6X12.7	0			1	1	1	1	1
HSS304.8X101.6X15.9	0			1	1	1	1	1
HSS304.8X101.6X4.8	0			1	1	1	1	1
HSS304.8X101.6X6.4	0			1	1	1	1	1
HSS304.8X101.6X7.9	0			1	1	1	1	1
HSS304.8X101.6X9.5	0			1	1	1	1	1
HSS304.8X152.4X12.7	0			1	1	1	1	1
HSS304.8X152.4X15.9	0			1	1	1	1	1
HSS304.8X152.4X4.8	0			1	1	1	1	1
HSS304.8X152.4X6.4	0			1	1	1	1	1
HSS304.8X152.4X7.9	0			1	1	1	1	1
HSS304.8X152.4X9.5	0			1	1	1	1	1
HSS304.8X203.2X12.7	0			1	1	1	1	1
HSS304.8X203.2X15.9	0			1	1	1	1	1
HSS304.8X203.2X4.8	0			1	1	1	1	1
HSS304.8X203.2X6.4	0			1	1	1	1	1
HSS304.8X203.2X7.9	0			1	1	1	1	1
HSS304.8X203.2X9.5	0			1	1	1	1	1
HSS304.8X254X12.7	0			1	1	1	1	1
HSS304.8X254X6.4	0			1	1	1	1	1
HSS304.8X254X7.9	0			1	1	1	1	1
HSS304.8X254X9.5	0			1	1	1	1	1
HSS304.8X304.8X12.7	0			1	1	1	1	1
HSS304.8X304.8X15.9	0			1	1	1	1	1
HSS304.8X304.8X4.8	0			1	1	1	1	1
HSS304.8X304.8X6.4	0			1	1	1	1	1
HSS304.8X304.8X7.9	0			1	1	1	1	1
HSS304.8X304.8X9.5	0			1	1	1	1	1
HSS304.8X50.8X4.8	0			1	1	1	1	1
HSS304.8X50.8X6.4	0			1	1	1	1	1
HSS304.8X50.8X7.9	0			1	1	1	1	1
HSS304.8X76.2X4.8	0			1	1	1	1	1
HSS304.8X76.2X6.4	0			1	1	1	1	1
HSS304.8X76.2X7.9	0			1	1	1	1	1
HSS304.8X88.9X7.9	0			1	1	1	1	1
HSS304.8X88.9X9.5	0			1	1	1	1	1
HSS355.6X101.6X12.7	0			1	1	1	1	1
HSS355.6X101.6X15.9	0			1	1	1	1	1
HSS355.6X101.6X4.8	0			1	1	1	1	1
HSS355.6X101.6X6.4	0			1	1	1	1	1
HSS355.6X101.6X7.9	0			1	1	1	1	1
HSS355.6X101.6X9.5	0			1	1	1	1	1
HSS355.6X152.4X12.7	0			1	1	1	1	1
HSS355.6X152.4X15.9	0			1	1	1	1	1
HSS355.6X152.4X4.8	0			1	1	1	1	1
HSS355.6X152.4X6.4	0			1	1	1	1	1
HSS355.6X152.4X7.9	0			1	1	1	1	1
HSS355.6X152.4X9.5	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
HSS355.6X254X12.7	0			1	1	1	1	1
HSS355.6X254X15.9	0			1	1	1	1	1
HSS355.6X254X6.4	0			1	1	1	1	1
HSS355.6X254X7.9	0			1	1	1	1	1
HSS355.6X254X9.5	0			1	1	1	1	1
HSS355.6X355.6X12.7	0.0003			1	1	1	1	1
HSS355.6X355.6X15.9	0			1	1	1	1	1
HSS355.6X355.6X7.9	0			1	1	1	1	1
HSS355.6X355.6X9.5	0			1	1	1	1	1
HSS406.4X101.6X12.7	0			1	1	1	1	1
HSS406.4X101.6X15.9	0			1	1	1	1	1
HSS406.4X101.6X4.8	2.26E-05			1	1	1	1	1
HSS406.4X101.6X6.4	2.39E-05			1	1	1	1	1
HSS406.4X101.6X7.9	0			1	1	1	1	1
HSS406.4X101.6X9.5	0			1	1	1	1	1
HSS406.4X203.2X12.7	0			1	1	1	1	1
HSS406.4X203.2X15.9	0			1	1	1	1	1
HSS406.4X203.2X6.4	0			1	1	1	1	1
HSS406.4X203.2X7.9	0			1	1	1	1	1
HSS406.4X203.2X9.5	0			1	1	1	1	1
HSS406.4X304.8X12.7	0			1	1	1	1	1
HSS406.4X304.8X15.9	0			1	1	1	1	1
HSS406.4X304.8X7.9	0			1	1	1	1	1
HSS406.4X304.8X9.5	0			1	1	1	1	1
HSS406.4X406.4X12.7	0			1	1	1	1	1
HSS406.4X406.4X15.9	0			1	1	1	1	1
HSS406.4X406.4X7.9	0			1	1	1	1	1
HSS406.4X406.4X9.5	0			1	1	1	1	1
HSS457.2X152.4X12.7	0			1	1	1	1	1
HSS457.2X152.4X15.9	0			1	1	1	1	1
HSS457.2X152.4X6.4	0			1	1	1	1	1
HSS457.2X152.4X7.9	0			1	1	1	1	1
HSS457.2X152.4X9.5	0			1	1	1	1	1
HSS50.8X25.4X3.2	0			1	1	1	1	1
HSS50.8X25.4X4.8	0			1	1	1	1	1
HSS50.8X38.1X3.2	0			1	1	1	1	1
HSS50.8X38.1X4.8	0			1	1	1	1	1
HSS50.8X50.8X3.2	0.0001			1	1	1	1	1
HSS50.8X50.8X4.8	0			1	1	1	1	1
HSS50.8X50.8X6.4	0			1	1	1	1	1
HSS508X101.6X12.7	0			1	1	1	1	1
HSS508X101.6X6.4	0			1	1	1	1	1
HSS508X101.6X7.9	0			1	1	1	1	1
HSS508X101.6X9.5	0			1	1	1	1	1
HSS508X203.2X12.7	0			1	1	1	1	1
HSS508X203.2X15.9	0			1	1	1	1	1
HSS508X203.2X7.9	0			1	1	1	1	1
HSS508X203.2X9.5	0			1	1	1	1	1
HSS508X304.8X12.7	0			1	1	1	1	1
HSS508X304.8X15.9	0			1	1	1	1	1
HSS508X304.8X7.9	0			1	1	1	1	1
HSS508X304.8X9.5	0			1	1	1	1	1
HSS57.2X50.8X3.2	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
HSS57.2X50.8X4.8	0			1	1	1	1	1
HSS57.2X57.2X3.2	0			1	1	1	1	1
HSS57.2X57.2X4.8	0			1	1	1	1	1
HSS57.2X57.2X6.4	0			1	1	1	1	1
HSS63.5X25.4X3.2	0			1	1	1	1	1
HSS63.5X25.4X4.8	0			1	1	1	1	1
HSS63.5X38.1X3.2	0			1	1	1	1	1
HSS63.5X38.1X4.8	0			1	1	1	1	1
HSS63.5X38.1X6.4	0			1	1	1	1	1
HSS63.5X50.8X3.2	0			1	1	1	1	1
HSS63.5X50.8X4.8	0			1	1	1	1	1
HSS63.5X50.8X6.4	0			1	1	1	1	1
HSS63.5X63.5X3.2	0			1	1	1	1	1
HSS63.5X63.5X4.8	0			1	1	1	1	1
HSS63.5X63.5X6.4	0			1	1	1	1	1
HSS63.5X63.5X7.9	0			1	1	1	1	1
HSS76.2X25.4X3.2	0			1	1	1	1	1
HSS76.2X25.4X4.8	0			1	1	1	1	1
HSS76.2X38.1X3.2	0			1	1	1	1	1
HSS76.2X38.1X4.8	0			1	1	1	1	1
HSS76.2X38.1X6.4	0			1	1	1	1	1
HSS76.2X50.8X3.2	0			1	1	1	1	1
HSS76.2X50.8X4.8	0			1	1	1	1	1
HSS76.2X50.8X6.4	0			1	1	1	1	1
HSS76.2X50.8X7.9	0			1	1	1	1	1
HSS76.2X63.5X3.2	0			1	1	1	1	1
HSS76.2X63.5X4.8	0			1	1	1	1	1
HSS76.2X63.5X6.4	0			1	1	1	1	1
HSS76.2X63.5X7.9	0			1	1	1	1	1
HSS76.2X76.2X3.2	0			1	1	1	1	1
HSS76.2X76.2X4.8	0			1	1	1	1	1
HSS76.2X76.2X6.4	0			1	1	1	1	1
HSS76.2X76.2X7.9	0			1	1	1	1	1
HSS76.2X76.2X9.5	0			1	1	1	1	1
HSS88.9X38.1X3.2	0			1	1	1	1	1
HSS88.9X38.1X4.8	0			1	1	1	1	1
HSS88.9X38.1X6.4	0			1	1	1	1	1
HSS88.9X50.8X3.2	0			1	1	1	1	1
HSS88.9X50.8X4.8	0			1	1	1	1	1
HSS88.9X50.8X6.4	0			1	1	1	1	1
HSS88.9X63.5X3.2	0			1	1	1	1	1
HSS88.9X63.5X4.8	0			1	1	1	1	1
HSS88.9X63.5X6.4	0			1	1	1	1	1
HSS88.9X63.5X7.9	0			1	1	1	1	1
HSS88.9X63.5X9.5	0			1	1	1	1	1
HSS88.9X88.9X3.2	0			1	1	1	1	1
HSS88.9X88.9X4.8	0			1	1	1	1	1
HSS88.9X88.9X6.4	0			1	1	1	1	1
HSS88.9X88.9X7.9	0			1	1	1	1	1
HSS88.9X88.9X9.5	0			1	1	1	1	1
L102X102X11.1	-1.5902	1.7646	-1.7646	1	1	1	1	1
L102X102X12.7	-1.5346	1.7253	-1.7253	1	1	1	1	1
L102X102X15.9	-1.4263	1.6434	-1.6434	1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
L102X102X19	-1.3218	1.5574	-1.5574	1	1	1	1	1
L102X102X6.4	-1.7613	1.8749	-1.8749	1	1	1	1	1
L102X102X7.9	-1.703	1.839	-1.839	1	1	1	1	1
L102X102X9.5	-1.6458	1.8021	-1.8021	1	1	1	1	1
L102X76X12.7	-1.2793	1.2325	-1.6763	1	1	1	1	1
L102X76X15.9	-1.2065	1.1597	-1.5661	1	1	1	1	1
L102X76X6.4	-1.4437	1.3706	-1.8632	1	1	1	1	1
L102X76X7.9	-1.3977	1.3369	-1.8204	1	1	1	1	1
L102X76X9.5	-1.3561	1.3025	-1.775	1	1	1	1	1
L102X89X12.7	-1.6533	1.4756	-1.7076	1	1	1	1	1
L102X89X6.4	-1.6486	1.619	-1.8706	1	1	1	1	1
L102X89X7.9	-1.6043	1.5843	-1.8323	1	1	1	1	1
L102X89X9.5	-1.7368	1.5487	-1.7923	1	1	1	1	1
L127X127X11.1	-2.0758	2.2617	-2.2617	1	1	1	1	1
L127X127X12.7	-2.0187	2.224	-2.224	1	1	1	1	1
L127X127X15.9	-1.9068	2.1459	-2.1459	1	1	1	1	1
L127X127X19	-1.7983	2.0644	-2.0644	1	1	1	1	1
L127X127X22.2	-1.6959	1.9822	-1.9822	1	1	1	1	1
L127X127X7.9	-2.1909	2.3337	-2.3337	1	1	1	1	1
L127X127X9.5	-2.1327	2.2979	-2.2979	1	1	1	1	1
L127X76X11.1	-1.3361	1.2748	-2.1842	1	1	1	1	1
L127X76X12.7	-1.2977	1.2412	-2.1217	1	1	1	1	1
L127X76X6.4	-1.4713	1.3728	-2.3441	1	1	1	1	1
L127X76X7.9	-1.4208	1.3402	-2.2949	1	1	1	1	1
L127X76X9.5	-1.3768	1.3074	-2.2413	1	1	1	1	1
L127X89X12.7	-1.5477	1.4878	-2.1706	1	1	1	1	1
L127X89X15.9	-1.4742	1.4179	-2.0612	1	1	1	1	1
L127X89X19	-1.4042	1.3466	-1.9414	1	1	1	1	1
L127X89X6.4	-1.7214	1.6219	-2.3558	1	1	1	1	1
L127X89X7.9	-1.6709	1.5889	-2.3134	1	1	1	1	1
L127X89X9.5	-1.6268	1.5555	-2.2684	1	1	1	1	1
L152X102X11.1	-1.8607	1.7822	-2.7168	1	1	1	1	1
L152X102X12.7	-1.8202	1.7485	-2.6672	1	1	1	1	1
L152X102X14.3	-1.7809	1.7145	-2.615	1	1	1	1	1
L152X102X15.9	-1.7455	1.6803	-2.5602	1	1	1	1	1
L152X102X19	-1.6733	1.6109	-2.4431	1	1	1	1	1
L152X102X22.2	-1.6076	1.5427	-2.3204	1	1	1	1	1
L152X102X7.9	-1.9523	1.8479	-2.8072	1	1	1	1	1
L152X102X9.5	-1.903	1.815	-2.763	1	1	1	1	1
L152X152X11.1	-2.5659	2.7572	-2.7572	1	1	1	1	1
L152X152X12.7	-2.5078	2.7205	-2.7205	1	1	1	1	1
L152X152X14.3	-2.4505	2.6831	-2.6831	1	1	1	1	1
L152X152X15.9	-2.3936	2.645	-2.645	1	1	1	1	1
L152X152X19	-2.2822	2.5666	-2.5666	1	1	1	1	1
L152X152X22.2	-2.1765	2.4879	-2.4879	1	1	1	1	1
L152X152X25.4	-2.0705	2.4039	-2.4039	1	1	1	1	1
L152X152X7.9	-2.6824	2.8275	-2.8275	1	1	1	1	1
L152X152X9.5	-2.6235	2.7924	-2.7924	1	1	1	1	1
L152X89X12.7	-1.5625	1.493	-2.6272	1	1	1	1	1
L152X89X7.9	-1.6945	1.5909	-2.7921	1	1	1	1	1
L152X89X9.5	-1.6452	1.5584	-2.7409	1	1	1	1	1
L178X102X11.1	-1.8608	1.785	-3.2069	1	1	1	1	1
L178X102X12.7	-1.8203	1.7521	-3.1503	1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
L178X102X15.9	-1.7456	1.686	-3.0265	1	1	1	1	1
L178X102X19	-1.6734	1.6191	-2.8887	1	1	1	1	1
L178X102X9.5	-1.9031	1.817	-3.259	1	1	1	1	1
L203X102X11.1	-1.8609	1.7864	-3.675	1	1	1	1	1
L203X102X12.7	-1.8204	1.7541	-3.6107	1	1	1	1	1
L203X102X14.3	-1.781	1.7216	-3.5417	1	1	1	1	1
L203X102X15.9	-1.7456	1.689	-3.4682	1	1	1	1	1
L203X102X19	-1.6734	1.6235	-3.3079	1	1	1	1	1
L203X102X22.2	-1.6077	1.5596	-3.1366	1	1	1	1	1
L203X102X25.4	-1.5391	1.4933	-2.9452	1	1	1	1	1
L203X152X11.1	-2.8451	2.7669	-3.7477	1	1	1	1	1
L203X152X12.7	-2.8046	2.7333	-3.7067	1	1	1	1	1
L203X152X14.3	-2.7653	2.6994	-3.6644	1	1	1	1	1
L203X152X15.9	-2.7298	2.6653	-3.6208	1	1	1	1	1
L203X152X19	-2.6577	2.5962	-3.5293	1	1	1	1	1
L203X152X22.2	-2.592	2.5281	-3.4355	1	1	1	1	1
L203X152X25.4	-2.5233	2.4568	-3.3333	1	1	1	1	1
L203X203X12.7	-3.5055	3.73	-3.73	1	1	1	1	1
L203X203X14.3	-3.4467	3.6941	-3.6941	1	1	1	1	1
L203X203X15.9	-3.3885	3.6577	-3.6577	1	1	1	1	1
L203X203X19	-3.2733	3.5833	-3.5833	1	1	1	1	1
L203X203X22.2	-3.1637	3.5092	-3.5092	1	1	1	1	1
L203X203X25.4	-3.0524	3.4305	-3.4305	1	1	1	1	1
L203X203X28.6	-2.9433	3.3497	-3.3497	1	1	1	1	1
L51X51X3.2	-0.8748	0.9334	-0.9334	1	1	1	1	1
L51X51X4.8	-0.8175	0.8972	-0.8972	1	1	1	1	1
L51X51X6.4	-0.7617	0.8586	-0.8586	1	1	1	1	1
L51X51X7.9	-0.7079	0.8179	-0.8179	1	1	1	1	1
L51X51X9.5	-0.6561	0.7752	-0.7752	1	1	1	1	1
L64X38X4.8	-0.6978	0.6538	-1.1208	1	1	1	1	1
L64X38X6.4	-0.6562	0.6206	-1.0609	1	1	1	1	1
L64X51X4.8	-0.9369	0.9017	-1.1418	1	1	1	1	1
L64X51X6.4	-0.8945	0.8669	-1.0988	1	1	1	1	1
L64X51X7.9	-0.8546	0.831	-1.0524	1	1	1	1	1
L64X51X9.5	-0.8158	0.7941	-1.0025	1	1	1	1	1
L64X64X12.7	-0.7968	0.9471	-0.9471	1	1	1	1	1
L64X64X4.8	-1.0665	1.149	-1.149	1	1	1	1	1
L64X64X6.4	-1.0094	1.112	-1.112	1	1	1	1	1
L64X64X7.9	-0.9538	1.0732	-1.0732	1	1	1	1	1
L64X64X9.5	-0.8998	1.0327	-1.0327	1	1	1	1	1
L76X51X12.7	-0.7727	0.732	-1.0937	1	1	1	1	1
L76X51X4.8	-0.9715	0.9037	-1.3852	1	1	1	1	1
L76X51X6.4	-0.924	0.8704	-1.3369	1	1	1	1	1
L76X51X7.9	-0.8829	0.8365	-1.2833	1	1	1	1	1
L76X51X9.5	-0.8448	0.8021	-1.2247	1	1	1	1	1
L76X64X11.1	-1.02	1.0089	-1.222	1	1	1	1	1
L76X64X12.7	-0.979	0.9704	-1.1717	1	1	1	1	1
L76X64X4.8	-1.1939	1.1522	-1.3957	1	1	1	1	1
L76X64X6.4	-1.1458	1.1176	-1.356	1	1	1	1	1
L76X64X7.9	-1.1018	1.0821	-1.3137	1	1	1	1	1
L76X64X9.5	-1.06	1.0457	-1.2688	1	1	1	1	1
L76X76X11.1	-1.0889	1.248	-1.248	1	1	1	1	1
L76X76X12.7	-1.1718	1.206	-1.206	1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
L76X76X4.8	-1.3685	1.4003	-1.4003	1	1	1	1	1
L76X76X6.4	-1.3274	1.3642	-1.3642	1	1	1	1	1
L76X76X7.9	-1.1978	1.3267	-1.3267	1	1	1	1	1
L76X76X9.5	-1.2478	1.2878	-1.2878	1	1	1	1	1
L89X64X12.7	-1.0287	0.9819	-1.3918	1	1	1	1	1
L89X64X6.4	-1.1931	1.1204	-1.5988	1	1	1	1	1
L89X64X7.9	-1.1472	1.0866	-1.5523	1	1	1	1	1
L89X64X9.5	-1.1056	1.0521	-1.5022	1	1	1	1	1
L89X76X11.1	-1.2699	1.2609	-1.4851	1	1	1	1	1
L89X76X12.7	-1.3956	1.223	-1.4389	1	1	1	1	1
L89X76X6.4	-1.5645	1.3683	-1.6102	1	1	1	1	1
L89X76X7.9	-1.3519	1.3332	-1.5703	1	1	1	1	1
L89X76X9.5	-1.4782	1.2972	-1.5284	1	1	1	1	1
L89X89X11.1	-1.3352	1.5029	-1.5029	1	1	1	1	1
L89X89X12.7	-1.4185	1.4624	-1.4624	1	1	1	1	1
L89X89X6.4	-1.577	1.6158	-1.6158	1	1	1	1	1
L89X89X7.9	-1.4463	1.5792	-1.5792	1	1	1	1	1
L89X89X9.5	-1.4962	1.5414	-1.5414	1	1	1	1	1
Rufter								
Truss								
W1000X222	0			1	1	1	1	1
W1000X249	0			1	1	1	1	1
W1000X272	0			1	1	1	1	1
W1000X296	0			1	1	1	1	1
W1000X314	0			1	1	1	1	1
W1000X321	0			1	1	1	1	1
W1000X350	0			1	1	1	1	1
W1000X371	0			1	1	1	1	1
W1000X393	0			1	1	1	1	1
W1000X412	0			1	1	1	1	1
W1000X415	0			1	1	1	1	1
W1000X438	0			1	1	1	1	1
W1000X443	0			1	1	1	1	1
W1000X483	0			1	1	1	1	1
W1000X486	0			1	1	1	1	1
W1000X494	0			1	1	1	1	1
W1000X539	0			1	1	1	1	1
W1000X554	0			1	1	1	1	1
W1000X584	0			1	1	1	1	1
W1000X591	0			1	1	1	1	1
W1000X642	0			1	1	1	1	1
W1000X748	0			1	1	1	1	1
W1000X883	0			1	1	1	1	1
W100X19.3	0			1	1	1	1	1
W1100X343	0			1	1	1	1	1
W1100X390	0			1	1	1	1	1
W1100X433	0			1	1	1	1	1
W1100X499	0			1	1	1	1	1
W130X23.8	0			1	1	1	1	1
W130X28.1	0			1	1	1	1	1
W150X13	0			1	1	1	1	1
W150X13.5	0			1	1	1	1	1
W150X18	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
W150X22.5	0			1	1	1	1	1
W150X24	0			1	1	1	1	1
W150X29.8	0			1	1	1	1	1
W150X37.1	0			1	1	1	1	1
W200X100	0			1	1	1	1	1
W200X15	0			1	1	1	1	1
W200X19.3	0			1	1	1	1	1
W200X22.5	0			1	1	1	1	1
W200X26.6	0			1	1	1	1	1
W200X31.3	0			1	1	1	1	1
W200X35.9	0			1	1	1	1	1
W200X41.7	0			1	1	1	1	1
W200X46.1	0			1	1	1	1	1
W200X52	0			1	1	1	1	1
W200X59	0			1	1	1	1	1
W200X71	0			1	1	1	1	1
W200X86	0			1	1	1	1	1
W250X101	0			1	1	1	1	1
W250X115	0			1	1	1	1	1
W250X131	0			1	1	1	1	1
W250X149	0			1	1	1	1	1
W250X167	0			1	1	1	1	1
W250X17.9	0			1	1	1	1	1
W250X22.3-1	0			1	1	1	1	1
W250X25.3-1	0			1	1	1	1	1
W250X28.4-1	0			1	1	1	1	1
W250X32.7	0			1	1	1	1	1
W250X38.5	0			1	1	1	1	1
W250X44.8	0			1	1	1	1	1
W250X49.1-1	0			1	1	1	1	1
W250X58-1	0			1	1	1	1	1
W250X67-1	0			1	1	1	1	1
W250X73	0			1	1	1	1	1
W250X80	0			1	1	1	1	1
W250X89-1	0			1	1	1	1	1
W310X107-1	0			1	1	1	1	1
W310X117-1	0			1	1	1	1	1
W310X129-1	0			1	1	1	1	1
W310X143-1	0			1	1	1	1	1
W310X158-1	0			1	1	1	1	1
W310X179-1	0			1	1	1	1	1
W310X202-1	0			1	1	1	1	1
W310X21-1	0			1	1	1	1	1
W310X226-1	0			1	1	1	1	1
W310X23.8-1	0			1	1	1	1	1
W310X253-1	0			1	1	1	1	1
W310X28.3-1	0			1	1	1	1	1
W310X283-1	0			1	1	1	1	1
W310X313-1	0			1	1	1	1	1
W310X32.7-1	0			1	1	1	1	1
W310X342	0			1	1	1	1	1
W310X375	0			1	1	1	1	1
W310X38.7-1	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
W310X415	0			1	1	1	1	1
W310X44.5-1	0			1	1	1	1	1
W310X454	0			1	1	1	1	1
W310X500	0			1	1	1	1	1
W310X52	0			1	1	1	1	1
W310X60	0			1	1	1	1	1
W310X67	0			1	1	1	1	1
W310X74	0			1	1	1	1	1
W310X79-1	0			1	1	1	1	1
W310X86-1	0			1	1	1	1	1
W310X97	0			1	1	1	1	1
W360X101-1	0			1	1	1	1	1
W360X1086	0			1	1	1	1	1
W360X110-1	0			1	1	1	1	1
W360X122-1	0			1	1	1	1	1
W360X134	0			1	1	1	1	1
W360X147	0			1	1	1	1	1
W360X162-1	0			1	1	1	1	1
W360X179-1	0			1	1	1	1	1
W360X196-1	0			1	1	1	1	1
W360X216-1	0			1	1	1	1	1
W360X237-1	0			1	1	1	1	1
W360X262-1	0			1	1	1	1	1
W360X287-1	0			1	1	1	1	1
W360X314-1	0			1	1	1	1	1
W360X32.9-1	0			1	1	1	1	1
W360X347-1	0			1	1	1	1	1
W360X382-1	0			1	1	1	1	1
W360X39-1	0			1	1	1	1	1
W360X421-1	0			1	1	1	1	1
W360X44-1	0			1	1	1	1	1
W360X463-1	0			1	1	1	1	1
W360X509-1	0			1	1	1	1	1
W360X51-1	0			1	1	1	1	1
W360X551-1	0			1	1	1	1	1
W360X57.8-1	0			1	1	1	1	1
W360X592-1	0			1	1	1	1	1
W360X634-1	0			1	1	1	1	1
W360X64	0			1	1	1	1	1
W360X677-1	0			1	1	1	1	1
W360X72	0			1	1	1	1	1
W360X744-1	0			1	1	1	1	1
W360X79	0			1	1	1	1	1
W360X818	0			1	1	1	1	1
W360X900	0			1	1	1	1	1
W360X91-1	0			1	1	1	1	1
W360X990	0			1	1	1	1	1
W410X100	0			1	1	1	1	1
W410X114	0			1	1	1	1	1
W410X132	0			1	1	1	1	1
W410X149	0			1	1	1	1	1
W410X38.8-1	0			1	1	1	1	1
W410X46.1-1	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
W410X53-1	0			1	1	1	1	1
W410X60	0			1	1	1	1	1
W410X67	0			1	1	1	1	1
W410X75	0			1	1	1	1	1
W410X85	0			1	1	1	1	1
W460X106	0			1	1	1	1	1
W460X113	0			1	1	1	1	1
W460X128	0			1	1	1	1	1
W460X144	0			1	1	1	1	1
W460X158	0			1	1	1	1	1
W460X177	0			1	1	1	1	1
W460X193	0			1	1	1	1	1
W460X213	0			1	1	1	1	1
W460X235	0			1	1	1	1	1
W460X260	0			1	1	1	1	1
W460X286	0			1	1	1	1	1
W460X315	0			1	1	1	1	1
W460X349	0			1	1	1	1	1
W460X384	0			1	1	1	1	1
W460X421	0			1	1	1	1	1
W460X464	0			1	1	1	1	1
W460X52-1	0			1	1	1	1	1
W460X60-1	0			1	1	1	1	1
W460X68-1	0			1	1	1	1	1
W460X74-1	0			1	1	1	1	1
W460X82-1	0			1	1	1	1	1
W460X89-1	0			1	1	1	1	1
W460X97	0			1	1	1	1	1
W530X101-1	0			1	1	1	1	1
W530X109	0			1	1	1	1	1
W530X123	0			1	1	1	1	1
W530X138	0			1	1	1	1	1
W530X150	0			1	1	1	1	1
W530X165	0			1	1	1	1	1
W530X182	0			1	1	1	1	1
W530X196	0			1	1	1	1	1
W530X219	0			1	1	1	1	1
W530X248	0			1	1	1	1	1
W530X272	0			1	1	1	1	1
W530X300	0			1	1	1	1	1
W530X66-1	0			1	1	1	1	1
W530X72	0			1	1	1	1	1
W530X74-1	0			1	1	1	1	1
W530X82-1	0			1	1	1	1	1
W530X85-1	0			1	1	1	1	1
W530X92-1	0			1	1	1	1	1
W610X101-1	0			1	1	1	1	1
W610X113-1	0			1	1	1	1	1
W610X125-1	0			1	1	1	1	1
W610X140-1	0			1	1	1	1	1
W610X153	0			1	1	1	1	1
W610X155	0			1	1	1	1	1
W610X174	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
W610X195	0			1	1	1	1	1
W610X217	0			1	1	1	1	1
W610X241	0			1	1	1	1	1
W610X262	0			1	1	1	1	1
W610X285	0			1	1	1	1	1
W610X307	0			1	1	1	1	1
W610X341	0			1	1	1	1	1
W610X372	0			1	1	1	1	1
W610X415	0			1	1	1	1	1
W610X455	0			1	1	1	1	1
W610X498	0			1	1	1	1	1
W610X551	0			1	1	1	1	1
W610X82-1	0			1	1	1	1	1
W610X92-1	0			1	1	1	1	1
W690X125-1	0			1	1	1	1	1
W690X140-1	0			1	1	1	1	1
W690X152-1	0			1	1	1	1	1
W690X170-1	0			1	1	1	1	1
W690X192-1	0			1	1	1	1	1
W690X217	0			1	1	1	1	1
W690X240	0			1	1	1	1	1
W690X265	0			1	1	1	1	1
W690X289	0			1	1	1	1	1
W690X323	0			1	1	1	1	1
W690X350	0			1	1	1	1	1
W690X384	0			1	1	1	1	1
W690X419	0			1	1	1	1	1
W690X457	0			1	1	1	1	1
W690X500	0			1	1	1	1	1
W690X548	0			1	1	1	1	1
W690X802	0			1	1	1	1	1
W760X134-1	0			1	1	1	1	1
W760X147-1	0			1	1	1	1	1
W760X161-1	0			1	1	1	1	1
W760X173-1	0			1	1	1	1	1
W760X185-1	0			1	1	1	1	1
W760X196-1	0			1	1	1	1	1
W760X220	0			1	1	1	1	1
W760X257	0			1	1	1	1	1
W760X284	0			1	1	1	1	1
W760X314	0			1	1	1	1	1
W760X350	0			1	1	1	1	1
W760X389	0			1	1	1	1	1
W760X434	0			1	1	1	1	1
W760X484	0			1	1	1	1	1
W760X531	0			1	1	1	1	1
W760X582	0			1	1	1	1	1
W840X176-1	0			1	1	1	1	1
W840X193-1	0			1	1	1	1	1
W840X210-1	0			1	1	1	1	1
W840X226-1	0			1	1	1	1	1
W840X251	0			1	1	1	1	1
W840X299	0			1	1	1	1	1

Name	PNA Offset 2 in	SC Offset 3 in	SC Offset 2 in	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier
W840X329	0			1	1	1	1	1
W840X359	0			1	1	1	1	1
W840X392	0			1	1	1	1	1
W840X433	0			1	1	1	1	1
W840X473	0			1	1	1	1	1
W840X527	0			1	1	1	1	1
W840X576	0			1	1	1	1	1
W920X201	0			1	1	1	1	1
W920X223-1	0			1	1	1	1	1
W920X238-1	0			1	1	1	1	1
W920X253-1	0			1	1	1	1	1
W920X271-1	0			1	1	1	1	1
W920X289	0			1	1	1	1	1
W920X313	0			1	1	1	1	1
W920X344	0			1	1	1	1	1
W920X345	0			1	1	1	1	1
W920X368	0			1	1	1	1	1
W920X381	0			1	1	1	1	1
W920X390	0			1	1	1	1	1
W920X420	0			1	1	1	1	1
W920X449	0			1	1	1	1	1
W920X491	0			1	1	1	1	1
W920X537	0			1	1	1	1	1
W920X588	0			1	1	1	1	1
W920X656	0			1	1	1	1	1
W920X725	0			1	1	1	1	1
W920X787	0			1	1	1	1	1
W920X970	0			1	1	1	1	1

Table 2.2 - Frame Section Property Definitions - Summary (Part 5 of 4)

Name	I22 Modifier	Mass Modifier	Weight Modifier
Column			
HSS101.6X101.6X12.7	1	1	1
HSS101.6X101.6X3.2	1	1	1
HSS101.6X101.6X4.8	1	1	1
HSS101.6X101.6X6.4	1	1	1
HSS101.6X101.6X7.9	1	1	1
HSS101.6X101.6X9.5	1	1	1
HSS101.6X50.8X3.2	1	1	1
HSS101.6X50.8X4.8	1	1	1
HSS101.6X50.8X6.4	1	1	1
HSS101.6X50.8X7.9	1	1	1
HSS101.6X50.8X9.5	1	1	1
HSS101.6X63.5X3.2	1	1	1
HSS101.6X63.5X4.8	1	1	1
HSS101.6X63.5X6.4	1	1	1
HSS101.6X63.5X7.9	1	1	1
HSS101.6X63.5X9.5	1	1	1
HSS101.6X76.2X3.2	1	1	1
HSS101.6X76.2X4.8	1	1	1
HSS101.6X76.2X6.4	1	1	1
HSS101.6X76.2X7.9	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS101.6X76.2X9.5	1	1	1
HSS114.3X114.3X12.7	1	1	1
HSS114.3X114.3X3.2	1	1	1
HSS114.3X114.3X4.8	1	1	1
HSS114.3X114.3X6.4	1	1	1
HSS114.3X114.3X7.9	1	1	1
HSS114.3X114.3X9.5	1	1	1
HSS127X101.6X12.7	1	1	1
HSS127X101.6X3.2	1	1	1
HSS127X101.6X4.8	1	1	1
HSS127X101.6X6.4	1	1	1
HSS127X101.6X7.9	1	1	1
HSS127X101.6X9.5	1	1	1
HSS127X127X12.7	1	1	1
HSS127X127X3.2	1	1	1
HSS127X127X4.8	1	1	1
HSS127X127X6.4	1	1	1
HSS127X127X7.9	1	1	1
HSS127X127X9.5	1	1	1
HSS127X50.8X3.2	1	1	1
HSS127X50.8X4.8	1	1	1
HSS127X50.8X6.4	1	1	1
HSS127X50.8X7.9	1	1	1
HSS127X50.8X9.5	1	1	1
HSS127X63.5X3.2	1	1	1
HSS127X63.5X4.8	1	1	1
HSS127X63.5X6.4	1	1	1
HSS127X76.2X12.7	1	1	1
HSS127X76.2X3.2	1	1	1
HSS127X76.2X4.8	1	1	1
HSS127X76.2X6.4	1	1	1
HSS127X76.2X7.9	1	1	1
HSS127X76.2X9.5	1	1	1
HSS139.7X139.7X3.2	1	1	1
HSS139.7X139.7X4.8	1	1	1
HSS139.7X139.7X6.4	1	1	1
HSS139.7X139.7X7.9	1	1	1
HSS139.7X139.7X9.5	1	1	1
HSS152.4X101.6X12.7	1	1	1
HSS152.4X101.6X3.2	1	1	1
HSS152.4X101.6X4.8	1	1	1
HSS152.4X101.6X6.4	1	1	1
HSS152.4X101.6X7.9	1	1	1
HSS152.4X101.6X9.5	1	1	1
HSS152.4X127X12.7	1	1	1
HSS152.4X127X3.2	1	1	1
HSS152.4X127X4.8	1	1	1
HSS152.4X127X6.4	1	1	1
HSS152.4X127X7.9	1	1	1
HSS152.4X127X9.5	1	1	1
HSS152.4X152.4X12.7	1	1	1
HSS152.4X152.4X15.9	1	1	1
HSS152.4X152.4X3.2	1	1	1
HSS152.4X152.4X4.8	1	1	1
HSS152.4X152.4X6.4	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS152.4X152.4X7.9	1	1	1
HSS152.4X152.4X9.5	1	1	1
HSS152.4X50.8X3.2	1	1	1
HSS152.4X50.8X4.8	1	1	1
HSS152.4X50.8X6.4	1	1	1
HSS152.4X50.8X7.9	1	1	1
HSS152.4X50.8X9.5	1	1	1
HSS152.4X76.2X12.7	1	1	1
HSS152.4X76.2X3.2	1	1	1
HSS152.4X76.2X4.8	1	1	1
HSS152.4X76.2X6.4	1	1	1
HSS152.4X76.2X7.9	1	1	1
HSS152.4X76.2X9.5	1	1	1
HSS177.8X101.6X12.7	1	1	1
HSS177.8X101.6X3.2	1	1	1
HSS177.8X101.6X4.8	1	1	1
HSS177.8X101.6X6.4	1	1	1
HSS177.8X101.6X7.9	1	1	1
HSS177.8X101.6X9.5	1	1	1
HSS177.8X127X12.7	1	1	1
HSS177.8X127X3.2	1	1	1
HSS177.8X127X4.8	1	1	1
HSS177.8X127X6.4	1	1	1
HSS177.8X127X7.9	1	1	1
HSS177.8X127X9.5	1	1	1
HSS177.8X177.8X12.7	1	1	1
HSS177.8X177.8X15.9	1	1	1
HSS177.8X177.8X3.2	1	1	1
HSS177.8X177.8X4.8	1	1	1
HSS177.8X177.8X6.4	1	1	1
HSS177.8X177.8X7.9	1	1	1
HSS177.8X177.8X9.5	1	1	1
HSS177.8X50.8X3.2	1	1	1
HSS177.8X50.8X4.8	1	1	1
HSS177.8X50.8X6.4	1	1	1
HSS177.8X76.2X12.7	1	1	1
HSS177.8X76.2X3.2	1	1	1
HSS177.8X76.2X4.8	1	1	1
HSS177.8X76.2X6.4	1	1	1
HSS177.8X76.2X7.9	1	1	1
HSS177.8X76.2X9.5	1	1	1
HSS203.2X101.6X12.7	1	1	1
HSS203.2X101.6X15.9	1	1	1
HSS203.2X101.6X3.2	1	1	1
HSS203.2X101.6X4.8	1	1	1
HSS203.2X101.6X6.4	1	1	1
HSS203.2X101.6X7.9	1	1	1
HSS203.2X101.6X9.5	1	1	1
HSS203.2X152.4X12.7	1	1	1
HSS203.2X152.4X15.9	1	1	1
HSS203.2X152.4X4.8	1	1	1
HSS203.2X152.4X6.4	1	1	1
HSS203.2X152.4X7.9	1	1	1
HSS203.2X152.4X9.5	1	1	1
HSS203.2X203.2X12.7	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS203.2X203.2X15.9	1	1	1
HSS203.2X203.2X3.2	1	1	1
HSS203.2X203.2X4.8	1	1	1
HSS203.2X203.2X6.4	1	1	1
HSS203.2X203.2X7.9	1	1	1
HSS203.2X203.2X9.5	1	1	1
HSS203.2X50.8X3.2	1	1	1
HSS203.2X50.8X4.8	1	1	1
HSS203.2X50.8X6.4	1	1	1
HSS203.2X50.8X7.9	1	1	1
HSS203.2X50.8X9.5	1	1	1
HSS203.2X76.2X12.7	1	1	1
HSS203.2X76.2X3.2	1	1	1
HSS203.2X76.2X4.8	1	1	1
HSS203.2X76.2X6.4	1	1	1
HSS203.2X76.2X7.9	1	1	1
HSS203.2X76.2X9.5	1	1	1
HSS228.6X127X12.7	1	1	1
HSS228.6X127X15.9	1	1	1
HSS228.6X127X4.8	1	1	1
HSS228.6X127X6.4	1	1	1
HSS228.6X127X7.9	1	1	1
HSS228.6X127X9.5	1	1	1
HSS228.6X177.8X12.7	1	1	1
HSS228.6X177.8X15.9	1	1	1
HSS228.6X177.8X4.8	1	1	1
HSS228.6X177.8X6.4	1	1	1
HSS228.6X177.8X7.9	1	1	1
HSS228.6X177.8X9.5	1	1	1
HSS228.6X228.6X12.7	1	1	1
HSS228.6X228.6X15.9	1	1	1
HSS228.6X228.6X3.2	1	1	1
HSS228.6X228.6X6.4	1	1	1
HSS228.6X228.6X7.9	1	1	1
HSS228.6X228.6X74.8	1	1	1
HSS228.6X228.6X9.5	1	1	1
HSS228.6X76.2X12.7	1	1	1
HSS228.6X76.2X4.8	1	1	1
HSS228.6X76.2X6.4	1	1	1
HSS228.6X76.2X7.9	1	1	1
HSS228.6X76.2X9.5	1	1	1
HSS254X101.6X12.7	1	1	1
HSS254X101.6X15.9	1	1	1
HSS254X101.6X3.2	1	1	1
HSS254X101.6X4.8	1	1	1
HSS254X101.6X6.4	1	1	1
HSS254X101.6X7.9	1	1	1
HSS254X101.6X9.5	1	1	1
HSS254X127X4.8	1	1	1
HSS254X127X6.4	1	1	1
HSS254X127X7.9	1	1	1
HSS254X127X9.5	1	1	1
HSS254X152.4X12.7	1	1	1
HSS254X152.4X15.9	1	1	1
HSS254X152.4X4.8	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS254X152.4X6.4	1	1	1
HSS254X152.4X7.9	1	1	1
HSS254X152.4X9.5	1	1	1
HSS254X203.2X12.7	1	1	1
HSS254X203.2X15.9	1	1	1
HSS254X203.2X4.8	1	1	1
HSS254X203.2X6.4	1	1	1
HSS254X203.2X7.9	1	1	1
HSS254X203.2X9.5	1	1	1
HSS254X254X12.7	1	1	1
HSS254X254X15.9	1	1	1
HSS254X254X4.8	1	1	1
HSS254X254X6.4	1	1	1
HSS254X254X7.9	1	1	1
HSS254X254X9.5	1	1	1
HSS254X50.8X3.2	1	1	1
HSS254X50.8X4.8	1	1	1
HSS254X50.8X6.4	1	1	1
HSS254X50.8X7.9	1	1	1
HSS254X50.8X9.5	1	1	1
HSS254X76.2X3.2	1	1	1
HSS254X76.2X4.8	1	1	1
HSS254X76.2X6.4	1	1	1
HSS254X76.2X7.9	1	1	1
HSS254X76.2X9.5	1	1	1
HSS254X88.9X12.7	1	1	1
HSS254X88.9X3.2	1	1	1
HSS254X88.9X4.8	1	1	1
HSS254X88.9X6.4	1	1	1
HSS254X88.9X7.9	1	1	1
HSS254X88.9X9.5	1	1	1
HSS304.8X101.6X12.7	1	1	1
HSS304.8X101.6X15.9	1	1	1
HSS304.8X101.6X4.8	1	1	1
HSS304.8X101.6X6.4	1	1	1
HSS304.8X101.6X7.9	1	1	1
HSS304.8X101.6X9.5	1	1	1
HSS304.8X152.4X12.7	1	1	1
HSS304.8X152.4X15.9	1	1	1
HSS304.8X152.4X4.8	1	1	1
HSS304.8X152.4X6.4	1	1	1
HSS304.8X152.4X7.9	1	1	1
HSS304.8X152.4X9.5	1	1	1
HSS304.8X203.2X12.7	1	1	1
HSS304.8X203.2X15.9	1	1	1
HSS304.8X203.2X4.8	1	1	1
HSS304.8X203.2X6.4	1	1	1
HSS304.8X203.2X7.9	1	1	1
HSS304.8X203.2X9.5	1	1	1
HSS304.8X254X12.7	1	1	1
HSS304.8X254X6.4	1	1	1
HSS304.8X254X7.9	1	1	1
HSS304.8X254X9.5	1	1	1
HSS304.8X304.8X12.7	1	1	1
HSS304.8X304.8X15.9	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS304.8X304.8X4.8	1	1	1
HSS304.8X304.8X6.4	1	1	1
HSS304.8X304.8X7.9	1	1	1
HSS304.8X304.8X9.5	1	1	1
HSS304.8X50.8X4.8	1	1	1
HSS304.8X50.8X6.4	1	1	1
HSS304.8X50.8X7.9	1	1	1
HSS304.8X76.2X4.8	1	1	1
HSS304.8X76.2X6.4	1	1	1
HSS304.8X76.2X7.9	1	1	1
HSS304.8X88.9X7.9	1	1	1
HSS304.8X88.9X9.5	1	1	1
HSS355.6X101.6X12.7	1	1	1
HSS355.6X101.6X15.9	1	1	1
HSS355.6X101.6X4.8	1	1	1
HSS355.6X101.6X6.4	1	1	1
HSS355.6X101.6X7.9	1	1	1
HSS355.6X101.6X9.5	1	1	1
HSS355.6X152.4X12.7	1	1	1
HSS355.6X152.4X15.9	1	1	1
HSS355.6X152.4X4.8	1	1	1
HSS355.6X152.4X6.4	1	1	1
HSS355.6X152.4X7.9	1	1	1
HSS355.6X152.4X9.5	1	1	1
HSS355.6X254X12.7	1	1	1
HSS355.6X254X15.9	1	1	1
HSS355.6X254X6.4	1	1	1
HSS355.6X254X7.9	1	1	1
HSS355.6X254X9.5	1	1	1
HSS355.6X355.6X12.7	1	1	1
HSS355.6X355.6X15.9	1	1	1
HSS355.6X355.6X7.9	1	1	1
HSS355.6X355.6X9.5	1	1	1
HSS406.4X101.6X12.7	1	1	1
HSS406.4X101.6X15.9	1	1	1
HSS406.4X101.6X4.8	1	1	1
HSS406.4X101.6X6.4	1	1	1
HSS406.4X101.6X7.9	1	1	1
HSS406.4X101.6X9.5	1	1	1
HSS406.4X203.2X12.7	1	1	1
HSS406.4X203.2X15.9	1	1	1
HSS406.4X203.2X6.4	1	1	1
HSS406.4X203.2X7.9	1	1	1
HSS406.4X203.2X9.5	1	1	1
HSS406.4X304.8X12.7	1	1	1
HSS406.4X304.8X15.9	1	1	1
HSS406.4X304.8X7.9	1	1	1
HSS406.4X304.8X9.5	1	1	1
HSS406.4X406.4X12.7	1	1	1
HSS406.4X406.4X15.9	1	1	1
HSS406.4X406.4X7.9	1	1	1
HSS406.4X406.4X9.5	1	1	1
HSS457.2X152.4X12.7	1	1	1
HSS457.2X152.4X15.9	1	1	1
HSS457.2X152.4X6.4	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS457.2X152.4X7.9	1	1	1
HSS457.2X152.4X9.5	1	1	1
HSS50.8X25.4X3.2	1	1	1
HSS50.8X25.4X4.8	1	1	1
HSS50.8X38.1X3.2	1	1	1
HSS50.8X38.1X4.8	1	1	1
HSS50.8X50.8X3.2	1	1	1
HSS50.8X50.8X4.8	1	1	1
HSS50.8X50.8X6.4	1	1	1
HSS508X101.6X12.7	1	1	1
HSS508X101.6X6.4	1	1	1
HSS508X101.6X7.9	1	1	1
HSS508X101.6X9.5	1	1	1
HSS508X203.2X12.7	1	1	1
HSS508X203.2X15.9	1	1	1
HSS508X203.2X7.9	1	1	1
HSS508X203.2X9.5	1	1	1
HSS508X304.8X12.7	1	1	1
HSS508X304.8X15.9	1	1	1
HSS508X304.8X7.9	1	1	1
HSS508X304.8X9.5	1	1	1
HSS57.2X50.8X3.2	1	1	1
HSS57.2X50.8X4.8	1	1	1
HSS57.2X57.2X3.2	1	1	1
HSS57.2X57.2X4.8	1	1	1
HSS57.2X57.2X6.4	1	1	1
HSS63.5X25.4X3.2	1	1	1
HSS63.5X25.4X4.8	1	1	1
HSS63.5X38.1X3.2	1	1	1
HSS63.5X38.1X4.8	1	1	1
HSS63.5X38.1X6.4	1	1	1
HSS63.5X50.8X3.2	1	1	1
HSS63.5X50.8X4.8	1	1	1
HSS63.5X50.8X6.4	1	1	1
HSS63.5X63.5X3.2	1	1	1
HSS63.5X63.5X4.8	1	1	1
HSS63.5X63.5X6.4	1	1	1
HSS63.5X63.5X7.9	1	1	1
HSS76.2X25.4X3.2	1	1	1
HSS76.2X25.4X4.8	1	1	1
HSS76.2X38.1X3.2	1	1	1
HSS76.2X38.1X4.8	1	1	1
HSS76.2X38.1X6.4	1	1	1
HSS76.2X50.8X3.2	1	1	1
HSS76.2X50.8X4.8	1	1	1
HSS76.2X50.8X6.4	1	1	1
HSS76.2X50.8X7.9	1	1	1
HSS76.2X63.5X3.2	1	1	1
HSS76.2X63.5X4.8	1	1	1
HSS76.2X63.5X6.4	1	1	1
HSS76.2X63.5X7.9	1	1	1
HSS76.2X76.2X3.2	1	1	1
HSS76.2X76.2X4.8	1	1	1
HSS76.2X76.2X6.4	1	1	1
HSS76.2X76.2X7.9	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
HSS76.2X76.2X9.5	1	1	1
HSS88.9X38.1X3.2	1	1	1
HSS88.9X38.1X4.8	1	1	1
HSS88.9X38.1X6.4	1	1	1
HSS88.9X50.8X3.2	1	1	1
HSS88.9X50.8X4.8	1	1	1
HSS88.9X50.8X6.4	1	1	1
HSS88.9X63.5X3.2	1	1	1
HSS88.9X63.5X4.8	1	1	1
HSS88.9X63.5X6.4	1	1	1
HSS88.9X63.5X7.9	1	1	1
HSS88.9X63.5X9.5	1	1	1
HSS88.9X88.9X3.2	1	1	1
HSS88.9X88.9X4.8	1	1	1
HSS88.9X88.9X6.4	1	1	1
HSS88.9X88.9X7.9	1	1	1
HSS88.9X88.9X9.5	1	1	1
L102X102X11.1	1	1	1
L102X102X12.7	1	1	1
L102X102X15.9	1	1	1
L102X102X19	1	1	1
L102X102X6.4	1	1	1
L102X102X7.9	1	1	1
L102X102X9.5	1	1	1
L102X76X12.7	1	1	1
L102X76X15.9	1	1	1
L102X76X6.4	1	1	1
L102X76X7.9	1	1	1
L102X76X9.5	1	1	1
L102X89X12.7	1	1	1
L102X89X6.4	1	1	1
L102X89X7.9	1	1	1
L102X89X9.5	1	1	1
L127X127X11.1	1	1	1
L127X127X12.7	1	1	1
L127X127X15.9	1	1	1
L127X127X19	1	1	1
L127X127X22.2	1	1	1
L127X127X7.9	1	1	1
L127X127X9.5	1	1	1
L127X76X11.1	1	1	1
L127X76X12.7	1	1	1
L127X76X6.4	1	1	1
L127X76X7.9	1	1	1
L127X76X9.5	1	1	1
L127X89X12.7	1	1	1
L127X89X15.9	1	1	1
L127X89X19	1	1	1
L127X89X6.4	1	1	1
L127X89X7.9	1	1	1
L127X89X9.5	1	1	1
L152X102X11.1	1	1	1
L152X102X12.7	1	1	1
L152X102X14.3	1	1	1
L152X102X15.9	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
L152X102X19	1	1	1
L152X102X22.2	1	1	1
L152X102X7.9	1	1	1
L152X102X9.5	1	1	1
L152X152X11.1	1	1	1
L152X152X12.7	1	1	1
L152X152X14.3	1	1	1
L152X152X15.9	1	1	1
L152X152X19	1	1	1
L152X152X22.2	1	1	1
L152X152X25.4	1	1	1
L152X152X7.9	1	1	1
L152X152X9.5	1	1	1
L152X89X12.7	1	1	1
L152X89X7.9	1	1	1
L152X89X9.5	1	1	1
L178X102X11.1	1	1	1
L178X102X12.7	1	1	1
L178X102X15.9	1	1	1
L178X102X19	1	1	1
L178X102X9.5	1	1	1
L203X102X11.1	1	1	1
L203X102X12.7	1	1	1
L203X102X14.3	1	1	1
L203X102X15.9	1	1	1
L203X102X19	1	1	1
L203X102X22.2	1	1	1
L203X102X25.4	1	1	1
L203X152X11.1	1	1	1
L203X152X12.7	1	1	1
L203X152X14.3	1	1	1
L203X152X15.9	1	1	1
L203X152X19	1	1	1
L203X152X22.2	1	1	1
L203X152X25.4	1	1	1
L203X203X12.7	1	1	1
L203X203X14.3	1	1	1
L203X203X15.9	1	1	1
L203X203X19	1	1	1
L203X203X22.2	1	1	1
L203X203X25.4	1	1	1
L203X203X28.6	1	1	1
L51X51X3.2	1	1	1
L51X51X4.8	1	1	1
L51X51X6.4	1	1	1
L51X51X7.9	1	1	1
L51X51X9.5	1	1	1
L64X38X4.8	1	1	1
L64X38X6.4	1	1	1
L64X51X4.8	1	1	1
L64X51X6.4	1	1	1
L64X51X7.9	1	1	1
L64X51X9.5	1	1	1
L64X64X12.7	1	1	1
L64X64X4.8	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
L64X64X6.4	1	1	1
L64X64X7.9	1	1	1
L64X64X9.5	1	1	1
L76X51X12.7	1	1	1
L76X51X4.8	1	1	1
L76X51X6.4	1	1	1
L76X51X7.9	1	1	1
L76X51X9.5	1	1	1
L76X64X11.1	1	1	1
L76X64X12.7	1	1	1
L76X64X4.8	1	1	1
L76X64X6.4	1	1	1
L76X64X7.9	1	1	1
L76X64X9.5	1	1	1
L76X76X11.1	1	1	1
L76X76X12.7	1	1	1
L76X76X4.8	1	1	1
L76X76X6.4	1	1	1
L76X76X7.9	1	1	1
L76X76X9.5	1	1	1
L89X64X12.7	1	1	1
L89X64X6.4	1	1	1
L89X64X7.9	1	1	1
L89X64X9.5	1	1	1
L89X76X11.1	1	1	1
L89X76X12.7	1	1	1
L89X76X6.4	1	1	1
L89X76X7.9	1	1	1
L89X76X9.5	1	1	1
L89X89X11.1	1	1	1
L89X89X12.7	1	1	1
L89X89X6.4	1	1	1
L89X89X7.9	1	1	1
L89X89X9.5	1	1	1
Rufter			
Truss			
W1000X222	1	1	1
W1000X249	1	1	1
W1000X272	1	1	1
W1000X296	1	1	1
W1000X314	1	1	1
W1000X321	1	1	1
W1000X350	1	1	1
W1000X371	1	1	1
W1000X393	1	1	1
W1000X412	1	1	1
W1000X415	1	1	1
W1000X438	1	1	1
W1000X443	1	1	1
W1000X483	1	1	1
W1000X486	1	1	1
W1000X494	1	1	1
W1000X539	1	1	1
W1000X554	1	1	1
W1000X584	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
W1000X591	1	1	1
W1000X642	1	1	1
W1000X748	1	1	1
W1000X883	1	1	1
W100X19.3	1	1	1
W1100X343	1	1	1
W1100X390	1	1	1
W1100X433	1	1	1
W1100X499	1	1	1
W130X23.8	1	1	1
W130X28.1	1	1	1
W150X13	1	1	1
W150X13.5	1	1	1
W150X18	1	1	1
W150X22.5	1	1	1
W150X24	1	1	1
W150X29.8	1	1	1
W150X37.1	1	1	1
W200X100	1	1	1
W200X15	1	1	1
W200X19.3	1	1	1
W200X22.5	1	1	1
W200X26.6	1	1	1
W200X31.3	1	1	1
W200X35.9	1	1	1
W200X41.7	1	1	1
W200X46.1	1	1	1
W200X52	1	1	1
W200X59	1	1	1
W200X71	1	1	1
W200X86	1	1	1
W250X101	1	1	1
W250X115	1	1	1
W250X131	1	1	1
W250X149	1	1	1
W250X167	1	1	1
W250X17.9	1	1	1
W250X22.3-1	1	1	1
W250X25.3-1	1	1	1
W250X28.4-1	1	1	1
W250X32.7	1	1	1
W250X38.5	1	1	1
W250X44.8	1	1	1
W250X49.1-1	1	1	1
W250X58-1	1	1	1
W250X67-1	1	1	1
W250X73	1	1	1
W250X80	1	1	1
W250X89-1	1	1	1
W310X107-1	1	1	1
W310X117-1	1	1	1
W310X129-1	1	1	1
W310X143-1	1	1	1
W310X158-1	1	1	1
W310X179-1	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
W310X202-1	1	1	1
W310X21-1	1	1	1
W310X226-1	1	1	1
W310X23.8-1	1	1	1
W310X253-1	1	1	1
W310X28.3-1	1	1	1
W310X283-1	1	1	1
W310X313-1	1	1	1
W310X32.7-1	1	1	1
W310X342	1	1	1
W310X375	1	1	1
W310X38.7-1	1	1	1
W310X415	1	1	1
W310X44.5-1	1	1	1
W310X454	1	1	1
W310X500	1	1	1
W310X52	1	1	1
W310X60	1	1	1
W310X67	1	1	1
W310X74	1	1	1
W310X79-1	1	1	1
W310X86-1	1	1	1
W310X97	1	1	1
W360X101-1	1	1	1
W360X1086	1	1	1
W360X110-1	1	1	1
W360X122-1	1	1	1
W360X134	1	1	1
W360X147	1	1	1
W360X162-1	1	1	1
W360X179-1	1	1	1
W360X196-1	1	1	1
W360X216-1	1	1	1
W360X237-1	1	1	1
W360X262-1	1	1	1
W360X287-1	1	1	1
W360X314-1	1	1	1
W360X32.9-1	1	1	1
W360X347-1	1	1	1
W360X382-1	1	1	1
W360X39-1	1	1	1
W360X421-1	1	1	1
W360X44-1	1	1	1
W360X463-1	1	1	1
W360X509-1	1	1	1
W360X51-1	1	1	1
W360X551-1	1	1	1
W360X57.8-1	1	1	1
W360X592-1	1	1	1
W360X634-1	1	1	1
W360X64	1	1	1
W360X677-1	1	1	1
W360X72	1	1	1
W360X744-1	1	1	1
W360X79	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
W360X818	1	1	1
W360X900	1	1	1
W360X91-1	1	1	1
W360X990	1	1	1
W410X100	1	1	1
W410X114	1	1	1
W410X132	1	1	1
W410X149	1	1	1
W410X38.8-1	1	1	1
W410X46.1-1	1	1	1
W410X53-1	1	1	1
W410X60	1	1	1
W410X67	1	1	1
W410X75	1	1	1
W410X85	1	1	1
W460X106	1	1	1
W460X113	1	1	1
W460X128	1	1	1
W460X144	1	1	1
W460X158	1	1	1
W460X177	1	1	1
W460X193	1	1	1
W460X213	1	1	1
W460X235	1	1	1
W460X260	1	1	1
W460X286	1	1	1
W460X315	1	1	1
W460X349	1	1	1
W460X384	1	1	1
W460X421	1	1	1
W460X464	1	1	1
W460X52-1	1	1	1
W460X60-1	1	1	1
W460X68-1	1	1	1
W460X74-1	1	1	1
W460X82-1	1	1	1
W460X89-1	1	1	1
W460X97	1	1	1
W530X101-1	1	1	1
W530X109	1	1	1
W530X123	1	1	1
W530X138	1	1	1
W530X150	1	1	1
W530X165	1	1	1
W530X182	1	1	1
W530X196	1	1	1
W530X219	1	1	1
W530X248	1	1	1
W530X272	1	1	1
W530X300	1	1	1
W530X66-1	1	1	1
W530X72	1	1	1
W530X74-1	1	1	1
W530X82-1	1	1	1
W530X85-1	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
W530X92-1	1	1	1
W610X101-1	1	1	1
W610X113-1	1	1	1
W610X125-1	1	1	1
W610X140-1	1	1	1
W610X153	1	1	1
W610X155	1	1	1
W610X174	1	1	1
W610X195	1	1	1
W610X217	1	1	1
W610X241	1	1	1
W610X262	1	1	1
W610X285	1	1	1
W610X307	1	1	1
W610X341	1	1	1
W610X372	1	1	1
W610X415	1	1	1
W610X455	1	1	1
W610X498	1	1	1
W610X551	1	1	1
W610X82-1	1	1	1
W610X92-1	1	1	1
W690X125-1	1	1	1
W690X140-1	1	1	1
W690X152-1	1	1	1
W690X170-1	1	1	1
W690X192-1	1	1	1
W690X217	1	1	1
W690X240	1	1	1
W690X265	1	1	1
W690X289	1	1	1
W690X323	1	1	1
W690X350	1	1	1
W690X384	1	1	1
W690X419	1	1	1
W690X457	1	1	1
W690X500	1	1	1
W690X548	1	1	1
W690X802	1	1	1
W760X134-1	1	1	1
W760X147-1	1	1	1
W760X161-1	1	1	1
W760X173-1	1	1	1
W760X185-1	1	1	1
W760X196-1	1	1	1
W760X220	1	1	1
W760X257	1	1	1
W760X284	1	1	1
W760X314	1	1	1
W760X350	1	1	1
W760X389	1	1	1
W760X434	1	1	1
W760X484	1	1	1
W760X531	1	1	1
W760X582	1	1	1

Name	I22 Modifier	Mass Modifier	Weight Modifier
W840X176-1	1	1	1
W840X193-1	1	1	1
W840X210-1	1	1	1
W840X226-1	1	1	1
W840X251	1	1	1
W840X299	1	1	1
W840X329	1	1	1
W840X359	1	1	1
W840X392	1	1	1
W840X433	1	1	1
W840X473	1	1	1
W840X527	1	1	1
W840X576	1	1	1
W920X201	1	1	1
W920X223-1	1	1	1
W920X238-1	1	1	1
W920X253-1	1	1	1
W920X271-1	1	1	1
W920X289	1	1	1
W920X313	1	1	1
W920X344	1	1	1
W920X345	1	1	1
W920X368	1	1	1
W920X381	1	1	1
W920X390	1	1	1
W920X420	1	1	1
W920X449	1	1	1
W920X491	1	1	1
W920X537	1	1	1
W920X588	1	1	1
W920X656	1	1	1
W920X725	1	1	1
W920X787	1	1	1
W920X970	1	1	1

2.3 Shell Sections

Table 2.3 - Area Section Property Definitions - Summary

Name	Type	Element Type	Material	Total Thickness in	Deck Material	Deck Depth in
Slab1	Slab	Membrane	Concrete	0.1		

2.4 Reinforcement Sizes

Table 2.4 - Reinforcing Bar Sizes

Name	Diameter in	Area in2
#6	0.75	0.44

3 Loads

This chapter provides loading information as applied to the model.

3.1 Load Patterns

Table 3.1 - Load Pattern Definitions

Name	Is Auto Load	Type	Self Weight Multiplier	Auto Load
~LLRF	Yes	Other	0	
Dead	No	Dead	1	
Ex	No	Seismic	0	ASCE 7-10
Ey	No	Seismic	0	ASCE 7-10
Live	No	Live	0	
Wx	No	Wind	0	ASCE 7-10
Wy	No	Wind	0	ASCE 7-10

3.2 Auto Wind Loading

ASCE 7-10 Auto Wind Load Calculation

This calculation presents the automatically generated lateral wind loads for load pattern Wx according to ASCE 7-10, as calculated by ETABS.

Exposure Parameters

Exposure From = Shell Objects

Exposure Category = B

Wind Direction = 0 degrees

Top Story = Story2

Bottom Story = Base

Include Parapet = No

Factors and Coefficients

Solid/Gross Area Ratio = 0.2

Gradient Height, z_g [ASCE Table 26.9-1]		$z_g = 1200$
Emperical Exponent, α [ASCE Table 26.9-1]		$\alpha = 7$
Velocity Pressure Exposure Coefficient, K_z; [ASCE Table 27.3-1]	$K_z = 2.01(\frac{z}{z_g}$ for 15ft $\leq z \leq z_g$	
Topographical Factor, K_{zt} [ASCE 26.8.2]	$K_z = 2.01(\frac{15}{z_g}$ for z 15ft	$K_{zt} = 1$
Directionality Factor, K_d [ASCE 26.6]		$K_d = 0.85$
Gust Effect Factor, G [ASCE 26.9]		$G = 0.85$

Lateral Loading

Velocity Pressure, q_z [ASCE 27.3.2 Eq. 27.3-1]	$q_z = 0.00256K_zK_{zt}K_dV^2$
Design Wind Pressure, p [ASCE 27.4.2 Eq. 27.4-2]	$p = qGC_p$

Applied Story Forces

Story	Elevation	X-Dir	Y-Dir
	ft	kip	kip
Story2	21	-0.763	0
Story1	15	-0.043	0
Base	0	0	0

ASCE 7-10 Auto Wind Load Calculation

This calculation presents the automatically generated lateral wind loads for load pattern Wy according to ASCE 7-10, as calculated by ETABS.

Exposure Parameters

Exposure From = Shell Objects

Exposure Category = B

Wind Direction = 0 degrees

Top Story = Story2

Bottom Story = Base

Include Parapet = No

Factors and Coefficients

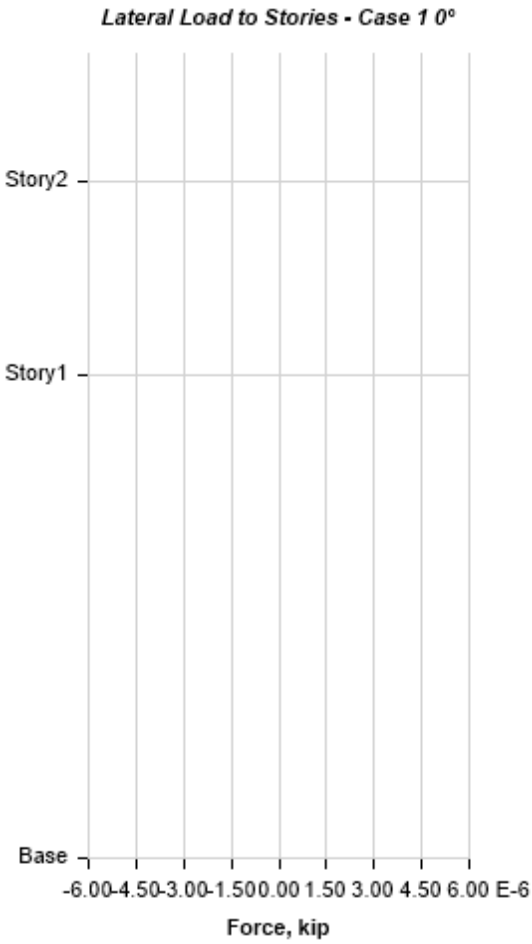
Solid/Gross Area Ratio = 0.2

Gradient Height, z_g [ASCE Table 26.9-1]		$z_g = 1200$
Emperical Exponent, α [ASCE Table 26.9-1]		$\alpha = 7$
Velocity Pressure Exposure Coefficient, $K_{z;}$ [ASCE Table 27.3-1]	$K_z = 2.01(\frac{z}{z_g}$ for 15ft $\leq z \leq z_g$	
Topographical Factor, K_{zt} [ASCE 26.8.2]	$K_z = 2.01(\frac{15}{z_g}$ for z 15ft	$K_{zt} = 1$
Directionality Factor, K_d [ASCE 26.6]		$K_d = 0.85$
Gust Effect Factor, G [ASCE 26.9]		$G = 0.85$

Lateral Loading

Velocity Pressure, q_z [ASCE 27.3.2 Eq. 27.3-1]	$q_z = 0.00256K_zK_{zt}K_dV^2$
Design Wind Pressure, p [ASCE 27.4.2 Eq. 27.4-2]	$p = qGC_p$

Applied Story Forces



Story	Elevation	X-Dir	Y-Dir
	ft	kip	kip
Story2	21	0	0
Story1	15	0	0
Base	0	0	0

3.3 Auto Seismic Loading

ASCE 7-10 Auto Seismic Load Calculation

This calculation presents the automatically generated lateral seismic loads for load pattern Ex according to ASCE 7-10, as calculated by ETABS.

Direction and Eccentricity

Direction = X

Structural Period

Period Calculation Method = Program Calculated

Coefficient, C_t [ASCE Table 12.8-2]	$C_t = 0.028ft$
Coefficient, x [ASCE Table 12.8-2]	$x = 0.8$
Structure Height Above Base, h_n	$h_n = 21$ ft
Long-Period Transition Period, T_L [ASCE 11.4.5]	$T_L = 8$ sec

Factors and Coefficients

Response Modification Factor, R [ASCE Table 12.2-1]	$R = 8$
System Overstrength Factor, Ω_0 [ASCE Table 12.2-1]	$\Omega_0 = 3$
Deflection Amplification Factor, C_d [ASCE Table 12.2-1]	$C_d = 5.5$
Importance Factor, I [ASCE Table 11.5-1]	$I = 1$

Ss and S1 Source = User Specified

Mapped MCE Spectral Response Acceleration, S_s [ASCE 11.4.1]	$S_s = 2.29g$
Mapped MCE Spectral Response Acceleration, S_1 [ASCE 11.4.1]	$S_1 = 0.869g$

Site Class [ASCE Table 20.3-1] = B - Rock

Site Coefficient, F_a [ASCE Table 11.4-1]	$F_a = 1$
Site Coefficient, F_v [ASCE Table 11.4-2]	$F_v = 1$

Seismic Response

MCE Spectral Response Acceleration, S_{MS} [ASCE 11.4.3, Eq. 11.4-1]	$S_{MS} = F_a S_S$	$S_{MS} = 2.29g$
MCE Spectral Response Acceleration, S_{M1} [ASCE 11.4.3, Eq. 11.4-2]	$S_{M1} = F_v S_1$	$S_{M1} = 0.869g$

Design Spectral Response Acceleration, S_{DS} [ASCE 11.4.4, Eq. 11.4-3] $S_{DS} = \frac{2}{3} S_{MS}$ $S_{DS} = 1.526667g$

Design Spectral Response Acceleration, S_{D1} [ASCE 11.4.4, Eq. 11.4-4] $S_{D1} = \frac{2}{3} S_{M1}$ $S_{D1} = 0.579333g$

Equivalent Lateral Forces

Seismic Response Coefficient, C_S [ASCE 12.8.1.1, Eq. 12.8-2]

$$C_S = \frac{S_{DS}}{(\frac{R}{I})}$$

[ASCE 12.8.1.1, Eq. 12.8-3]

$$C_{S,max} = \frac{S_{D1}}{T(\frac{R}{I})}$$

[ASCE 12.8.1.1, Eq. 12.8-5]

$$C_{S,min} = \max(0.044 S_{DS} I, 0.01) = 0.067173$$

[ASCE 12.8.1.1, Eq. 12.8-6]

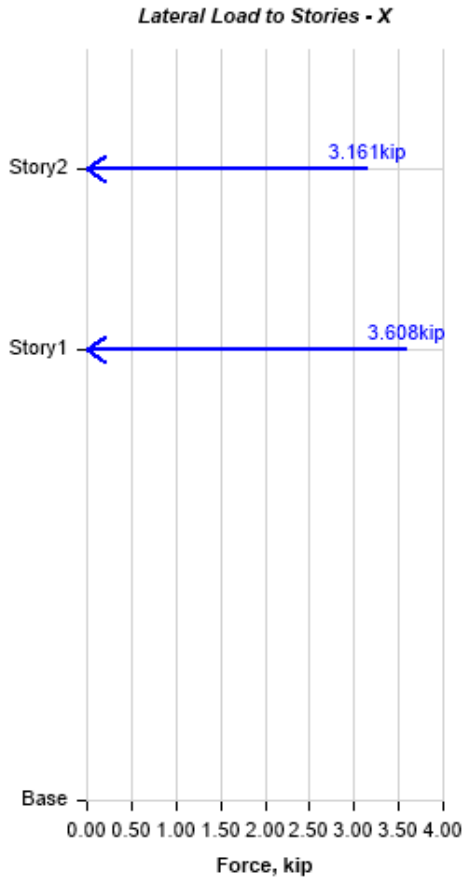
$$C_{S,min} = 0.5 \frac{S_1}{(\frac{R}{I})} \text{for} S_1 = 0.6g$$

$$C_{S,min} \leq C_s \leq C_{S,max}$$

Calculated Base Shear

Direction	Period Used (sec)	C_s	W (kip)	V (kip)
X	0.144	0.190833	35.4707	6.769

Applied Story Forces



Story	Elevation	X-Dir	Y-Dir
	ft	kip	kip
Story2	21	3.161	0
Story1	15	3.608	0
Base	0	0	0

ASCE 7-10 Auto Seismic Load Calculation

This calculation presents the automatically generated lateral seismic loads for load pattern Ey according to ASCE 7-10, as calculated by ETABS.

Direction and Eccentricity

Direction = Y

Structural Period

Period Calculation Method = Program Calculated

Coefficient, C_t [ASCE Table 12.8-2]	$C_t = 0.028ft$
Coefficient, x [ASCE Table 12.8-2]	$x = 0.8$
Structure Height Above Base, h_n	$h_n = 21$ ft
Long-Period Transition Period, T_L [ASCE 11.4.5]	$T_L = 8$ sec

Factors and Coefficients

Response Modification Factor, R [ASCE Table 12.2-1]	$R = 8$
System Overstrength Factor, Ω_0 [ASCE Table 12.2-1]	$\Omega_0 = 3$
Deflection Amplification Factor, C_d [ASCE Table 12.2-1]	$C_d = 5.5$
Importance Factor, I [ASCE Table 11.5-1]	$I = 1$

Ss and S1 Source = User Specified

Mapped MCE Spectral Response Acceleration, S_s [ASCE 11.4.1]	$S_s = 2.29g$
Mapped MCE Spectral Response Acceleration, S_1 [ASCE 11.4.1]	$S_1 = 0.869g$

Site Class [ASCE Table 20.3-1] = B - Rock

Site Coefficient, F_a [ASCE Table 11.4-1]	$F_a = 1$
Site Coefficient, F_v [ASCE Table 11.4-2]	$F_v = 1$

Seismic Response

MCE Spectral Response Acceleration, S_{MS} [ASCE 11.4.3, Eq. 11.4-1]	$S_{MS} = F_a S_S$	$S_{MS} = 2.29g$
MCE Spectral Response Acceleration, S_{M1} [ASCE 11.4.3, Eq. 11.4-2]	$S_{M1} = F_v S_1$	$S_{M1} = 0.869g$

Design Spectral Response Acceleration, S_{DS} [ASCE 11.4.4, Eq. 11.4-3] $S_{DS} = \frac{2}{3} S_{MS}$ $S_{DS} = 1.526667g$

Design Spectral Response Acceleration, S_{D1} [ASCE 11.4.4, Eq. 11.4-4] $S_{D1} = \frac{2}{3} S_{M1}$ $S_{D1} = 0.579333g$

Equivalent Lateral Forces

Seismic Response Coefficient, C_S [ASCE 12.8.1.1, Eq. 12.8-2]

$$C_S = \frac{S_{DS}}{(\frac{R}{I})}$$

[ASCE 12.8.1.1, Eq. 12.8-3]

$$C_{S,max} = \frac{S_{D1}}{T(\frac{R}{I})}$$

[ASCE 12.8.1.1, Eq. 12.8-5]

$$C_{S,min} = \max(0.044 S_{DS} I, 0.01) = 0.067173$$

[ASCE 12.8.1.1, Eq. 12.8-6]

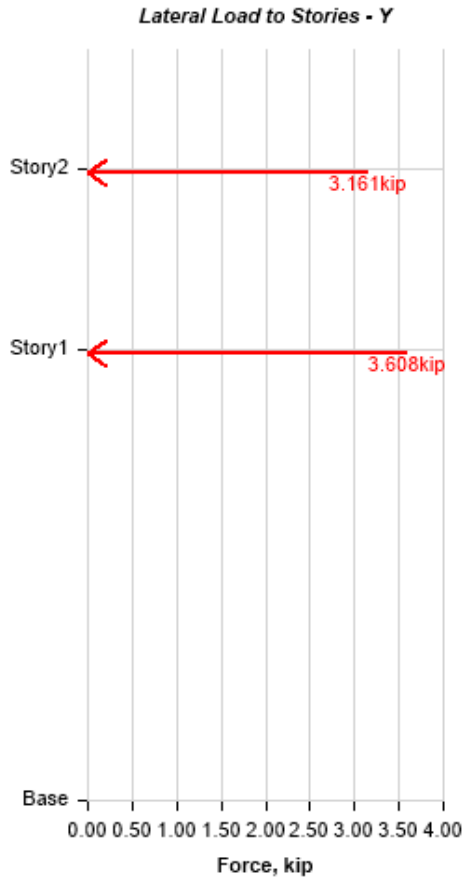
$$C_{S,min} = 0.5 \frac{S_1}{(\frac{R}{I})} \text{for} S_1 = 0.6g$$

$$C_{S,min} \leq C_s \leq C_{S,max}$$

Calculated Base Shear

Direction	Period Used (sec)	C_s	W (kip)	V (kip)
Y	0.194	0.190833	35.4707	6.769

Applied Story Forces



Story	Elevation	X-Dir	Y-Dir
	ft	kip	kip
Story2	21	0	3.161
Story1	15	0	3.608
Base	0	0	0

3.4 Load Cases

Table 3.6 - Load Case Definitions - Summary

Name	Type	Notes
Dead	Linear Static	
Live	Linear Static	
Modal	Modal - Eigen	
Ex	Linear Static	
Ey	Linear Static	
Wx	Linear Static	
Wy	Linear Static	

3.5 Load Combinations

Table 3.7 - Load Combination Definitions

Name	Type	Is Auto	Load Name	Mode	SF	Notes
1.2 DL + 1.6LL	Linear Add	No	Dead		1.2	Dead + Live [Strength]
1.2 DL + 1.6LL			Live		1.6	
DConS1	Linear Add	Yes	Dead		1.4	Dead [Strength]
DConS2	Linear Add	Yes	Dead		1.2	Dead + Live [Strength]
DConS2			Live		1.6	
DConS3	Linear Add	Yes	Dead		1.2	Dead + Live + Wind + Snow [Strength]
DConS3			Live		1	
DConS3			Wx		1	
DConS4	Linear Add	Yes	Dead		1.2	Dead + Live - Wind + Snow [Strength]
DConS4			Live		1	
DConS4			Wx		-1	
DConS5	Linear Add	Yes	Dead		1.2	Dead + Live + Wind + Snow [Strength]
DConS5			Live		1	
DConS5			Wy		1	
DConS6	Linear Add	Yes	Dead		1.2	Dead + Live - Wind + Snow [Strength]
DConS6			Live		1	
DConS6			Wy		-1	
DConS7	Linear Add	Yes	Dead		0.9	Dead (min) + Wind [Strength]
DConS7			Wx		1	
DConS8	Linear Add	Yes	Dead		0.9	Dead (min) - Wind [Strength]
DConS8			Wx		-1	
DConS9	Linear Add	Yes	Dead		0.9	Dead (min) + Wind [Strength]
DConS9			Wy		1	
DConS10	Linear Add	Yes	Dead		0.9	Dead (min) - Wind [Strength]
DConS10			Wy		-1	
DConS11	Linear Add	Yes	Dead		1.3	Dead + Live + Static Earthquake [Strength]
DConS11			Live		1	
DConS11			Ex		1	
DConS12	Linear Add	Yes	Dead		1.3	Dead + Live - Static Earthquake [Strength]
DConS12			Live		1	
DConS12			Ex		-1	
DConS13	Linear Add	Yes	Dead		1.3	Dead + Live + Static Earthquake [Strength]
DConS13			Live		1	
DConS13			Ey		1	
DConS14	Linear Add	Yes	Dead		1.3	Dead + Live - Static Earthquake [Strength]
DConS14			Live		1	
DConS14			Ey		-1	
DConS15	Linear Add	Yes	Dead		0.8	Dead (min) + Static Earthquake [Strength]
DConS15			Ex		1	
DConS16	Linear Add	Yes	Dead		0.8	Dead (min) - Static Earthquake [Strength]

Name	Type	Is Auto	Load Name	Mode	SF	Notes
DConS16			Ex		-1	
DConS17	Linear Add	Yes	Dead		0.8	Dead (min) + Static Earthquake [Strength]
DConS17			Ey		1	
DConS18	Linear Add	Yes	Dead		0.8	Dead (min) - Static Earthquake [Strength]
DConS18			Ey		-1	
UDCmpC1	Linear Add	No	Dead		1.4	Dead [Construction]
UDCmpC2	Linear Add	No	Dead		1.2	Dead + Construction Live [Construction]
UDCmpD1	Linear Add	No	Dead		1	Dead [Deflections]
UDCmpD2	Linear Add	No	Dead		1	Dead + Live [Deflections]
UDCmpD2			Live		1	
UDCmpS1	Linear Add	No	Dead		1.4	Dead [Strength]
UDCmpS2	Linear Add	No	Dead		1.2	Dead + Live [Strength]
UDCmpS2			Live		1.6	
UDStlD1	Linear Add	No	Dead		1	Dead [Deflections]
UDStlD2	Linear Add	No	Dead		1	Dead + Live [Deflections]
UDStlD2			Live		1	
UDStlS1	Linear Add	No	Dead		1.4	Dead [Strength]
UDStlS3	Linear Add	No	Dead		1.2	Dead + Live + Wind + Snow [Strength]
UDStlS3			Live		1	
UDStlS3			Wx		1	
UDStlS4	Linear Add	No	Dead		1.2	Dead + Live - Wind + Snow [Strength]
UDStlS4			Live		1	
UDStlS4			Wx		-1	
UDStlS5	Linear Add	No	Dead		1.2	Dead + Live + Wind + Snow [Strength]
UDStlS5			Live		1	
UDStlS5			Wy		1	
UDStlS6	Linear Add	No	Dead		1.2	Dead + Live - Wind + Snow [Strength]
UDStlS6			Live		1	
UDStlS6			Wy		-1	
UDStlS7	Linear Add	No	Dead		0.9	Dead (min) + Wind [Strength]
UDStlS7			Wx		1	
UDStlS8	Linear Add	No	Dead		0.9	Dead (min) - Wind [Strength]
UDStlS8			Wx		-1	
UDStlS9	Linear Add	No	Dead		0.9	Dead (min) + Wind [Strength]
UDStlS9			Wy		1	
UDStlS10	Linear Add	No	Dead		0.9	Dead (min) - Wind [Strength]
UDStlS10			Wy		-1	
UDStlS11	Linear Add	No	Dead		1.3	Dead + Live + Static Earthquake [Strength]
UDStlS11			Live		1	
UDStlS11			Ex		1	
UDStlS12	Linear Add	No	Dead		1.3	Dead + Live - Static Earthquake [Strength]
UDStlS12			Live		1	
UDStlS12			Ex		-1	
UDStlS13	Linear Add	No	Dead		1.3	Dead + Live + Static Earthquake [Strength]
UDStlS13			Live		1	
UDStlS13			Ey		1	
UDStlS14	Linear Add	No	Dead		1.3	Dead + Live - Static Earthquake [Strength]
UDStlS14			Live		1	
UDStlS14			Ey		-1	
UDStlS15	Linear Add	No	Dead		0.8	Dead (min) + Static Earthquake [Strength]
UDStlS15			Ex		1	
UDStlS16	Linear Add	No	Dead		0.8	Dead (min) - Static Earthquake [Strength]
UDStlS16			Ex		-1	
UDStlS17	Linear Add	No	Dead		0.8	Dead (min) + Static Earthquake [Strength]
UDStlS17			Ey		1	

Name	Type	Is Auto	Load Name	Mode	SF	Notes
UDStlS18	Linear Add	No	Dead		0.8	Dead (min) - Static Earthquake [Strength]
UDStlS18			Ey		-1	

4 Analysis Results

This chapter provides analysis results.

4.1 Structure Results

Table 4.1 - Base Reactions (Part 1 of 2)

Output Case	Case Type	Step Type	Step Number	Step Label	FX kip	FY kip	FZ kip	MX kip-ft	MY kip-ft
Dead	LinStatic				0	0	153.205	4596.1368	-3676.9094
Live	LinStatic				0	0	172.8	5184	-4147.2
Ex	LinStatic				-6.769	0	0	0	-120.5011
Ey	LinStatic				0	-6.769	0	120.5011	0
Wx	LinStatic				0.806	0	15.399	461.9777	-391.8083
Wy	LinStatic				0	0	0	0	0
UDStlS1	Combination				0	0	214.486	6434.5915	-5147.6732
1.2 DL + 1.6LL	Combination				0	0	460.325	13809.7642	-11047.8113
UDStlS3	Combination				0.806	0	372.045	11161.3419	-8951.2996
UDStlS4	Combination				-0.806	0	341.246	10237.3864	-8167.6831
UDStlS5	Combination				0	0	356.645	10699.3642	-8559.4913
UDStlS6	Combination				0	0	356.645	10699.3642	-8559.4913
UDStlS7	Combination				0.806	0	153.283	4598.5008	-3701.0268
UDStlS8	Combination				-0.806	0	122.485	3674.5454	-2917.4102
UDStlS9	Combination				0	0	137.884	4136.5231	-3309.2185
UDStlS10	Combination				0	0	137.884	4136.5231	-3309.2185
UDStlS11	Combination				-6.769	0	371.966	11158.9778	-9047.6834
UDStlS12	Combination				6.769	0	371.966	11158.9778	-8806.6812
UDStlS13	Combination				0	-6.769	371.966	11279.4789	-8927.1823
UDStlS14	Combination				0	6.769	371.966	11038.4767	-8927.1823
UDStlS15	Combination				-6.769	0	122.564	3676.9094	-3062.0286
UDStlS16	Combination				6.769	0	122.564	3676.9094	-2821.0265
UDStlS17	Combination				0	-6.769	122.564	3797.4105	-2941.5276
UDStlS18	Combination				0	6.769	122.564	3556.4083	-2941.5276
UDStlD1	Combination				0	0	153.205	4596.1368	-3676.9094
UDStlD2	Combination				0	0	326.005	9780.1368	-7824.1094
UDCmpD1	Combination				0	0	153.205	4596.1368	-3676.9094
UDCmpD2	Combination				0	0	326.005	9780.1368	-7824.1094
UDCmpC1	Combination				0	0	214.486	6434.5915	-5147.6732
UDCmpC2	Combination				0	0	183.845	5515.3642	-4412.2913
UDCmpS1	Combination				0	0	214.486	6434.5915	-5147.6732
UDCmpS2	Combination				0	0	460.325	13809.7642	-11047.8113
DConS1	Combination				0	0	214.486	6434.5915	-5147.6732
DConS2	Combination				0	0	460.325	13809.7642	-11047.8113
DConS3	Combination				0.806	0	372.045	11161.3419	-8951.2996
DConS4	Combination				-0.806	0	341.246	10237.3864	-8167.6831
DConS5	Combination				0	0	356.645	10699.3642	-8559.4913
DConS6	Combination				0	0	356.645	10699.3642	-8559.4913
DConS7	Combination				0.806	0	153.283	4598.5008	-3701.0268
DConS8	Combination				-0.806	0	122.485	3674.5454	-2917.4102
DConS9	Combination				0	0	137.884	4136.5231	-3309.2185
DConS10	Combination				0	0	137.884	4136.5231	-3309.2185
DConS11	Combination				-6.769	0	371.966	11158.9778	-9047.6834
DConS12	Combination				6.769	0	371.966	11158.9778	-8806.6812
DConS13	Combination				0	-6.769	371.966	11279.4789	-8927.1823
DConS14	Combination				0	6.769	371.966	11038.4767	-8927.1823
DConS15	Combination				-6.769	0	122.564	3676.9094	-3062.0286
DConS16	Combination				6.769	0	122.564	3676.9094	-2821.0265
DConS17	Combination				0	-6.769	122.564	3797.4105	-2941.5276

Output Case	Case Type	Step Type	Step Number	Step Label	FX kip	FY kip	FZ kip	MX kip-ft	MY kip-ft
DConS18	Combination				0	6.769	122.564	3556.4083	-2941.5276

Table 4.1 - Base Reactions (Part 2 of 2)

MZ kip-ft	X ft	Y ft	Z ft
0	0	0	0
0	0	0	0
203.0695	0	0	0
-162.4556	0	0	0
-24.1785	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
-24.1785	0	0	0
24.1785	0	0	0
0	0	0	0
0	0	0	0
-24.1785	0	0	0
24.1785	0	0	0
0	0	0	0
0	0	0	0
203.0695	0	0	0
-203.0695	0	0	0
-162.4556	0	0	0
162.4556	0	0	0
203.0695	0	0	0
-203.0695	0	0	0
-162.4556	0	0	0
162.4556	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0
-24.1785	0	0	0
24.1785	0	0	0
0	0	0	0
0	0	0	0
-24.1785	0	0	0
24.1785	0	0	0
0	0	0	0
0	0	0	0
203.0695	0	0	0
-203.0695	0	0	0
-162.4556	0	0	0
162.4556	0	0	0
203.0695	0	0	0
-203.0695	0	0	0
-162.4556	0	0	0
162.4556	0	0	0

4.2 Story Results

Table 4.2 - Story Drifts

Story	Output Case	Case Type	Step Type	Step Number	Step Label	Direction	Drift	Label	X ft	Y ft	Z ft
Story2	Dead	LinStatic				X	0	72	24	36	21
Story2	Dead	LinStatic				Y	0.000448	118	24	60	21
Story2	Live	LinStatic				X	0	72	24	36	21
Story2	Live	LinStatic				Y	0.000574	118	24	60	21
Story2	Ex	LinStatic				X	5E-06	49	24	24	21
Story2	Ey	LinStatic				Y	0.000228	3	24	0	21
Story2	Wx	LinStatic				X	5E-06	72	24	36	21
Story2	UDStlS1	Combination				X	0	72	24	36	21
Story2	UDStlS1	Combination				Y	0.000627	118	24	60	21
Story2	1.2 DL + 1.6LL	Combination				X	0	72	24	36	21
Story2	1.2 DL + 1.6LL	Combination				Y	0.001455	118	24	60	21
Story2	UDStlS3	Combination				X	5E-06	72	24	36	21
Story2	UDStlS4	Combination				X	5E-06	49	24	24	21
Story2	UDStlS5	Combination				X	0	72	24	36	21
Story2	UDStlS5	Combination				Y	0.001111	118	24	60	21
Story2	UDStlS6	Combination				X	0	72	24	36	21
Story2	UDStlS6	Combination				Y	0.001111	118	24	60	21
Story2	UDStlS7	Combination				X	5E-06	72	24	36	21
Story2	UDStlS8	Combination				X	5E-06	49	24	24	21
Story2	UDStlS9	Combination				X	0	72	24	36	21
Story2	UDStlS9	Combination				Y	0.000403	118	24	60	21
Story2	UDStlS10	Combination				X	0	72	24	36	21
Story2	UDStlS10	Combination				Y	0.000403	118	24	60	21
Story2	UDStlS11	Combination				X	5E-06	49	24	24	21
Story2	UDStlS12	Combination				X	5E-06	72	24	36	21
Story2	UDStlS13	Combination				Y	0.001384	118	24	60	21
Story2	UDStlS14	Combination				Y	0.001384	3	24	0	21
Story2	UDStlS15	Combination				X	5E-06	49	24	24	21
Story2	UDStlS16	Combination				X	5E-06	72	24	36	21
Story2	UDStlS17	Combination				Y	0.000586	118	24	60	21
Story2	UDStlS18	Combination				Y	0.000586	3	24	0	21
Story2	UDStlD1	Combination				X	0	72	24	36	21
Story2	UDStlD1	Combination				Y	0.000448	118	24	60	21
Story2	UDStlD2	Combination				X	0	72	24	36	21
Story2	UDStlD2	Combination				Y	0.001022	118	24	60	21
Story2	UDCmpD1	Combination				X	0	72	24	36	21
Story2	UDCmpD1	Combination				Y	0.000448	118	24	60	21
Story2	UDCmpD2	Combination				X	0	72	24	36	21
Story2	UDCmpD2	Combination				Y	0.001022	118	24	60	21
Story2	UDCmpC1	Combination				X	0	72	24	36	21
Story2	UDCmpC1	Combination				Y	0.000627	118	24	60	21
Story2	UDCmpC2	Combination				X	0	72	24	36	21
Story2	UDCmpC2	Combination				Y	0.000537	118	24	60	21
Story2	UDCmpS1	Combination				X	0	72	24	36	21
Story2	UDCmpS1	Combination				Y	0.000627	118	24	60	21
Story2	UDCmpS2	Combination				X	0	72	24	36	21
Story2	UDCmpS2	Combination				Y	0.001455	118	24	60	21
Story2	DConS1	Combination				X	0	72	24	36	21
Story2	DConS1	Combination				Y	0.000627	118	24	60	21
Story2	DConS2	Combination				X	0	72	24	36	21
Story2	DConS2	Combination				Y	0.001455	118	24	60	21

Story	Output Case	Case Type	Step Type	Step Number	Step Label	Direction	Drift	Label	X ft	Y ft	Z ft
Story2	DConS3	Combination				X	5E-06	72	24	36	21
Story2	DConS4	Combination				X	5E-06	49	24	24	21
Story2	DConS5	Combination				X	0	72	24	36	21
Story2	DConS5	Combination				Y	0.001111	118	24	60	21
Story2	DConS6	Combination				X	0	72	24	36	21
Story2	DConS6	Combination				Y	0.001111	118	24	60	21
Story2	DConS7	Combination				X	5E-06	72	24	36	21
Story2	DConS8	Combination				X	5E-06	49	24	24	21
Story2	DConS9	Combination				X	0	72	24	36	21
Story2	DConS9	Combination				Y	0.000403	118	24	60	21
Story2	DConS10	Combination				X	0	72	24	36	21
Story2	DConS10	Combination				Y	0.000403	118	24	60	21
Story2	DConS11	Combination				X	5E-06	49	24	24	21
Story2	DConS12	Combination				X	5E-06	72	24	36	21
Story2	DConS13	Combination				Y	0.001384	118	24	60	21
Story2	DConS14	Combination				Y	0.001384	3	24	0	21
Story2	DConS15	Combination				X	5E-06	49	24	24	21
Story2	DConS16	Combination				X	5E-06	72	24	36	21
Story2	DConS17	Combination				Y	0.000586	118	24	60	21
Story2	DConS18	Combination				Y	0.000586	3	24	0	21
Story1	Dead	LinStatic				X	0.000122	48	48	24	15
Story1	Live	LinStatic				X	0.000145	48	48	24	15
Story1	Ex	LinStatic				X	0.000214	71	48	36	15
Story1	Ey	LinStatic				Y	0.000344	70	0	36	15
Story1	Wx	LinStatic				X	4.8E-05	70	0	36	15
Story1	UDStlS1	Combination				X	0.000171	48	48	24	15
Story1	1.2 DL + 1.6LL	Combination				X	0.000378	48	48	24	15
Story1	UDStlS3	Combination				X	0.00034	70	0	36	15
Story1	UDStlS4	Combination				X	0.000315	48	48	24	15
Story1	UDStlS5	Combination				X	0.000291	48	48	24	15
Story1	UDStlS6	Combination				X	0.000291	48	48	24	15
Story1	UDStlS7	Combination				X	0.000158	70	0	36	15
Story1	UDStlS8	Combination				X	0.000134	48	48	24	15
Story1	UDStlS9	Combination				X	0.00011	48	48	24	15
Story1	UDStlS10	Combination				X	0.00011	48	48	24	15
Story1	UDStlS11	Combination				X	0.000518	71	48	36	15
Story1	UDStlS12	Combination				X	0.000518	70	0	36	15
Story1	UDStlS13	Combination				Y	0.000357	117	48	60	15
Story1	UDStlS14	Combination				Y	0.000357	1	0	0	15
Story1	UDStlS15	Combination				X	0.000312	71	48	36	15
Story1	UDStlS16	Combination				X	0.000312	70	0	36	15
Story1	UDStlS17	Combination				Y	0.000347	117	48	60	15
Story1	UDStlS18	Combination				Y	0.000347	1	0	0	15
Story1	UDStlD1	Combination				X	0.000122	48	48	24	15
Story1	UDStlD2	Combination				X	0.000267	48	48	24	15
Story1	UDCmpD1	Combination				X	0.000122	48	48	24	15
Story1	UDCmpD2	Combination				X	0.000267	48	48	24	15
Story1	UDCmpC1	Combination				X	0.000171	48	48	24	15
Story1	UDCmpC2	Combination				X	0.000147	48	48	24	15
Story1	UDCmpS1	Combination				X	0.000171	48	48	24	15
Story1	UDCmpS2	Combination				X	0.000378	48	48	24	15
Story1	DConS1	Combination				X	0.000171	48	48	24	15
Story1	DConS2	Combination				X	0.000378	48	48	24	15
Story1	DConS3	Combination				X	0.00034	70	0	36	15
Story1	DConS4	Combination				X	0.000315	48	48	24	15

Story	Output Case	Case Type	Step Type	Step Number	Step Label	Direction	Drift	Label	X ft	Y ft	Z ft
Story1	DConS5	Combination				X	0.000291	48	48	24	15
Story1	DConS6	Combination				X	0.000291	48	48	24	15
Story1	DConS7	Combination				X	0.000158	70	0	36	15
Story1	DConS8	Combination				X	0.000134	48	48	24	15
Story1	DConS9	Combination				X	0.00011	48	48	24	15
Story1	DConS10	Combination				X	0.00011	48	48	24	15
Story1	DConS11	Combination				X	0.000518	71	48	36	15
Story1	DConS12	Combination				X	0.000518	70	0	36	15
Story1	DConS13	Combination				Y	0.000357	117	48	60	15
Story1	DConS14	Combination				Y	0.000357	1	0	0	15
Story1	DConS15	Combination				X	0.000312	71	48	36	15
Story1	DConS16	Combination				X	0.000312	70	0	36	15
Story1	DConS17	Combination				Y	0.000347	117	48	60	15
Story1	DConS18	Combination				Y	0.000347	1	0	0	15

Table 4.3 - Story Max Over Avg Drifts

Story	Output Case	Case Type	Step Type	Step Number	Step Label	Direction	Max Drift in	Avg Drift in	Ratio
Story2	Dead	LinStatic				X	0	0	3.046
Story2	Dead	LinStatic				Y	0.032231	0	74390000000000
Story2	Live	LinStatic				X	0	0	3.059
Story2	Live	LinStatic				Y	0.041321	0	81440000000000
Story2	Ex	LinStatic				X	0.000395	0.000306	1.289
Story2	Ey	LinStatic				Y	0.016439	0.016314	1.008
Story2	Wx	LinStatic				X	0.000373	0.000293	1.275
Story2	UDStlS1	Combination				X	0	0	3.046
Story2	UDStlS1	Combination				Y	0.045124	0	74590000000000
Story2	1.2 DL + 1.6LL	Combination				X	0	0	3.054
Story2	1.2 DL + 1.6LL	Combination				Y	0.104791	0	78960000000000
Story2	UDStlS3	Combination				X	0.000373	0.000293	1.275
Story2	UDStlS4	Combination				X	0.000373	0.000293	1.275
Story2	UDStlS5	Combination				X	0	0	3.052
Story2	UDStlS5	Combination				Y	0.079998	0	77640000000000
Story2	UDStlS6	Combination				X	0	0	3.052
Story2	UDStlS6	Combination				Y	0.079998	0	77640000000000
Story2	UDStlS7	Combination				X	0.000373	0.000293	1.275
Story2	UDStlS8	Combination				X	0.000373	0.000293	1.275
Story2	UDStlS9	Combination				X	0	0	3.046
Story2	UDStlS9	Combination				Y	0.029008	0	74320000000000
Story2	UDStlS10	Combination				X	0	0	3.046
Story2	UDStlS10	Combination				Y	0.029008	0	74320000000000
Story2	UDStlS11	Combination				X	0.000395	0.000306	1.289
Story2	UDStlS12	Combination				X	0.000395	0.000306	1.289
Story2	UDStlS13	Combination				Y	0.09966	0.016439	6.062
Story2	UDStlS14	Combination				Y	0.09966	0.016439	6.062
Story2	UDStlS15	Combination				X	0.000395	0.000306	1.289
Story2	UDStlS16	Combination				X	0.000395	0.000306	1.289
Story2	UDStlS17	Combination				Y	0.042224	0.016439	2.569
Story2	UDStlS18	Combination				Y	0.042224	0.016439	2.569
Story2	UDStlD1	Combination				X	0	0	3.046
Story2	UDStlD1	Combination				Y	0.032231	0	74390000000000
Story2	UDStlD2	Combination				X	0	0	3.053
Story2	UDStlD2	Combination				Y	0.073552	0	77870000000000
Story2	UDCmpD1	Combination				X	0	0	3.046

Story	Output Case	Case Type	Step Type	Step Number	Step Label	Direction	Max Drift in	Avg Drift in	Ratio
Story2	UDCmpD1	Combination				Y	0.032231	0	74390000000000
Story2	UDCmpD2	Combination				X	0	0	3.053
Story2	UDCmpD2	Combination				Y	0.073552	0	77870000000000
Story2	UDCmpC1	Combination				X	0	0	3.046
Story2	UDCmpC1	Combination				Y	0.045124	0	74590000000000
Story2	UDCmpC2	Combination				X	0	0	3.046
Story2	UDCmpC2	Combination				Y	0.038677	0	73950000000000
Story2	UDCmpS1	Combination				X	0	0	3.046
Story2	UDCmpS1	Combination				Y	0.045124	0	74590000000000
Story2	UDCmpS2	Combination				X	0	0	3.054
Story2	UDCmpS2	Combination				Y	0.104791	0	78960000000000
Story2	DConS1	Combination				X	0	0	3.046
Story2	DConS1	Combination				Y	0.045124	0	74590000000000
Story2	DConS2	Combination				X	0	0	3.054
Story2	DConS2	Combination				Y	0.104791	0	78960000000000
Story2	DConS3	Combination				X	0.000373	0.000293	1.275
Story2	DConS4	Combination				X	0.000373	0.000293	1.275
Story2	DConS5	Combination				X	0	0	3.052
Story2	DConS5	Combination				Y	0.079998	0	77640000000000
Story2	DConS6	Combination				X	0	0	3.052
Story2	DConS6	Combination				Y	0.079998	0	77640000000000
Story2	DConS7	Combination				X	0.000373	0.000293	1.275
Story2	DConS8	Combination				X	0.000373	0.000293	1.275
Story2	DConS9	Combination				X	0	0	3.046
Story2	DConS9	Combination				Y	0.029008	0	74320000000000
Story2	DConS10	Combination				X	0	0	3.046
Story2	DConS10	Combination				Y	0.029008	0	74320000000000
Story2	DConS11	Combination				X	0.000395	0.000306	1.289
Story2	DConS12	Combination				X	0.000395	0.000306	1.289
Story2	DConS13	Combination				Y	0.09966	0.016439	6.062
Story2	DConS14	Combination				Y	0.09966	0.016439	6.062
Story2	DConS15	Combination				X	0.000395	0.000306	1.289
Story2	DConS16	Combination				X	0.000395	0.000306	1.289
Story2	DConS17	Combination				Y	0.042224	0.016439	2.569
Story2	DConS18	Combination				Y	0.042224	0.016439	2.569
Story1	Dead	LinStatic				X	0.021979	0	18580000000000
Story1	Live	LinStatic				X	0.026062	0	17120000000000
Story1	Ex	LinStatic				X	0.038583	0.038302	1.007
Story1	Ey	LinStatic				Y	0.061863	0.061801	1.001
Story1	Wx	LinStatic				X	0.008711	0.006507	1.339
Story1	UDStlS1	Combination				X	0.03077	0	18600000000000
Story1	1.2 DL + 1.6LL	Combination				X	0.068073	0	17680000000000
Story1	UDStlS3	Combination				X	0.061147	0.006507	9.397
Story1	UDStlS4	Combination				X	0.056739	0.006507	8.72
Story1	UDStlS5	Combination				X	0.052436	0	17790000000000
Story1	UDStlS6	Combination				X	0.052436	0	17790000000000
Story1	UDStlS7	Combination				X	0.028492	0.006507	4.379
Story1	UDStlS8	Combination				X	0.024083	0.006507	3.701
Story1	UDStlS9	Combination				X	0.019781	0	18560000000000
Story1	UDStlS10	Combination				X	0.019781	0	18560000000000
Story1	UDStlS11	Combination				X	0.093217	0.038583	2.416
Story1	UDStlS12	Combination				X	0.093217	0.038583	2.416
Story1	UDStlS13	Combination				Y	0.064348	0.061739	1.042
Story1	UDStlS14	Combination				Y	0.064348	0.061739	1.042

Story	Output Case	Case Type	Step Type	Step Number	Step Label	Direction	Max Drift in	Avg Drift in	Ratio
Story1	UDStlS15	Combination				X	0.056166	0.038583	1.456
Story1	UDStlS16	Combination				X	0.056166	0.038583	1.456
Story1	UDStlS17	Combination				Y	0.062549	0.061739	1.013
Story1	UDStlS18	Combination				Y	0.062549	0.061739	1.013
Story1	UDStlD1	Combination				X	0.021979	0	18580000000000
Story1	UDStlD2	Combination				X	0.04804	0	17770000000000
Story1	UDCmpD1	Combination				X	0.021979	0	18580000000000
Story1	UDCmpD2	Combination				X	0.04804	0	17770000000000
Story1	UDCmpC1	Combination				X	0.03077	0	18600000000000
Story1	UDCmpC2	Combination				X	0.026374	0	18580000000000
Story1	UDCmpS1	Combination				X	0.03077	0	18600000000000
Story1	UDCmpS2	Combination				X	0.068073	0	17680000000000
Story1	DConS1	Combination				X	0.03077	0	18600000000000
Story1	DConS2	Combination				X	0.068073	0	17680000000000
Story1	DConS3	Combination				X	0.061147	0.006507	9.397
Story1	DConS4	Combination				X	0.056739	0.006507	8.72
Story1	DConS5	Combination				X	0.052436	0	17790000000000
Story1	DConS6	Combination				X	0.052436	0	17790000000000
Story1	DConS7	Combination				X	0.028492	0.006507	4.379
Story1	DConS8	Combination				X	0.024083	0.006507	3.701
Story1	DConS9	Combination				X	0.019781	0	18560000000000
Story1	DConS10	Combination				X	0.019781	0	18560000000000
Story1	DConS11	Combination				X	0.093217	0.038583	2.416
Story1	DConS12	Combination				X	0.093217	0.038583	2.416
Story1	DConS13	Combination				Y	0.064348	0.061739	1.042
Story1	DConS14	Combination				Y	0.064348	0.061739	1.042
Story1	DConS15	Combination				X	0.056166	0.038583	1.456
Story1	DConS16	Combination				X	0.056166	0.038583	1.456
Story1	DConS17	Combination				Y	0.062549	0.061739	1.013
Story1	DConS18	Combination				Y	0.062549	0.061739	1.013

Table 4.4 - Story Forces (Part 1 of 2)

Story	Location	Output Case	Case Type	Step Type	Step Number	Step Label	P kip	VX kip	VY kip	T kip-ft
Story2	Top	Dead	LinStatic				8.409	0	0	0
Story2	Bottom	Dead	LinStatic				134.035	0	0	0
Story2	Top	Live	LinStatic				10.8	0	0	0
Story2	Bottom	Live	LinStatic				162	0	0	0
Story2	Top	Ex	LinStatic				0	-3.161	0	94.8316
Story2	Bottom	Ex	LinStatic				0	-3.161	0	94.8316
Story2	Top	Ey	LinStatic				0	0	-3.161	-75.8653
Story2	Bottom	Ey	LinStatic				0	0	-3.161	-75.8653
Story2	Top	Wx	LinStatic				0.989	0.057	0	-1.7122
Story2	Bottom	Wx	LinStatic				14.466	0.763	0	-22.8849
Story2	Top	Wy	LinStatic				0	0	0	0
Story2	Bottom	Wy	LinStatic				0	0	0	0
Story2	Top	UDStlS1	Combination				11.773	0	0	0
Story2	Bottom	UDStlS1	Combination				187.648	0	0	0
Story2	Top	1.2 DL + 1.6LL	Combination				27.371	0	0	0
Story2	Bottom	1.2 DL + 1.6LL	Combination				420.041	0	0	0
Story2	Top	UDStlS3	Combination				21.88	0.057	0	-1.7122
Story2	Bottom	UDStlS3	Combination				337.307	0.763	0	-22.8849
Story2	Top	UDStlS4	Combination				19.902	-0.057	0	1.7122
Story2	Bottom	UDStlS4	Combination				308.376	-0.763	0	22.8849

Story	Location	Output Case	Case Type	Step Type	Step Number	Step Label	P kip	VX kip	VY kip	T kip-ft
Story2	Top	UDStlS5	Combination				20.891	0	0	0
Story2	Bottom	UDStlS5	Combination				322.841	0	0	0
Story2	Top	UDStlS6	Combination				20.891	0	0	0
Story2	Bottom	UDStlS6	Combination				322.841	0	0	0
Story2	Top	UDStlS7	Combination				8.558	0.057	0	-1.7122
Story2	Bottom	UDStlS7	Combination				135.097	0.763	0	-22.8849
Story2	Top	UDStlS8	Combination				6.579	-0.057	0	1.7122
Story2	Bottom	UDStlS8	Combination				106.165	-0.763	0	22.8849
Story2	Top	UDStlS9	Combination				7.568	0	0	0
Story2	Bottom	UDStlS9	Combination				120.631	0	0	0
Story2	Top	UDStlS10	Combination				7.568	0	0	0
Story2	Bottom	UDStlS10	Combination				120.631	0	0	0
Story2	Top	UDStlS11	Combination				21.732	-3.161	0	94.8316
Story2	Bottom	UDStlS11	Combination				336.245	-3.161	0	94.8316
Story2	Top	UDStlS12	Combination				21.732	3.161	0	-94.8316
Story2	Bottom	UDStlS12	Combination				336.245	3.161	0	-94.8316
Story2	Top	UDStlS13	Combination				21.732	0	-3.161	-75.8653
Story2	Bottom	UDStlS13	Combination				336.245	0	-3.161	-75.8653
Story2	Top	UDStlS14	Combination				21.732	0	3.161	75.8653
Story2	Bottom	UDStlS14	Combination				336.245	0	3.161	75.8653
Story2	Top	UDStlS15	Combination				6.727	-3.161	0	94.8316
Story2	Bottom	UDStlS15	Combination				107.228	-3.161	0	94.8316
Story2	Top	UDStlS16	Combination				6.727	3.161	0	-94.8316
Story2	Bottom	UDStlS16	Combination				107.228	3.161	0	-94.8316
Story2	Top	UDStlS17	Combination				6.727	0	-3.161	-75.8653
Story2	Bottom	UDStlS17	Combination				107.228	0	-3.161	-75.8653
Story2	Top	UDStlS18	Combination				6.727	0	3.161	75.8653
Story2	Bottom	UDStlS18	Combination				107.228	0	3.161	75.8653
Story2	Top	UDStlD1	Combination				8.409	0	0	0
Story2	Bottom	UDStlD1	Combination				134.035	0	0	0
Story2	Top	UDStlD2	Combination				19.209	0	0	0
Story2	Bottom	UDStlD2	Combination				296.035	0	0	0
Story2	Top	UDCmpD1	Combination				8.409	0	0	0
Story2	Bottom	UDCmpD1	Combination				134.035	0	0	0
Story2	Top	UDCmpD2	Combination				19.209	0	0	0
Story2	Bottom	UDCmpD2	Combination				296.035	0	0	0
Story2	Top	UDCmpC1	Combination				11.773	0	0	0
Story2	Bottom	UDCmpC1	Combination				187.648	0	0	0
Story2	Top	UDCmpC2	Combination				10.091	0	0	0
Story2	Bottom	UDCmpC2	Combination				160.841	0	0	0
Story2	Top	UDCmpS1	Combination				11.773	0	0	0
Story2	Bottom	UDCmpS1	Combination				187.648	0	0	0
Story2	Top	UDCmpS2	Combination				27.371	0	0	0
Story2	Bottom	UDCmpS2	Combination				420.041	0	0	0
Story2	Top	DConS1	Combination				11.773	0	0	0
Story2	Bottom	DConS1	Combination				187.648	0	0	0
Story2	Top	DConS2	Combination				27.371	0	0	0
Story2	Bottom	DConS2	Combination				420.041	0	0	0
Story2	Top	DConS3	Combination				21.88	0.057	0	-1.7122
Story2	Bottom	DConS3	Combination				337.307	0.763	0	-22.8849
Story2	Top	DConS4	Combination				19.902	-0.057	0	1.7122
Story2	Bottom	DConS4	Combination				308.376	-0.763	0	22.8849
Story2	Top	DConS5	Combination				20.891	0	0	0
Story2	Bottom	DConS5	Combination				322.841	0	0	0
Story2	Top	DConS6	Combination				20.891	0	0	0

Story	Location	Output Case	Case Type	Step Type	Step Number	Step Label	P kip	VX kip	VY kip	T kip-ft
Story2	Bottom	DConS6	Combination				322.841	0	0	0
Story2	Top	DConS7	Combination				8.558	0.057	0	-1.7122
Story2	Bottom	DConS7	Combination				135.097	0.763	0	-22.8849
Story2	Top	DConS8	Combination				6.579	-0.057	0	1.7122
Story2	Bottom	DConS8	Combination				106.165	-0.763	0	22.8849
Story2	Top	DConS9	Combination				7.568	0	0	0
Story2	Bottom	DConS9	Combination				120.631	0	0	0
Story2	Top	DConS10	Combination				7.568	0	0	0
Story2	Bottom	DConS10	Combination				120.631	0	0	0
Story2	Top	DConS11	Combination				21.732	-3.161	0	94.8316
Story2	Bottom	DConS11	Combination				336.245	-3.161	0	94.8316
Story2	Top	DConS12	Combination				21.732	3.161	0	-94.8316
Story2	Bottom	DConS12	Combination				336.245	3.161	0	-94.8316
Story2	Top	DConS13	Combination				21.732	0	-3.161	-75.8653
Story2	Bottom	DConS13	Combination				336.245	0	-3.161	-75.8653
Story2	Top	DConS14	Combination				21.732	0	3.161	75.8653
Story2	Bottom	DConS14	Combination				336.245	0	3.161	75.8653
Story2	Top	DConS15	Combination				6.727	-3.161	0	94.8316
Story2	Bottom	DConS15	Combination				107.228	-3.161	0	94.8316
Story2	Top	DConS16	Combination				6.727	3.161	0	-94.8316
Story2	Bottom	DConS16	Combination				107.228	3.161	0	-94.8316
Story2	Top	DConS17	Combination				6.727	0	-3.161	-75.8653
Story2	Bottom	DConS17	Combination				107.228	0	-3.161	-75.8653
Story2	Top	DConS18	Combination				6.727	0	3.161	75.8653
Story2	Bottom	DConS18	Combination				107.228	0	3.161	75.8653
Story1	Top	Dead	LinStatic				147.915	0	0	0
Story1	Bottom	Dead	LinStatic				153.205	0	0	0
Story1	Top	Live	LinStatic				172.8	0	0	0
Story1	Bottom	Live	LinStatic				172.8	0	0	0
Story1	Top	Ex	LinStatic				0	-6.769	0	203.0695
Story1	Bottom	Ex	LinStatic				0	-6.769	0	203.0695
Story1	Top	Ey	LinStatic				0	0	-6.769	-162.4556
Story1	Bottom	Ey	LinStatic				0	0	-6.769	-162.4556
Story1	Top	Wx	LinStatic				15.399	0.806	0	-24.1785
Story1	Bottom	Wx	LinStatic				15.399	0.806	0	-24.1785
Story1	Top	Wy	LinStatic				0	0	0	0
Story1	Bottom	Wy	LinStatic				0	0	0	0
Story1	Top	UDStlS1	Combination				207.081	0	0	0
Story1	Bottom	UDStlS1	Combination				214.486	0	0	0
Story1	Top	1.2 DL + 1.6LL	Combination				453.978	0	0	0
Story1	Bottom	1.2 DL + 1.6LL	Combination				460.325	0	0	0
Story1	Top	UDStlS3	Combination				365.697	0.806	0	-24.1785
Story1	Bottom	UDStlS3	Combination				372.045	0.806	0	-24.1785
Story1	Top	UDStlS4	Combination				334.899	-0.806	0	24.1785
Story1	Bottom	UDStlS4	Combination				341.246	-0.806	0	24.1785
Story1	Top	UDStlS5	Combination				350.298	0	0	0
Story1	Bottom	UDStlS5	Combination				356.645	0	0	0
Story1	Top	UDStlS6	Combination				350.298	0	0	0
Story1	Bottom	UDStlS6	Combination				356.645	0	0	0
Story1	Top	UDStlS7	Combination				148.523	0.806	0	-24.1785
Story1	Bottom	UDStlS7	Combination				153.283	0.806	0	-24.1785
Story1	Top	UDStlS8	Combination				117.724	-0.806	0	24.1785
Story1	Bottom	UDStlS8	Combination				122.485	-0.806	0	24.1785
Story1	Top	UDStlS9	Combination				133.124	0	0	0
Story1	Bottom	UDStlS9	Combination				137.884	0	0	0

Story	Location	Output Case	Case Type	Step Type	Step Number	Step Label	P kip	VX kip	VY kip	T kip-ft
Story1	Top	UDStlS10	Combination				133.124	0	0	0
Story1	Bottom	UDStlS10	Combination				137.884	0	0	0
Story1	Top	UDStlS11	Combination				365.09	-6.769	0	203.0695
Story1	Bottom	UDStlS11	Combination				371.966	-6.769	0	203.0695
Story1	Top	UDStlS12	Combination				365.09	6.769	0	-203.0695
Story1	Bottom	UDStlS12	Combination				371.966	6.769	0	-203.0695
Story1	Top	UDStlS13	Combination				365.09	0	-6.769	-162.4556
Story1	Bottom	UDStlS13	Combination				371.966	0	-6.769	-162.4556
Story1	Top	UDStlS14	Combination				365.09	0	6.769	162.4556
Story1	Bottom	UDStlS14	Combination				371.966	0	6.769	162.4556
Story1	Top	UDStlS15	Combination				118.332	-6.769	0	203.0695
Story1	Bottom	UDStlS15	Combination				122.564	-6.769	0	203.0695
Story1	Top	UDStlS16	Combination				118.332	6.769	0	-203.0695
Story1	Bottom	UDStlS16	Combination				122.564	6.769	0	-203.0695
Story1	Top	UDStlS17	Combination				118.332	0	-6.769	-162.4556
Story1	Bottom	UDStlS17	Combination				122.564	0	-6.769	-162.4556
Story1	Top	UDStlS18	Combination				118.332	0	6.769	162.4556
Story1	Bottom	UDStlS18	Combination				122.564	0	6.769	162.4556
Story1	Top	UDStlD1	Combination				147.915	0	0	0
Story1	Bottom	UDStlD1	Combination				153.205	0	0	0
Story1	Top	UDStlD2	Combination				320.715	0	0	0
Story1	Bottom	UDStlD2	Combination				326.005	0	0	0
Story1	Top	UDCmpD1	Combination				147.915	0	0	0
Story1	Bottom	UDCmpD1	Combination				153.205	0	0	0
Story1	Top	UDCmpD2	Combination				320.715	0	0	0
Story1	Bottom	UDCmpD2	Combination				326.005	0	0	0
Story1	Top	UDCmpC1	Combination				207.081	0	0	0
Story1	Bottom	UDCmpC1	Combination				214.486	0	0	0
Story1	Top	UDCmpC2	Combination				177.498	0	0	0
Story1	Bottom	UDCmpC2	Combination				183.845	0	0	0
Story1	Top	UDCmpS1	Combination				207.081	0	0	0
Story1	Bottom	UDCmpS1	Combination				214.486	0	0	0
Story1	Top	UDCmpS2	Combination				453.978	0	0	0
Story1	Bottom	UDCmpS2	Combination				460.325	0	0	0
Story1	Top	DConS1	Combination				207.081	0	0	0
Story1	Bottom	DConS1	Combination				214.486	0	0	0
Story1	Top	DConS2	Combination				453.978	0	0	0
Story1	Bottom	DConS2	Combination				460.325	0	0	0
Story1	Top	DConS3	Combination				365.697	0.806	0	-24.1785
Story1	Bottom	DConS3	Combination				372.045	0.806	0	-24.1785
Story1	Top	DConS4	Combination				334.899	-0.806	0	24.1785
Story1	Bottom	DConS4	Combination				341.246	-0.806	0	24.1785
Story1	Top	DConS5	Combination				350.298	0	0	0
Story1	Bottom	DConS5	Combination				356.645	0	0	0
Story1	Top	DConS6	Combination				350.298	0	0	0
Story1	Bottom	DConS6	Combination				356.645	0	0	0
Story1	Top	DConS7	Combination				148.523	0.806	0	-24.1785
Story1	Bottom	DConS7	Combination				153.283	0.806	0	-24.1785
Story1	Top	DConS8	Combination				117.724	-0.806	0	24.1785
Story1	Bottom	DConS8	Combination				122.485	-0.806	0	24.1785
Story1	Top	DConS9	Combination				133.124	0	0	0
Story1	Bottom	DConS9	Combination				137.884	0	0	0
Story1	Top	DConS10	Combination				133.124	0	0	0
Story1	Bottom	DConS10	Combination				137.884	0	0	0
Story1	Top	DConS11	Combination				365.09	-6.769	0	203.0695

Story	Location	Output Case	Case Type	Step Type	Step Number	Step Label	P kip	VX kip	VY kip	T kip-ft
Story1	Bottom	DConS11	Combination				371.966	-6.769	0	203.0695
Story1	Top	DConS12	Combination				365.09	6.769	0	-203.0695
Story1	Bottom	DConS12	Combination				371.966	6.769	0	-203.0695
Story1	Top	DConS13	Combination				365.09	0	-6.769	-162.4556
Story1	Bottom	DConS13	Combination				371.966	0	-6.769	-162.4556
Story1	Top	DConS14	Combination				365.09	0	6.769	162.4556
Story1	Bottom	DConS14	Combination				371.966	0	6.769	162.4556
Story1	Top	DConS15	Combination				118.332	-6.769	0	203.0695
Story1	Bottom	DConS15	Combination				122.564	-6.769	0	203.0695
Story1	Top	DConS16	Combination				118.332	6.769	0	-203.0695
Story1	Bottom	DConS16	Combination				122.564	6.769	0	-203.0695
Story1	Top	DConS17	Combination				118.332	0	-6.769	-162.4556
Story1	Bottom	DConS17	Combination				122.564	0	-6.769	-162.4556
Story1	Top	DConS18	Combination				118.332	0	6.769	162.4556
Story1	Bottom	DConS18	Combination				122.564	0	6.769	162.4556

Table 4.4 - Story Forces (Part 2 of 2)

Story	Location	MX kip-ft	MY kip-ft
Story2	Top	252.2761	-201.8209
Story2	Bottom	4021.035	-3216.828
Story2	Top	324	-259.2
Story2	Bottom	4860	-3888
Story2	Top	0	0
Story2	Bottom	0	-18.9663
Story2	Top	0	0
Story2	Bottom	18.9663	0
Story2	Top	29.6777	-23.7421
Story2	Bottom	433.974	-377.3548
Story2	Top	0	0
Story2	Bottom	0	0
Story2	Top	353.1865	-282.5492
Story2	Bottom	5629.449	-4503.5592
Story2	Top	821.1313	-656.905
Story2	Bottom	12601.242	-10080.9936
Story2	Top	656.409	-525.1272
Story2	Bottom	10119.216	-8125.5484
Story2	Top	597.0536	-477.6429
Story2	Bottom	9251.268	-7370.8388
Story2	Top	626.7313	-501.385
Story2	Bottom	9685.242	-7748.1936
Story2	Top	626.7313	-501.385
Story2	Bottom	9685.242	-7748.1936
Story2	Top	256.7261	-205.3809
Story2	Bottom	4052.9055	-3272.5
Story2	Top	197.3708	-157.8966
Story2	Bottom	3184.9575	-2517.7904
Story2	Top	227.0485	-181.6388
Story2	Bottom	3618.9315	-2895.1452
Story2	Top	227.0485	-181.6388
Story2	Bottom	3618.9315	-2895.1452
Story2	Top	651.9589	-521.5671
Story2	Bottom	10087.3455	-8088.8427
Story2	Top	651.9589	-521.5671
Story2	Bottom	10087.3455	-8050.9101

Story	Location	MX kip-ft	MY kip-ft
Story2	Top	651.9589	-521.5671
Story2	Bottom	10106.3119	-8069.8764
Story2	Top	651.9589	-521.5671
Story2	Bottom	10068.3792	-8069.8764
Story2	Top	201.8209	-161.4567
Story2	Bottom	3216.828	-2592.4287
Story2	Top	201.8209	-161.4567
Story2	Bottom	3216.828	-2554.4961
Story2	Top	201.8209	-161.4567
Story2	Bottom	3235.7943	-2573.4624
Story2	Top	201.8209	-161.4567
Story2	Bottom	3197.8617	-2573.4624
Story2	Top	252.2761	-201.8209
Story2	Bottom	4021.035	-3216.828
Story2	Top	576.2761	-461.0209
Story2	Bottom	8881.035	-7104.828
Story2	Top	252.2761	-201.8209
Story2	Bottom	4021.035	-3216.828
Story2	Top	576.2761	-461.0209
Story2	Bottom	8881.035	-7104.828
Story2	Top	353.1865	-282.5492
Story2	Bottom	5629.449	-4503.5592
Story2	Top	302.7313	-242.185
Story2	Bottom	4825.242	-3860.1936
Story2	Top	353.1865	-282.5492
Story2	Bottom	5629.449	-4503.5592
Story2	Top	821.1313	-656.905
Story2	Bottom	12601.242	-10080.9936
Story2	Top	353.1865	-282.5492
Story2	Bottom	5629.449	-4503.5592
Story2	Top	821.1313	-656.905
Story2	Bottom	12601.242	-10080.9936
Story2	Top	656.409	-525.1272
Story2	Bottom	10119.216	-8125.5484
Story2	Top	597.0536	-477.6429
Story2	Bottom	9251.268	-7370.8388
Story2	Top	626.7313	-501.385
Story2	Bottom	9685.242	-7748.1936
Story2	Top	626.7313	-501.385
Story2	Bottom	9685.242	-7748.1936
Story2	Top	256.7261	-205.3809
Story2	Bottom	4052.9055	-3272.5
Story2	Top	197.3708	-157.8966
Story2	Bottom	3184.9575	-2517.7904
Story2	Top	227.0485	-181.6388
Story2	Bottom	3618.9315	-2895.1452
Story2	Top	227.0485	-181.6388
Story2	Bottom	3618.9315	-2895.1452
Story2	Top	651.9589	-521.5671
Story2	Bottom	10087.3455	-8088.8427
Story2	Top	651.9589	-521.5671
Story2	Bottom	10087.3455	-8050.9101
Story2	Top	651.9589	-521.5671
Story2	Bottom	10106.3119	-8069.8764
Story2	Top	651.9589	-521.5671

Story	Location	MX kip-ft	MY kip-ft
Story2	Bottom	10068.3792	-8069.8764
Story2	Top	201.8209	-161.4567
Story2	Bottom	3216.828	-2592.4287
Story2	Top	201.8209	-161.4567
Story2	Bottom	3216.828	-2554.4961
Story2	Top	201.8209	-161.4567
Story2	Bottom	3235.7943	-2573.4624
Story2	Top	201.8209	-161.4567
Story2	Bottom	3197.8617	-2573.4624
Story1	Top	4437.4552	-3549.9642
Story1	Bottom	4596.1368	-3676.9094
Story1	Top	5184	-4147.2
Story1	Bottom	5184	-4147.2
Story1	Top	0	-18.9663
Story1	Bottom	0	-120.5011
Story1	Top	18.9663	0
Story1	Bottom	120.5011	0
Story1	Top	461.9777	-403.8975
Story1	Bottom	461.9777	-391.8083
Story1	Top	0	0
Story1	Bottom	0	0
Story1	Top	6212.4373	-4969.9499
Story1	Bottom	6434.5915	-5147.6732
Story1	Top	13619.3463	-10895.477
Story1	Bottom	13809.7642	-11047.8113
Story1	Top	10970.924	-8811.0546
Story1	Bottom	11161.3419	-8951.2996
Story1	Top	10046.9686	-8003.2595
Story1	Bottom	10237.3864	-8167.6831
Story1	Top	10508.9463	-8407.157
Story1	Bottom	10699.3642	-8559.4913
Story1	Top	10508.9463	-8407.157
Story1	Bottom	10699.3642	-8559.4913
Story1	Top	4455.6874	-3598.8653
Story1	Bottom	4598.5008	-3701.0268
Story1	Top	3531.732	-2791.0702
Story1	Bottom	3674.5454	-2917.4102
Story1	Top	3993.7097	-3194.9678
Story1	Bottom	4136.5231	-3309.2185
Story1	Top	3993.7097	-3194.9678
Story1	Bottom	4136.5231	-3309.2185
Story1	Top	10952.6918	-8781.1198
Story1	Bottom	11158.9778	-9047.6834
Story1	Top	10952.6918	-8743.1871
Story1	Bottom	11158.9778	-8806.6812
Story1	Top	10971.6581	-8762.1534
Story1	Bottom	11279.4789	-8927.1823
Story1	Top	10933.7255	-8762.1534
Story1	Bottom	11038.4767	-8927.1823
Story1	Top	3549.9642	-2858.9377
Story1	Bottom	3676.9094	-3062.0286
Story1	Top	3549.9642	-2821.005
Story1	Bottom	3676.9094	-2821.0265
Story1	Top	3568.9305	-2839.9713
Story1	Bottom	3797.4105	-2941.5276

Story	Location	MX kip-ft	MY kip-ft
Story1	Top	3530.9979	-2839.9713
Story1	Bottom	3556.4083	-2941.5276
Story1	Top	4437.4552	-3549.9642
Story1	Bottom	4596.1368	-3676.9094
Story1	Top	9621.4552	-7697.1642
Story1	Bottom	9780.1368	-7824.1094
Story1	Top	4437.4552	-3549.9642
Story1	Bottom	4596.1368	-3676.9094
Story1	Top	9621.4552	-7697.1642
Story1	Bottom	9780.1368	-7824.1094
Story1	Top	6212.4373	-4969.9499
Story1	Bottom	6434.5915	-5147.6732
Story1	Top	5324.9463	-4259.957
Story1	Bottom	5515.3642	-4412.2913
Story1	Top	6212.4373	-4969.9499
Story1	Bottom	6434.5915	-5147.6732
Story1	Top	13619.3463	-10895.477
Story1	Bottom	13809.7642	-11047.8113
Story1	Top	6212.4373	-4969.9499
Story1	Bottom	6434.5915	-5147.6732
Story1	Top	13619.3463	-10895.477
Story1	Bottom	13809.7642	-11047.8113
Story1	Top	10970.924	-8811.0546
Story1	Bottom	11161.3419	-8951.2996
Story1	Top	10046.9686	-8003.2595
Story1	Bottom	10237.3864	-8167.6831
Story1	Top	10508.9463	-8407.157
Story1	Bottom	10699.3642	-8559.4913
Story1	Top	10508.9463	-8407.157
Story1	Bottom	10699.3642	-8559.4913
Story1	Top	4455.6874	-3598.8653
Story1	Bottom	4598.5008	-3701.0268
Story1	Top	3531.732	-2791.0702
Story1	Bottom	3674.5454	-2917.4102
Story1	Top	3993.7097	-3194.9678
Story1	Bottom	4136.5231	-3309.2185
Story1	Top	3993.7097	-3194.9678
Story1	Bottom	4136.5231	-3309.2185
Story1	Top	10952.6918	-8781.1198
Story1	Bottom	11158.9778	-9047.6834
Story1	Top	10952.6918	-8743.1871
Story1	Bottom	11158.9778	-8806.6812
Story1	Top	10971.6581	-8762.1534
Story1	Bottom	11279.4789	-8927.1823
Story1	Top	10933.7255	-8762.1534
Story1	Bottom	11038.4767	-8927.1823
Story1	Top	3549.9642	-2858.9377
Story1	Bottom	3676.9094	-3062.0286
Story1	Top	3549.9642	-2821.005
Story1	Bottom	3676.9094	-2821.0265
Story1	Top	3568.9305	-2839.9713
Story1	Bottom	3797.4105	-2941.5276
Story1	Top	3530.9979	-2839.9713
Story1	Bottom	3556.4083	-2941.5276

4.3 Modal Results

Table 4.5 - Modal Periods And Frequencies

Case	Mode	Period sec	Frequency cyc/sec	CircFreq rad/sec	Eigenvalue rad2/sec2
Modal	1	0.194	5.166	32.4615	1053.748
Modal	2	0.144	6.964	43.7543	1914.4419
Modal	3	0.124	8.08	50.7672	2577.3098
Modal	4	0.101	9.891	62.1497	3862.5846
Modal	5	0.1	10.035	63.0504	3975.3491
Modal	6	0.094	10.678	67.0901	4501.079
Modal	7	0.093	10.81	67.9189	4612.9809
Modal	8	0.092	10.884	68.385	4676.5102
Modal	9	0.075	13.26	83.3171	6941.7322
Modal	10	0.069	14.524	91.2572	8327.8685
Modal	11	0.068	14.624	91.8879	8443.3782
Modal	12	0.062	16.023	100.6746	10135.3752

Table 4.6 - Modal Participating Mass Ratios (Part 1 of 2)

Case	Mode	Period sec	UX	UY	UZ	SumUX	SumUY	SumUZ	RX	RY	RZ
Modal	1	0.194	0	0.9856	0	0	0.9856	0	0.1154	0	0
Modal	2	0.144	0.9999	0	0	0.9999	0.9856	0	0	0.143	0
Modal	3	0.124	0	0	0	0.9999	0.9856	0	0	0	0.9932
Modal	4	0.101	0	0	0	0.9999	0.9856	0	0	0	0
Modal	5	0.1	0	0.0006	0	0.9999	0.9862	0	0.0068	0	0
Modal	6	0.094	0	0	0	0.9999	0.9862	0	0	0	0
Modal	7	0.093	0	0.0001	0	0.9999	0.9863	0	0.0011	0	0
Modal	8	0.092	0	0	0	0.9999	0.9863	0	0	0	0
Modal	9	0.075	0	0.0129	0	0.9999	0.9992	0	0.2762	0	0
Modal	10	0.069	0	0	0	0.9999	0.9992	0	0	0	0
Modal	11	0.068	0	0	0	0.9999	0.9992	0	0	0	0.0018
Modal	12	0.062	0	0	0	0.9999	0.9992	0	0	0	0

Table 4.6 - Modal Participating Mass Ratios (Part 2 of 2)

Case	Mode	SumRX	SumRY	SumRZ
Modal	1	0.1154	0	0
Modal	2	0.1154	0.143	0
Modal	3	0.1154	0.143	0.9932
Modal	4	0.1154	0.143	0.9932
Modal	5	0.1222	0.143	0.9932
Modal	6	0.1222	0.143	0.9932
Modal	7	0.1233	0.143	0.9932
Modal	8	0.1233	0.143	0.9932
Modal	9	0.3995	0.143	0.9932
Modal	10	0.3995	0.143	0.9932
Modal	11	0.3995	0.143	0.995
Modal	12	0.3995	0.143	0.995

Table 4.7 - Modal Load Participation Ratios

Case	ItemType	Item	Static %	Dynamic %
Modal	Acceleration	UX	100	99.99
Modal	Acceleration	UY	100	99.92
Modal	Acceleration	UZ	0	0

Table 4.8 - Modal Direction Factors

Case	Mode	Period sec	UX	UY	UZ	RZ
Modal	1	0.194	0	1	0	0
Modal	2	0.144	1	0	0	0
Modal	3	0.124	0	0	0	1
Modal	4	0.101	0	0.007	0	0.993
Modal	5	0.1	0	1	0	0
Modal	6	0.094	0.001	0	0	0.998
Modal	7	0.093	0	1	0	0
Modal	8	0.092	0	0.006	0	0.993
Modal	9	0.075	0	1	0	0
Modal	10	0.069	1	0	0	0
Modal	11	0.068	0	0	0	1
Modal	12	0.062	1	0	0	0

5 Design Data

This chapter provides design data and results.

5.1 Steel Frame Design

Table 5.1 - Steel Frame Design Preferences - AISC 360-10

Item	Value
Multi-Response Design	Step-by-Step - All
Frame Type	SMF
Seismic Design Category	D
Importance Factor	1
Design System Rho	1
Design System Sds	0.5
Design System R	8
Design System Omega0	3
Design System Cd	5.5
Design Provision	LRFD
Analysis Method	Direct Analysis
Second Order Method	General 2nd Order
Stiffness Reduction Method	Tau-b Fixed
Add Notional Load Case	No
Beta Factor	1.3
Beta Omega Factor	1.6
Phi (Bending)	0.9
Phi (Compression)	0.9
Phi (Tension-Yielding)	0.9
Phi (Tension-Fracture)	0.75
Phi (Shear)	0.9
Phi (Shear-Short Webbed Rolled I)	1
Phi (Torsion)	0.9
Omega (Bending)	
Omega (Compression)	
Omega (Tension-Yielding)	
Omega (Tension-Fracture)	
Omega (Shear)	
Omega (Shear-Short Webbed Rolled I)	
Omega (Torsion)	
Ignore Seismic Code?	No
Ignore Special Seismic Load?	No
Doubler Plate Plug-Welded?	Yes
HSS Welding Type	ERW
Reduced HSS Thickness	No
Consider Deflection?	Yes
DL Ratio	120
SDL+LL Ratio	120
LL Ratio	360
Total Ratio	240
Total Camber Limit	240
Pattern Live Load Factor	0.75
D/C Ratio Limit	0.95
Maximum Iterations	1